

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

APR 08 2016

Well File No.

28599

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

ND Oil & Gas Division

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed March 22, 2016
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Chalmers 5301 44-24 3BR					
Footages 926 F S L 245 F E L		Qtr-Qtr SESE	Section 24	Township 153 N	Range 101 W
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 03/22/2016 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 8218.46'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 8054.41'

Company Oasis Petroleum North America LLC		Telephone Number 281 404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date April 7, 2016	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date 4/26/2016	
By 	
Title TAYLOR ROTH Engineering Technician	



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28599

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed January 30, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other Well on pump	

Well Name and Number Chalmers 5301 44-24 3BR					
Footages 926 F S L 245 F E L		Qtr-Qtr SESE	Section 24	Township 153 N	Range 101 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 1/30/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 9999'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 9758'

Company Oasis Petroleum North America LLC		Telephone Number 281 404-9652	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>VS</i>		Printed Name Victoria Siemieniewski	
Title Regulatory Specialist		Date March 19, 2015	
Email Address vsiemieniewski@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date 4-1-2015	
By <i>[Signature]</i>	
Title JARED THUNE Engineering Technician	

Industrial Commission of North Dakota
Oil and Gas Division

Well or Facility No
28599

Verbal Approval To Purchase and Transport Oil Tight Hole **Yes**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	Kelly Johnson	(701) 580-0524

WELL INFORMATION

Well Name					Inspector
CHALMERS 5301 44-24 3BR					Richard Dunn
Well Location	QQ	Sec	Twp	Rng	County
	SESE	24	153 N	101 W	MCKENZIE
Footages	926	Feet From the	S	Line	Field
	245	Feet From the	E	Line	BAKER
					Pool
					BAKKEN
Date of First Production Through Permanent Wellhead	12/12/2014	This Is Not The First Sales			

PURCHASER / TRANSPORTER

Purchaser	Transporter
OASIS PETROLEUM MARKETING LLC	PRAIRIE FIELD SERVICES, LLC

TANK BATTERY

Central Tank Battery Number :

SALES INFORMATION This Is Not The First Sales

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD		DATE
5000	BBLS	240	BBLS	12/12/2014
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	

DETAILS

--

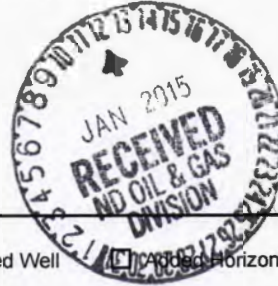
Start Date **12/12/2014**
Date Approved **12/12/2014**
Approved By **Jessica Gilkey**



WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)

Well File No.
28599



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Chalmers 5301 44-24 3BR		Spacing Unit Description Sec. 19 & 20 T 153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		State TX	
Zip Code 77002		Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 926 F S L	245 F E L	Qtr-Qtr SESE	Section 24	Township 153 N	Range 101 W	County MCKENZIE
Spud Date June 25, 2014	Date TD Reached September 17, 2014	Drilling Contractor and Rig Number NABORS B25		KB Elevation (Ft) 1995	Graded Elevation (Ft) 1970	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD; CBL from intermediate TD to surface						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	Type	String Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	6016	13 1/2	36			967	0
Vertical Hole	Intermediate	7	0	10916	8 3/4	32			1341	6009
Lateral1	Liner	4 1/2	10063	20808	6	13.5				

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20833	Perforations	10916 20833	10121		10/30/2014			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 10916-20833						Name of Zone (If Different from Pool Name)				
Date Well Completed (SEE INSTRUCTIONS) December 12, 2014			Producing Method Flowing		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In) Producing		
Date of Test 12/12/2014	Hours Tested 15	Choke Size 36 /64	Production for Test		Oil (Bbls) 864	Gas (MCF) 414	Water (Bbls) 3690	Oil Gravity-API (Corr.) °	Disposition of Gas Sold	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI) 2100		Calculated 24-Hour Rate	Oil (Bbls) 1382	Gas (MCF) 662	Water (Bbls) 5904	Gas-Oil Ratio 479		

PLUG BACK INFORMATION

[illegible]

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 10/30/2014	Stimulated Formation Bakken	Top (Ft) 10910	Bottom (Ft) 20833	Stimulation Stages 36	Volume 206015	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 4076819	Maximum Treatment Pressure (PSI) 9177		Maximum Treatment Rate (BBLS/Min) 75.0	
Details 40/70 Ceramic: 1513408 30/50 Ceramic: 2259661 100 mesh: 303750						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

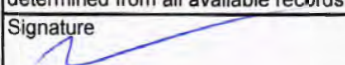
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

--	--	--

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jswenson@oasispetroleum.com	Date 01/08/2015
	Signature 	Printed Name Jennifer Swenson

**AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE – Form 8**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No. 28599
NDIC CTB No. 220407

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number CHALMERS 5301 44-24 3BR	Qtr-Qtr SESE	Section 24	Township 153	Range 101	County Mckenzie
--	------------------------	----------------------	------------------------	---------------------	---------------------------

Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
--	---	-----------------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective December 12, 2014
---	---	----------------------------	--

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Field Address	City	State	Zip Code
---------------	------	-------	----------

Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective December 12, 2014
---	---	-----------------------------	--

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
---------------------------------	---------------------	--------------------	--------------------------

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other Transporters Transporting From This Lease	% Transported	Date Effective
---	---------------	----------------

Prairie Field Services, LLC	25%	December 12, 2014
------------------------------------	------------	--------------------------

Other Transporters Transporting From This Lease	% Transported	Date Effective
---	---------------	----------------

		December 12, 2014
--	--	--------------------------

Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Date December 18, 2014
--	----------------------------------

Signature 	Printed Name Brianna Salinas	Title Marketing Assistant
---------------	--	-------------------------------------

Above Signature Witnessed By:	Printed Name	Title
-------------------------------	--------------	-------

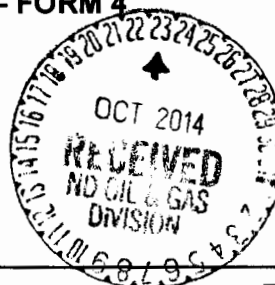
Signature 	Printed Name Dina Barron	Title Mktg. Contracts Administrator
---------------	------------------------------------	---

FOR STATE USE ONLY

Date Approved JAN 16 2015
By
Title Oil & Gas Production Analyst

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28599

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 22, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Change well status to CONFIDENTIAL

Well Name and Number Chalmers 5301 44-24 3BR							
Footages		Qtr-Qtr	Section	Township	Range		
926 F S L 245 F E L		SESE	24	153 N	101 W		
Field Baker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective immediately, we request CONFIDENTIAL STATUS for the above referenced well.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Assistant		Date October 22, 2014	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 10/24/14	
By 	
Title Engineering Technician	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



CTB
Well File No.
220407-01

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 15, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Central production facility-commingle prod

Well Name and Number (see details)					
Footages		Qtr-Qtr	Section	Township	Range
F	L	F	L	20	153 N 100 W
Field		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests approval to commingle oil and gas in a central production facility known as: 5300 19-20 CTB 2 with common ownership for the following wells:

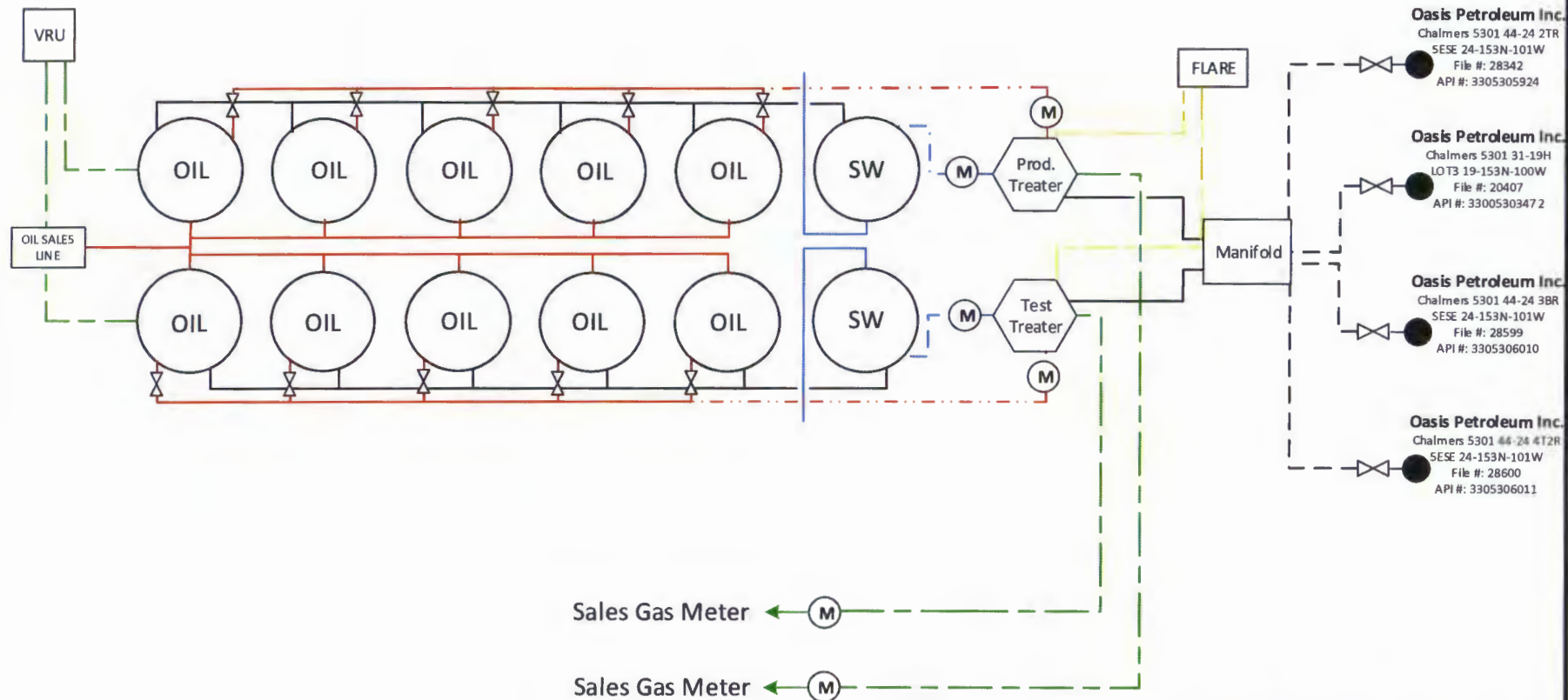
Well File #28342 Chalmes 5301 44-24 2TR SESE Sec 24-153N-R101W API 3305305924
Well File #20407 Chalmers 5301 31-19H LOT3 Sec 19-153N-101W API 33005303472
Well File #28599 Chalmers 5301 44-24 3BR SESE Sec 24 153N-101W API 3305306010
Well File #28600 Chalmers 5301 44-24 4T2R SESE Sec 21 153N-101W API 3305306011

Please find the following attachments:

1. A schematic drawing of the facility which diagrms the testing, treating, routing, and transferring of production. 2. A plat showing the location of the central facility 3. Affidavit of title indicating common ownership. Oasis will allocate production measured at the central production facility to the various wells on the basis of isolated production tests utilizing oil, gas and water meters on a test seperator at the central production facility. Oasis will measure the production from each well separately each month for a minimum of three days. Oasis believes that such allocation will result in an accurate determination of production from each well. Tank vapor gas is being recovered and burned by a 98% DRF enclosed combuster

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9591	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Chammoniston</i>	Printed Name Chelsea Covington		
Title Regulatory Assistant	Date October 13, 2014		
Email Address ccovington@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 10-17-2014	
By <i>David Tabon</i>	
Title PETROLEUM ENGINEER	



				
CHALMERS 5300 19-20 CENTRAL TANK BATTERY 2				
DATE	REV.	BY	APPR.	SCALE
SEPTEMBER 23, 2014	0	LEE		NA
LOCATION		FIELD		
NORTH DAKOTA		BAKER		

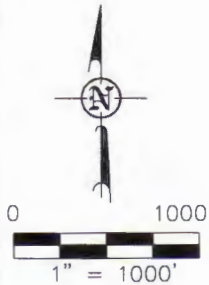
BATTERY LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"5300 19-20 CTB(A)"

SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL D.
KASEMAN, PLS, REGISTRATION
NUMBER 3880 LS ON 10/7/14
AND THE ORIGINAL DOCUMENTS ARE
STORED AT THE OFFICES OF
INTERSTATE ENGINEERING, INC.



24

FOUND REBAR
W/ 2" AC
LS 2352

R101W
R100W

AZ 90°03'35"

FOUND REBAR
W/ 2" AC
LS 2352

2630.15'

LOT 1

2631.88'

LOT 2

AZ 0°04'59"

2631.68'

LOT 3

AZ 0°05'03"

LOT 4

CHALMERS 5301 44-24 2TR
CHALMERS 5301 44-24 3BR
CHALMERS 5301 44-24 4T2R

FOUND STONE
W/ AC

AZ 90°44'54"

5267.09'

FOUND STONE
& REBAR

AZ 90°03'03"

2651.37'

FOUND STONE
& REBAR

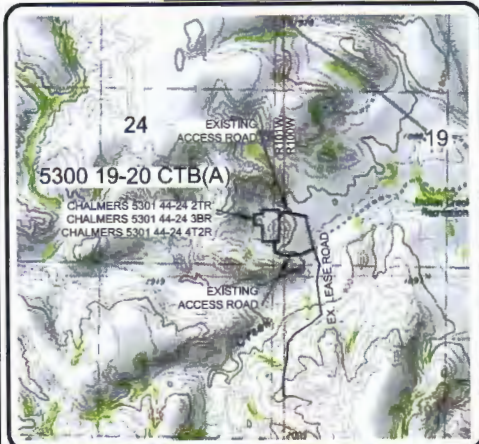
LOT 6

LOT 7

EDGE OF
LAKE

19

VICINITY MAP



MONUMENT - RECOVERED

MONUMENT - NOT RECOVERED

STAKED ON 9/3/2013

VERTICAL CONTROL DATUM WAS BASED UPON
GPS CONTROL POINT 16 WITH AN ELEVATION OF 2014.2'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS
PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY
SUPERVISION AND IS TRUE AND CORRECT TO
THE BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]



© 2014, INTERSTATE ENGINEERING, INC.

DARYL D. KASEMAN LS-3880

1/5

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

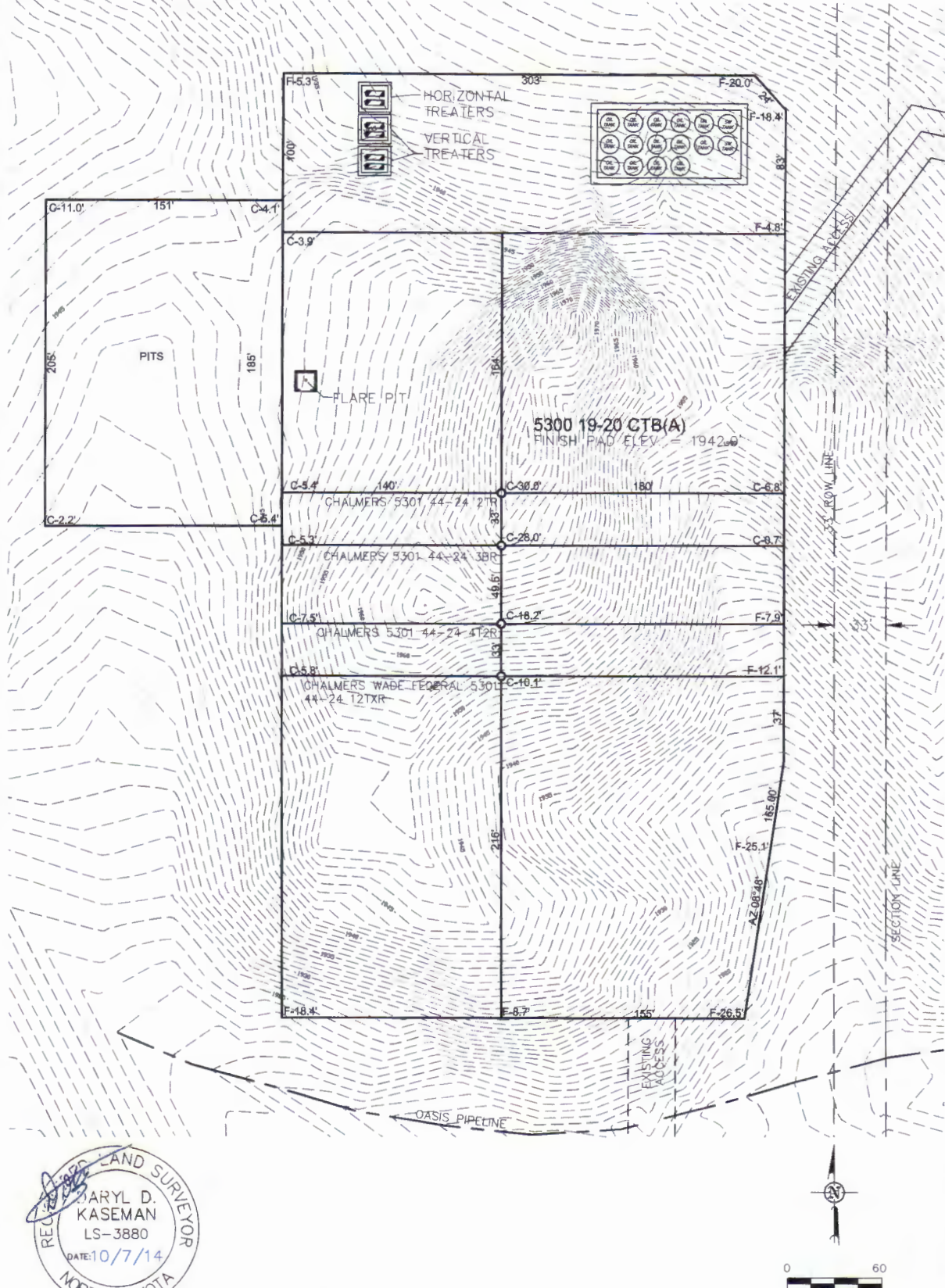
OASIS PETROLEUM NORTH AMERICA, LLC
BATTERY LOCATION PLAT
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No.: S14-08-241
Checked By: D.D.K. Date: OCT. 2014

Revision No.	Date	By	Description

THIS DOCUMENT WAS ORIGINALLY ISSUED
AND SEALED BY DARYL D. KASEMAN,
P.L.S., REGISTRATION NUMBER 3880 ON
10/7/14 AND THE ORIGINAL
DOCUMENTS ARE STORED AT THE
OFFICES OF INTERSTATE ENGINEERING,
INC.

PAD LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"5300 19-20 CTB(A)"
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



RECEIVED LAND SURVEYOR
DARYL D. KASEMAN
LS-3880
DATE: 10/7/14
NORTH DAKOTA

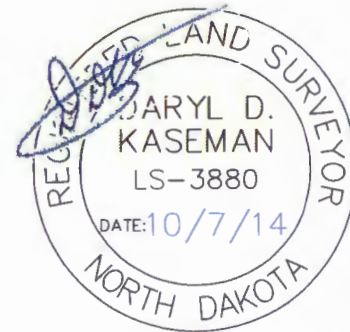
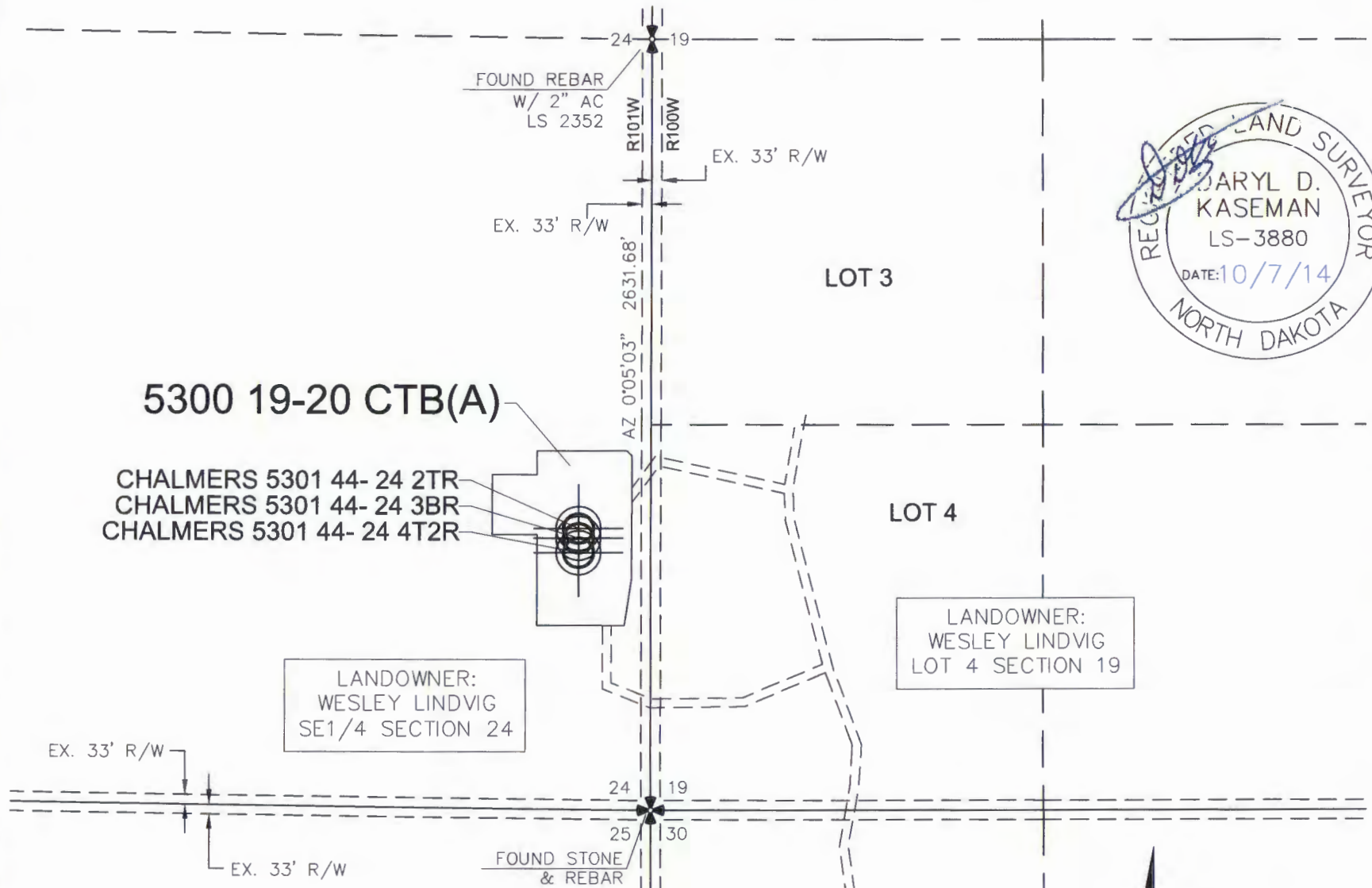


NOTE: All utilities shown are preliminary only, a complete
utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.

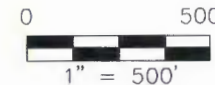
2/5 SHEET NO.	INTERSTATE ENGINEERING Professionals you need, people you trust	Interstate Engineering, Inc. P.O. Box 648 425 East Main Street Sidney, Montana 59270 Ph (406) 433-5617 Fax (406) 433-5618 www.interstateeng.com Other offices in Minnesota, North Dakota and South Dakota	OASIS PETROLEUM NORTH AMERICA, LLC PAD LAYOUT SECTION 24, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA	<table border="1"> <thead> <tr> <th>Revision No.</th> <th>Date</th> <th>By</th> <th>Description</th> </tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> </tbody> </table>	Revision No.	Date	By	Description												
			Revision No.	Date	By	Description														
Drawn By: J.D.M. Project No.: 514-09-241 Checked By: D.D.K. Date: OCT 2014																				

ACCESS APPROACH
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "5300 19-20 CTB(A)"
 SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 10/7/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



© 2014, INTERSTATE ENGINEERING, INC.

3/5 SHEET NO.	INTERSTATE ENGINEERING Professionals you need, people you trust	Interstate Engineering, Inc. P.O. Box 648 425 East Main Street Sidney, Montana 59270 Ph (406) 433-5617 Fax (406) 433-5618 www.interstateeng.com Other offices in Minnesota, North Dakota and South Dakota	OASIS PETROLEUM NORTH AMERICA, LLC ACCESS APPROACH SECTION 24, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA Drawn By: J.D.M. Checked By: D.D.K. Project No: S14-09-241 Date: OCT. 2014	Date By: _____ Description: _____ Revision No.: _____
		DARYL D. KASEMAN LS-3880 DATE: 10/7/14		



OASIS PETROLEUM NORTH AMERICA, LLC
 5300 19-20 CTB(A)
 QUAD LOCATION MAP
 SECTION 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

5300 19-20 CTB(A)

CHALMERS 5301 44-24 2TR
 CHALMERS 5301 44-24 3BR
 CHALMERS 5301 44-24 4T2R

EXISTING
ACCESS ROAD

EXISTING
ACCESS ROAD

EX LEASE ROAD

Indian Creek Bay
Recreation Area

MISSOURI



© 2014, INTERSTATE ENGINEERING, INC.

4/5

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No.: S14-09-251
 Checked By: D.D.K. Date: OCT 2014

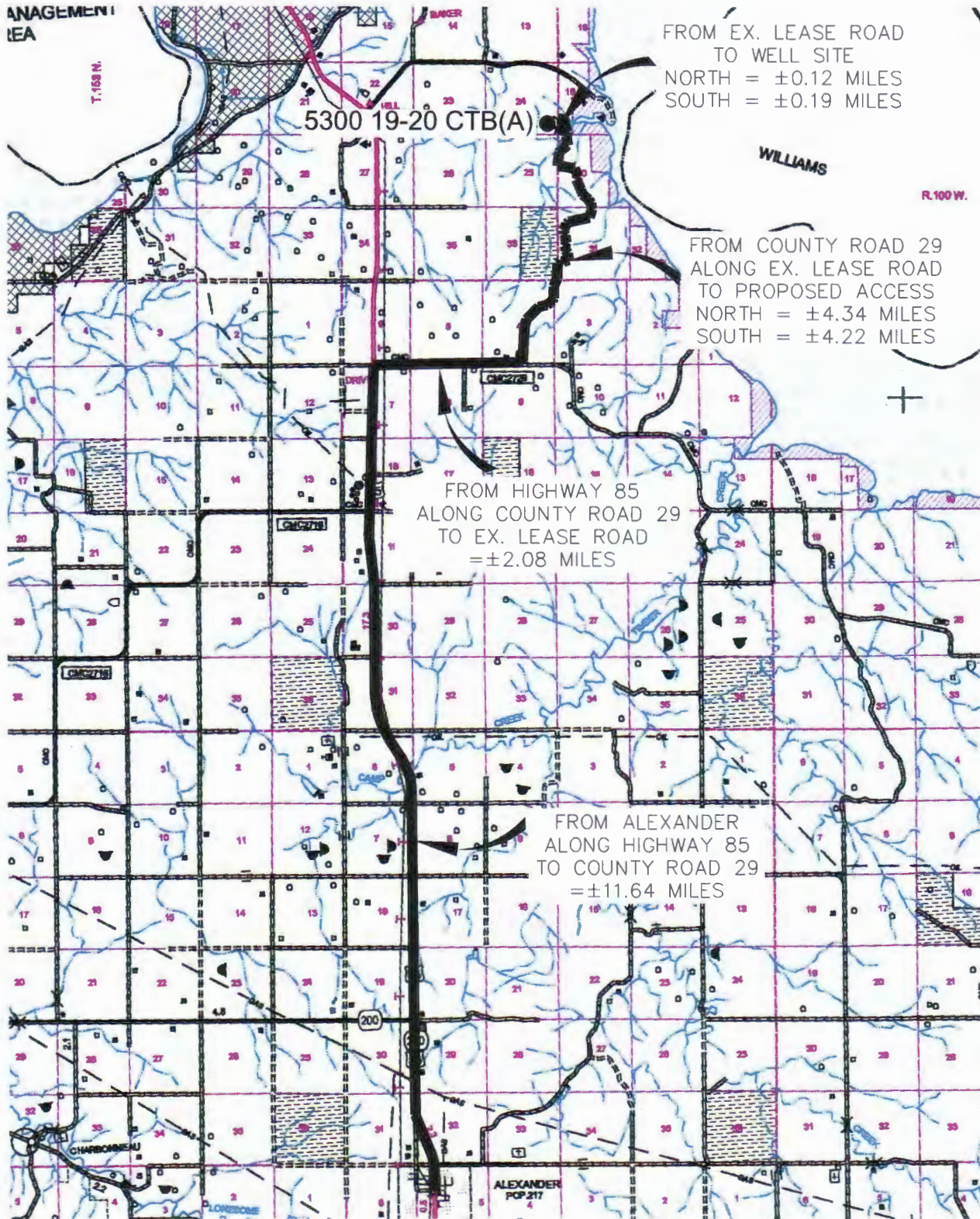
Revision No.	Date	By	Description

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"5300 19-20 CTB(A)"

SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

5/5

SHEET NO.

INTERSTATE ENGINEERING
Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

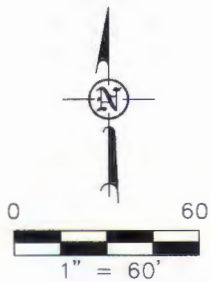
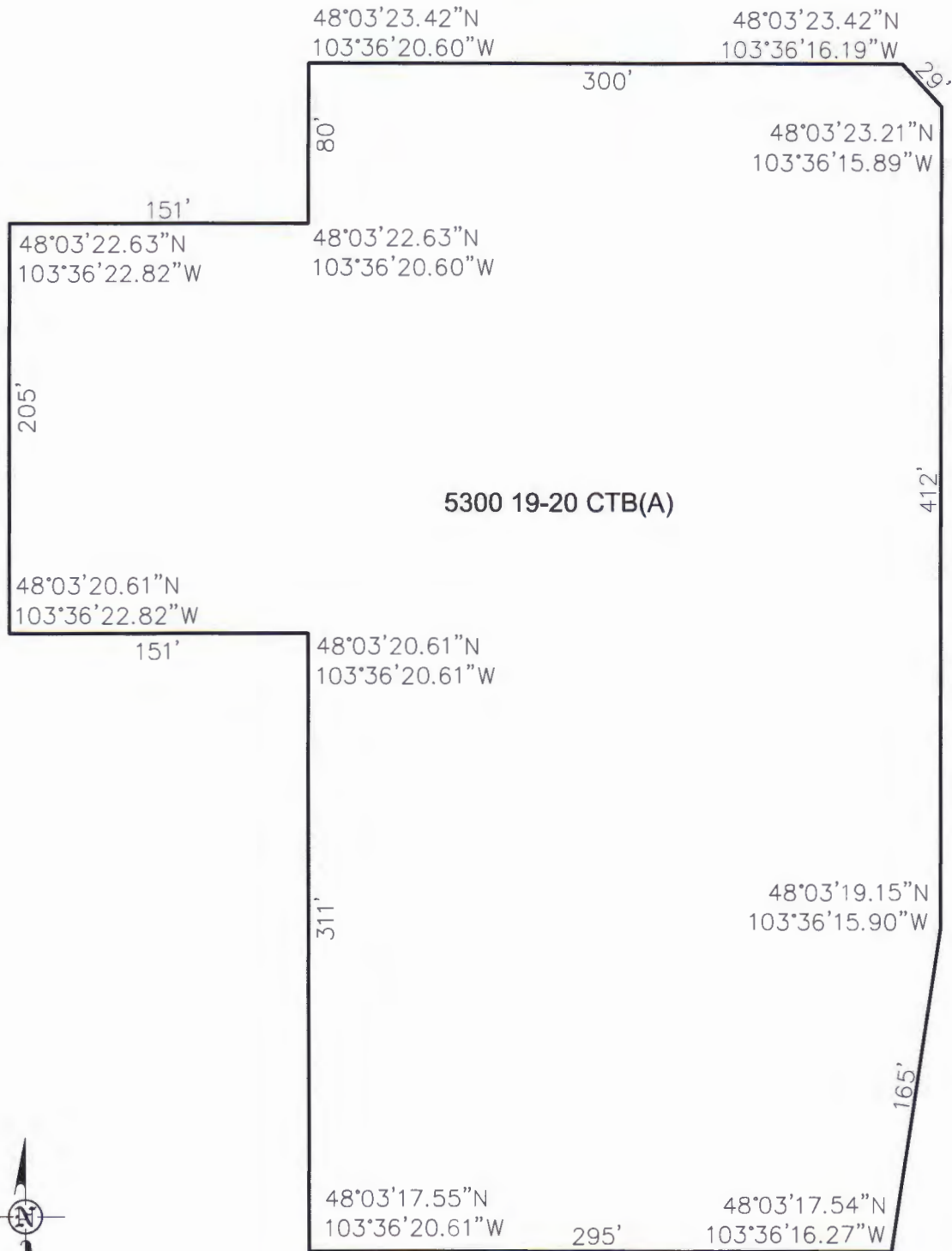
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No.: S14-09-241
Checked By: D.D.K. Date: OCT 2014

Revision No.	Date	By	Description

LAT/LONG PAD CORNERS



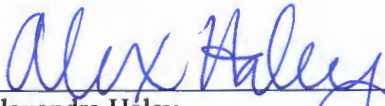
COMMINGLING AFFIDAVIT

STATE OF NORTH DAKOTA)
) ss.
COUNTY OF WILLIAMS)

The under signed, Alexandra Haley, of lawful age, being first duly sworn on her oath states that she is a duly authorized agent of Oasis Petroleum North America LLC, and that she has personal knowledge of the facts hereinafter set forth to make this Affidavit.

1. Sections 19 & 20, Township 153 North, Range 100 West, Williams County North Dakota constitute a spacing unit in accordance with the applicable orders for the Bakken pool.
2. Three wells have been drilled in the spacing unit, which are known as the Chalmers 5300 44-24 2TR, Chalmers 5300 44-24 3BR, and Chalmers 5300 44-24 4T2R.
3. By NDIC Order 19005 dated May 11, 2012 and recorded in Williams County as Document No. 737164, all oil and gas interest within the aforementioned spacing unit were pooled.
4. All Working Interests, Royalty Interests and Overriding Royalty Interests in the Chalmers 5300 44-24 2TR, Chalmers 5300 44-24 3BR, and Chalmers 5300 44-24 4T2R will be in common.

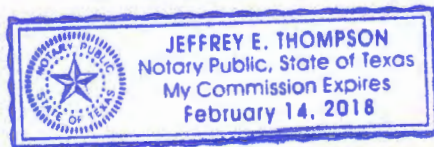
Dated this 15th day of September, 2014

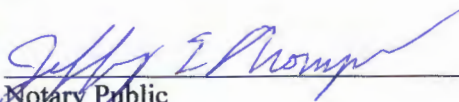


Alexandra Haley
Landman

STATE OF TEXAS)
) ss.
COUNTY OF HARRIS)

Subscribed to and sworn before me this 15th day of September, 2014





Notary Public
State of Texas
My Commission Expires: 2-14-18



Oasis Petroleum North America, LLC

Chalmers 5301 44-24 3BR

926' FSL & 245' FSL

SE SE Section 24, T153N, R101W

Baker Field / Middle Bakken

McKenzie County, North Dakota

BOTTOM HOLE LOCATION:

307' south & 10,439' east of surface location or approx.

619' FSL & 320' FEL, SE SE Section 20, T153N, R100W

Prepared for:

Clay Hargett
Oasis Petroleum North America, LLC
1001 Fannin Street, Suite 1500
Houston, Texas 77002

Prepared by:

Hannah Thatcher, Daniel Haynes, Kate Schwehr
PO Box 80507; Billings, MT 59108
(406) 259-4124
geology@sunburstconsulting.com
www.sunburstconsulting.com

WELL EVALUATION

Chalmers 5301 44-24 3BR



**Figure 1: Nabors B25 drills the Oasis Petroleum North America Chalmers 5301 44-24 3BR on July 3, 2014, south of Williston in McKenzie County, North Dakota.
(Kate Schwehr, wellsite geologist)**

INTRODUCTION

The Oasis Petroleum North America, LLC Chalmers 5301 44-24 3BR well [SE SE Section 24, T153N, R101W] is located approximately 30 miles south of Williston, North Dakota. The horizontal well was spud on June 11, 2014 and represents a continuation of Oasis Petroleum's development of the Bakken Formation within Baker Field. The Chalmers 3BR was planned to drill an approximately 9,953' lateral along a proposed azimuth of 91.52°. The well bore will be enhanced for production by multistage fracture stimulation.

crew in 30' intervals from 8,200' MD to 10,780' MD, 10' intervals from 10,780' MD to 10,939' MD, and 30' intervals from 10,939' MD to 20,833' TD. Wet and dry cuttings were examined under a tri-ocular microscope and complete lithologic descriptions and sampling intervals are provided in the lithology document within this evaluation. The set of dry cuttings collected during the duration of the well were sent to the State of North Dakota. Evidence of light hydrocarbons present in the drilling fluid was reported by the gas chromatography equipment and documented on the mud log presented with this report.

Zones of Interest:

The Mission Canyon Formation (Mississippian; Madison Group) was drilled at 9,291' TVD (-7,323') and is comprised largely of light gray lime mudstone followed by gray to tan earthy textured argillaceous lime mudstone. Shows within the Mission Canyon ranged from 20 to 94 units against 9.8 ppg mud.



Figure 3 & 4: Wet cutting of the Lodgepole Formation (left) and False Bakken sub-interval (right) at 10X.

The top of the Lodgepole Formation was logged at 9,849' TVD (-7,881'). In general the Lodgepole can be described as a medium to dark gray brown argillaceous lime mudstone with a crystalline texture and trace amounts of disseminated pyrite (Figure 3). The False Bakken (Carrington Shale), drilled at 10,584' TVD (-8,616'), is comprised of very dark brown to black, slightly pyritic shale with an earthy texture, and was found to be soft to firm (Figure 4). Strong hydrocarbon shows in the lower 100' of the Lodgepole were as high as 920 total gas units; suggest that some of the oil and gas from the Upper Bakken Shale may be exploiting fractures thought to exist in the Lower Lodgepole.

The Bakken Formation (Devonian – Mississippian) has four formal members, an upper and lower black, fissile, organic-rich pyritic shale, separated by an arenaceous limestone, siltstone and silty sandstone middle member. These three members overlay a silty shale or siltstone of the Pronghorn Member toward the basin depositional center. The Upper Bakken Shale was drilled at 10,592' TVD (-8,624') with sample returns typically described as black, carbonaceous, *petroliferous* shale with trace amounts of disseminated pyrite (Figure 5). The Middle Bakken, penetrated at 10,612' TVD (-8,644'), consists of a varying thickness and sequence of interbedded

siltstone, limestone and silty sandstone (Figures 6 & 7). *Trace spotty light brown oil staining* was common.



Figure 5, 6 & 7: Wet cuttings of the Upper Bakken Shale (left), Middle Bakken siltstone (center) and Middle Bakken silty sandstone (right) at 10X (photos by Kate Schwehr, wellsite geologist).

Geo-steering:

Kick-off point for the curve was established from the isopach of the “base last salt” marker to the Middle Bakken “target” in the offset wells. The Chalmers 5301 44-24 2TR was used as the primary offset through the vertical and curve sections. While drilling the curve, measured gamma ray signatures were compared to those of the three offsets and aided in the landing of the curve. The landing target was confirmed by the depth of the False Bakken, which was consistent with the offset wells. The curve was successfully landed within the Middle Bakken Member at a depth of 10,939’ MD (10,621’ TVD) placing the well bore approximately 9’ below the top of the Middle Bakken Formation. Directional tools were then pulled out of the hole and a string of 7” casing was set (10,898’ MD) and then cemented by Schlumberger.

Samples from the target zone varied in porosity, oil staining, color, cementation, and pyrite content. Stratigraphic location in the target zone was based on these sample observations along with gas shows, drill rates and gamma ray values. Severe doglegs were to be avoided so as to obtain the desired vertical section and aid in a successful completion liner run at TD.

The Oasis Petroleum North America, LLC prospect geologist defined the an initial target zone as an 13’ zone that began 11’ below the base of the Upper Bakken Shale and ended 16’ above the top of the Lower Bakken Shale. The target zone consisted of an upper silty sandstone interval reading 90-140 count gamma (A marker). The center of the target interval was comprised of slightly less silty sandstone reading 80-140 count gamma (B marker). Underlying the B Marker was another sandstone layer with gamma readings of 80-120 (C marker). The A-C gamma markers were used for determining depth within the target interval and plotted on the Chalmers 5301 44-24 3BR dip profile (Figure 9).

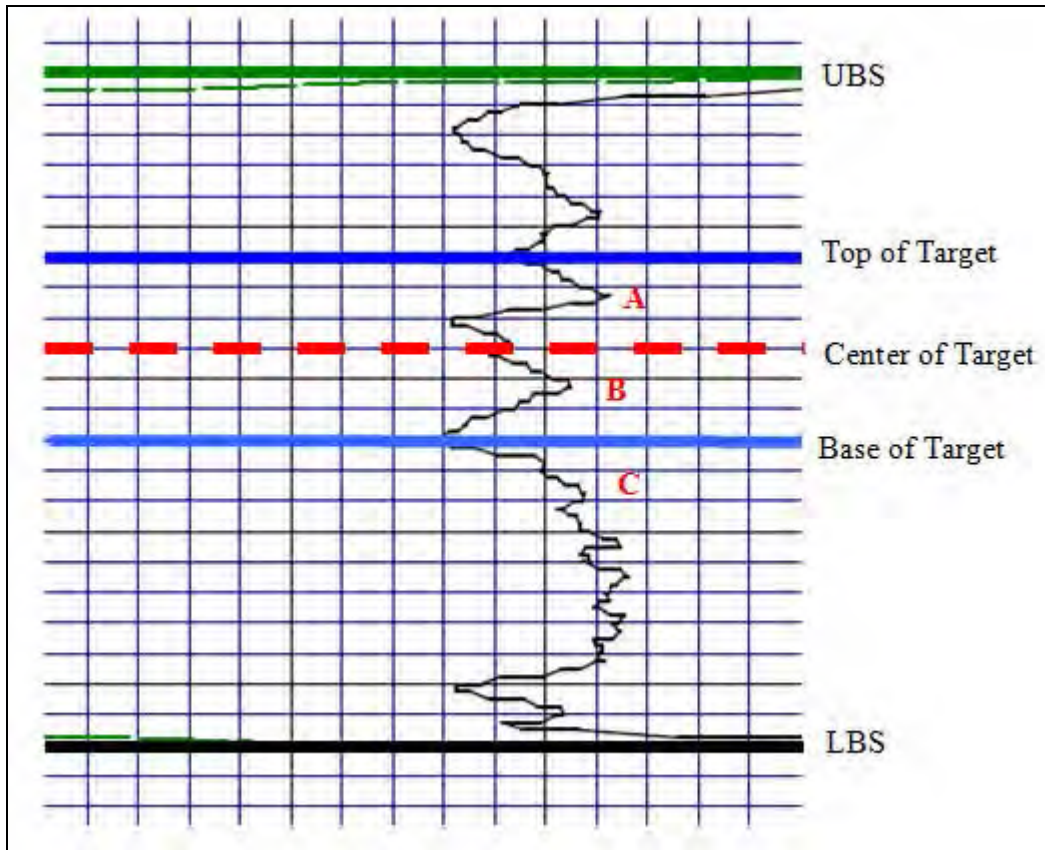


Figure 8: Offset gamma ray profile of the Middle Bakken (0-100 api. scale) *See dip profile (Figure 9) for marker presentation*

Using the aforementioned information gathered during drilling, offset log information and the structural data provided from Oasis Petroleum North America, LLC., well site geologists were able to anticipate, and interpret the local apparent dip during the drilling of the Chalmers 5301 44-24 3BR well. A total depth of 20,833' MD was reached on September 16, 2014 at 12:30 CDT. The target resultant was 100% within the Middle Bakken Member. The resulting structure of the Middle Bakken was a drop in TVD of 87.90' over 9,983' MD; resulting in an overall 0.50 down dip as portrayed on the Chalmers 5301 44-24 3BR dip profile (Figure 9).

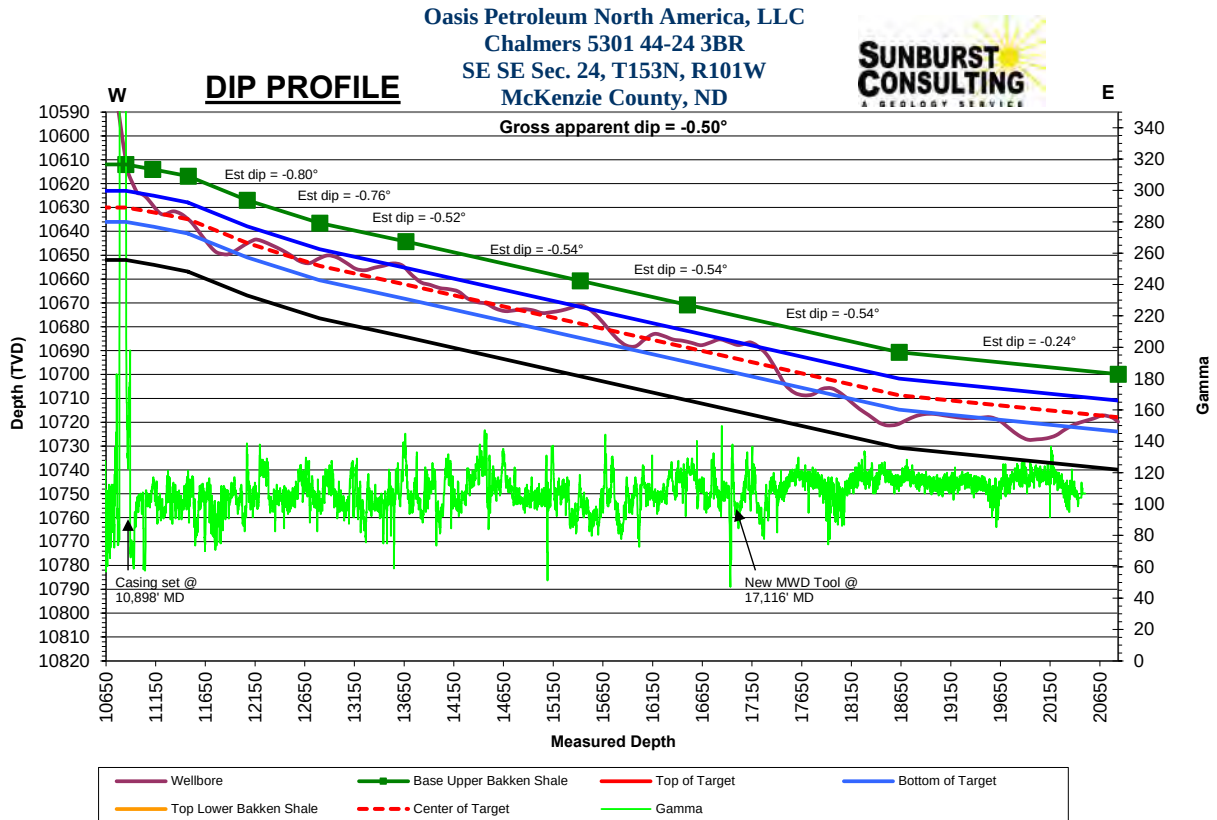


Figure 9: Well profile representing estimated dip value & gamma ray for the Chalmers 5301 44-24 3BR lateral.

Hydrocarbons:



Figures 10 & 11: 30' flare observed during Chalmers 5301 44-24 3BR lateral.

Gas monitoring and fluid gains were monitored to evaluate the viability of this reservoir during the drilling of the Chalmers 5301 44-24 3BR well. In the invert mud system, hydrostatic conditions were maintained near balance. This allowed for gas and fluid gains from the well to be monitored. Gas shows were encountered in vertical, curve and lateral drilling of the well.

Background gas observed during the drilling of the lateral ranged from 1,000 to 3,000 units. Invert drilling fluid was used throughout the drilling of the vertical and curve weighing 9.8-11.7 ppg. The lateral was drilled with saline drilling fluid with a mud weight of 9.7-10.15 ppg. Gas shows ranged up to 3,728 units and connection gases were observed up to 3,620 units. C1-C4 gas components were observed throughout the course of the lateral. Trip gases were observed as high as 5,976 units. Oil shows were very light throughout the lateral ranging from 0-3% in sample. When present it was a *light brown spotty oil stain* that yielded a *slow to moderate streaming to diffuse pale yellow cut fluorescence*.

SUMMARY

The Nabors B25 drilling rig successfully drilled a two-section horizontal well bore within the Three Forks Formation at the Chalmers 5301 44-24 3BR. A net of 9,894' was drilled within the Middle Bakken. A mud program consisting of diesel invert (9.8–11.7 ppg), during the vertical and curve build sections, and saline based mud (9.7-10.15 ppg), during the lateral maintained stable hole conditions and permitted adequate analysis of gas concentrations.

Projections of dip were successfully used to maintain the well bore in the Middle Bakken Member target for 85% of the lateral. Samples from the target consisted of a light to medium gray, light to medium gray-brown silty sandstone with trace to occasional amounts of disseminated pyrite and calcite cement. Intergranular porosity was generally seen throughout the entire lateral. Hydrocarbon shows in the target zone were high throughout the lateral. Samples from the ideal zone contained a spotty light brown oil stain.

The Chalmers 5301 44-24 3BR will be fitted with a 4 ½" production liner and swell packers in preparation for a fracture stimulation that will determine the true commercial value of the well. The well currently awaits fracture stimulation.

Respectfully submitted,

Hannah Thatcher

Kate Schwehr

Sunburst Consulting, Inc.

September 16, 2014

WELL DATA SUMMARY

<u>OPERATOR:</u>	Oasis Petroleum North America, LLC
<u>ADDRESS:</u>	1001 Fannin Street, Suite 1500 Houston, Texas 77002
<u>WELL NAME:</u>	Chalmers 5301 44-24 3BR
<u>API #:</u>	33-053-06010-00-00
<u>WELL FILE #:</u>	28599
<u>SURFACE LOCATION:</u>	926' FSL & 245' FSL SE SE Section 24, T153N, R101W
<u>FIELD/ OBJECTIVE:</u>	Baker Field / Middle Bakken
<u>COUNTY, STATE</u>	McKenzie County, North Dakota
<u>BASIN:</u>	Williston
<u>WELL TYPE:</u>	Middle Bakken Horizontal
<u>ELEVATION:</u>	GL: 1,943' KB:1,968'
<u>SPUD/ RE-ENTRY DATE:</u>	June 11, 2014
<u>BOTTOM HOLE LOCATION:</u>	307' south & 10,439' east of surface location or approx. 619' FSL & 320' FEL, SE SE Section 20, T153N, R100W
<u>CLOSURE COORDINATES:</u>	Closure Azimuth: 91.69° Closure Distance: 10,443.29'
<u>TOTAL DEPTH / DATE:</u>	20,833' on September 16, 2014 85% within target interval
<u>TOTAL DRILLING DAYS:</u>	19 days
<u>CONTRACTOR:</u>	Nabors B25

<u>PUMPS:</u>	H &H Triplex (stroke length - 12")
<u>TOOLPUSHERS:</u>	Casey Pippenger, Bruce Walter
<u>FIELD SUPERVISORS:</u>	Mike Crow, Travis Handran
<u>CHEMICAL COMPANY:</u>	Fluid Control
<u>MUD ENGINEER:</u>	Keith McCarty, Warren Carlson
<u>MUD TYPE:</u>	Fresh water in surface hole Diesel invert in curve; Salt water in lateral
<u>MUD LOSSES:</u>	Invert Mud: 1,186 bbls., Salt Water: Not tracked
<u>PROSPECT GEOLOGIST:</u>	Clay Hargett
<u>WELLSITE GEOLOGISTS:</u>	Hannah Thatcher, Daniel Haynes, Kate Schwehr
<u>GEOSTEERING SYSTEM:</u>	Sunburst Digital Wellsite Geological System
<u>ROCK SAMPLING:</u>	30' from 8,200' - 10,780' 10' from 10,780' -10,939' 30' from 10,939' - 20,833' (TD)
<u>SAMPLE EXAMINATION:</u>	Trinocular microscope & fluoroscope
<u>SAMPLE CUTS:</u>	Trichloroethylene
<u>GAS DETECTION:</u>	MSI (Mudlogging Systems, Inc.) TGC - total gas with chromatograph Serial Number(s): ML-382
<u>DIRECTIONAL DRILLERS:</u>	RPM, Inc. Mike Crow, Travis Handran, Will Wright
<u>MWD:</u>	Ryan Directional Service Inc. Dave Unger, Daniel Ogden
<u>CASING:</u>	Surface: 13 3/4" 55# J-55 set to 2,095' Intermediate Surface: 9 5/8" 40# HCL-110 set to 5,982' Intermediate: 7" 32# HCP-110 set 10,898'
<u>SAFETY/ H₂S MONITORING:</u>	Oilind Safety

KEY OFFSET WELLS:

Oasis Petroleum North America

Chalmers 5301 44-24 2TR

SE SE Sec. 24, T153N, R101W

McKenzie Co., ND

KB: 1,968'

Oasis Petroleum North America

Chalmers 5300 31-19H

NW SW Sec. 19, T153N, R100W

McKenzie Co., ND

KB: 1,929'

Missouri Basin Well

Lynn 1

SW NW Sec. 23, T153N, R101W

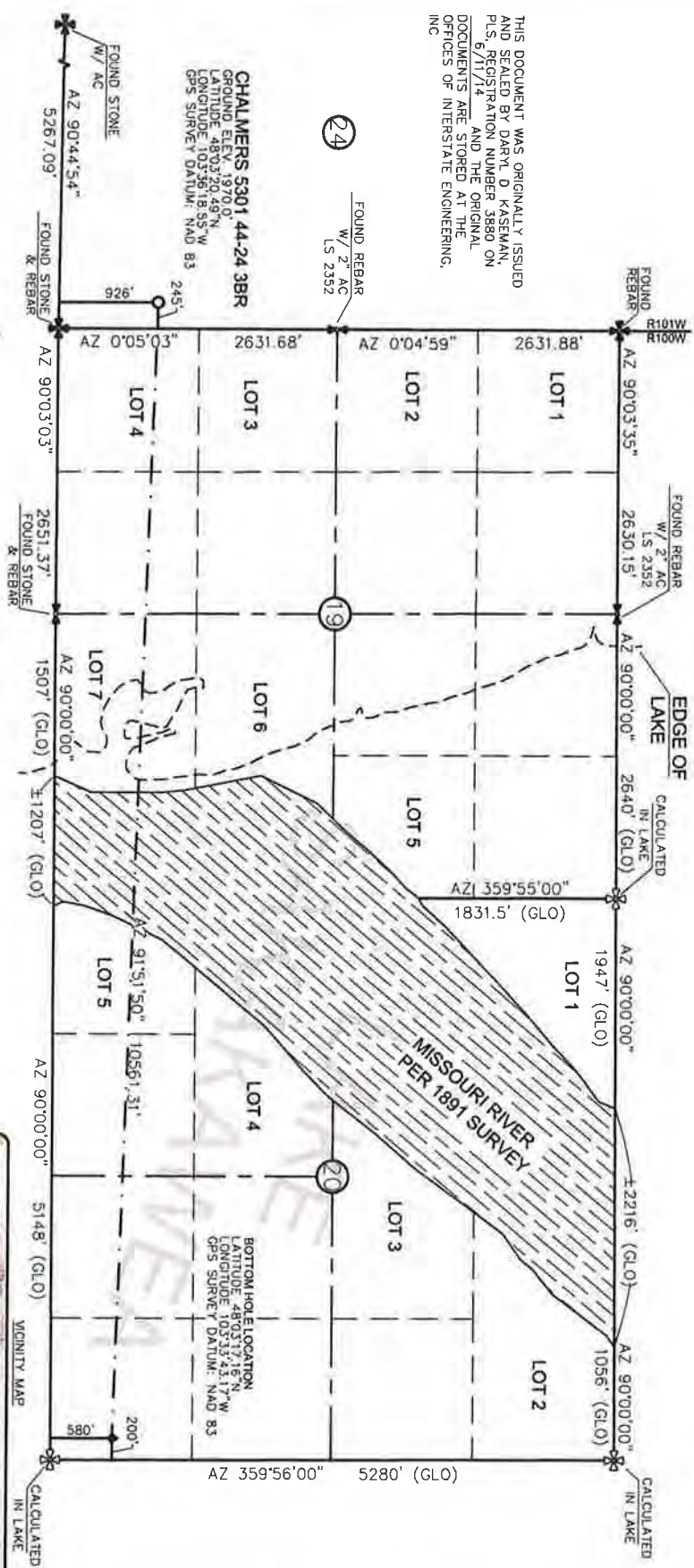
McKenzie Co., ND

KB: 2,142'

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS. 6/11/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5301 44-24 3BR"

826 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
 SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



MONUMENT - RECOVERED
 MONUMENT - NOT RECOVERED



STAKED ON 9/3/13
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 16 WITH AN ELEVATION OF 2014.2'
 THIS SURVEY AND PLAT IS BEING PROVIDED AT THE
 REQUEST OF ERIC BAYES OF OASIS PETROLEUM
 THAT THIS PLAT CORRECTLY REPRESENTS WORK
 PERFORMED BY ME OR UNDER MY SUPERVISION
 AND IS TRUE AND CORRECT TO THE BEST OF MY
 KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880



© 2014, INTERSTATE ENGINEERING, INC.

1/8

INTERSTATE ENGINEERING

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph: (406) 433-5017
 Fax: (406) 433-5018
 www.interstateeng.com

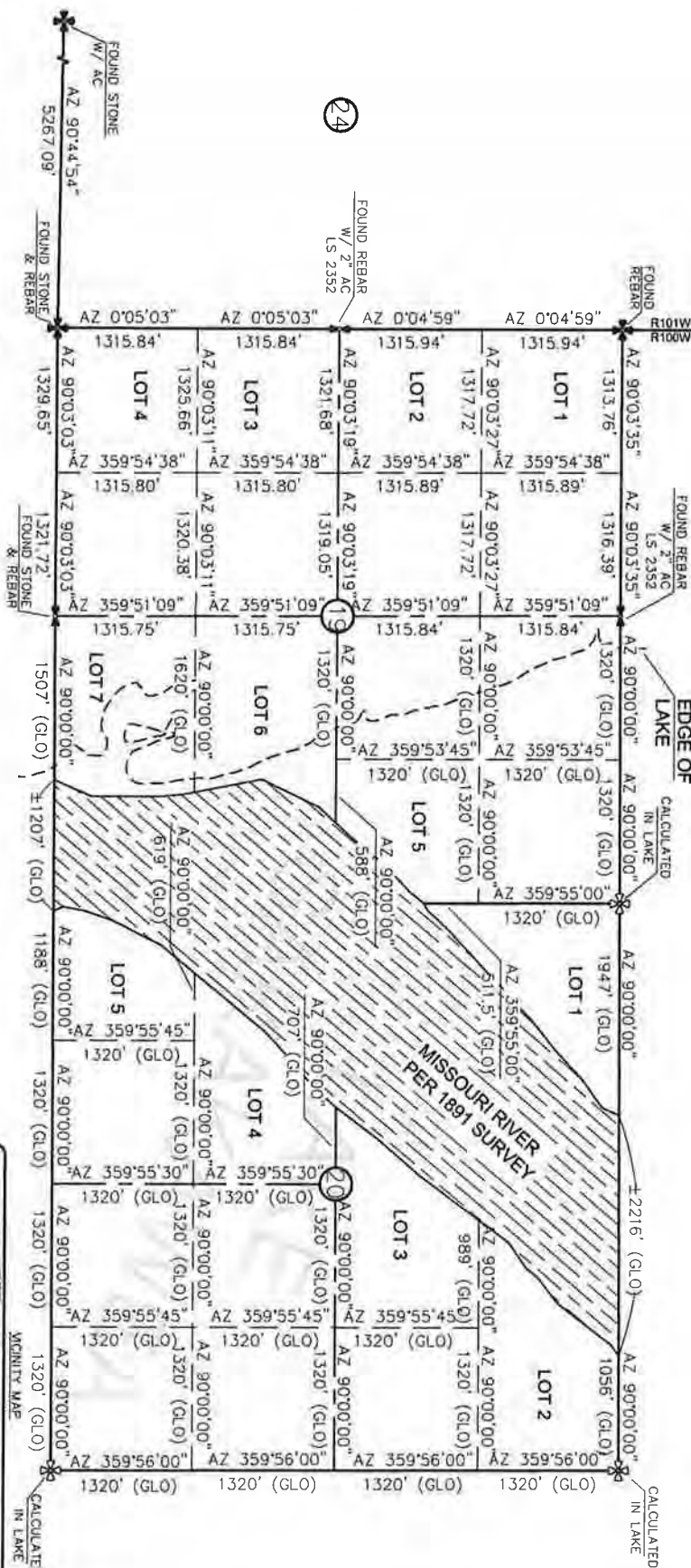
OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

Revision No.	Date	By	Description
REV 1	1/28/11	BNH	CHANGES AND LAYOUT
REV 2	5/26/11	JAN	ADDED WELL TO PAD
REV 3	5/26/11	JAN	CHANGED NAME, W2ED WELL
REV 4	5/27/11	BNH	ADDED DIMENSIONS

SECTION BREAKDOWN

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
CHALMERS 5301 44-24 3BR

SECTION 24, T153N, R101W, & SECTION 19 & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA
928 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE



MONUMENT - RECOVERED
MONUMENT - NOT RECOVERED

0 1000
1" = 1000'

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS., REGISTRATION NUMBER 3880 ON 6/11/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.



© 2014, INTERSTATE ENGINEERING, INC.

2/8



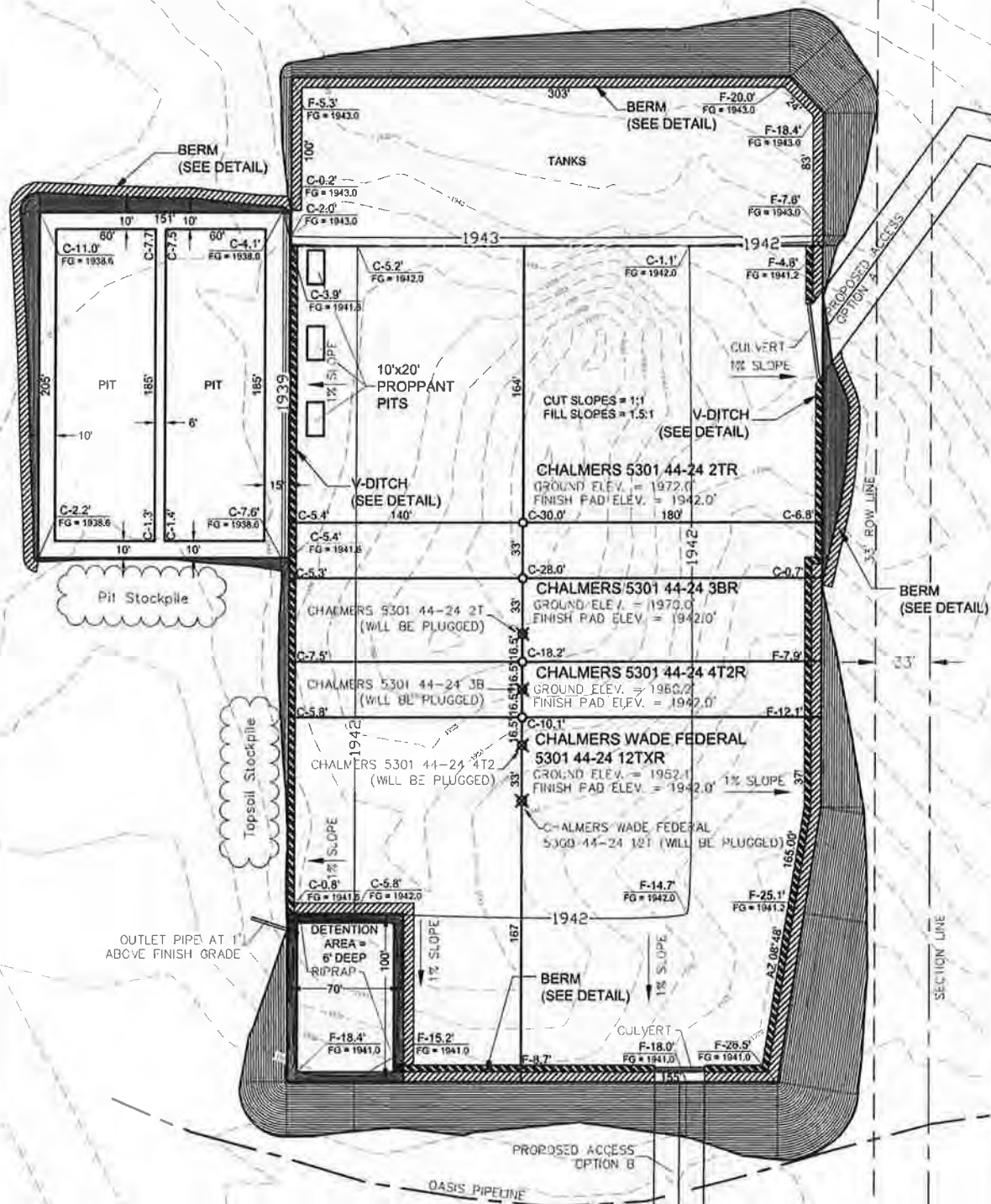
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sikeston, Missouri 65770
Ph: (408) 433-5617
Fax: (408) 433-5618
www.interstateeng.com

OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTION 24, T153N, R101W, & SECTIONS 19 & 20, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: J.L.S. Project No.: 513-09-23501
Checked By: J.D.K. Date: MAY 2014

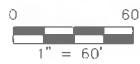
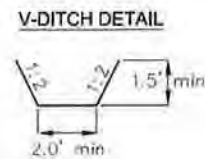
Rev	Date	By	Description
REV 1	1/20/14	BLA	CHANGED PAD LAYOUT
REV 2	1/20/14	BLA	ADDED WELL TO PAD
REV 3	1/20/14	BLA	CHANGED NAME, MOVED WELL
REV 4	6/11/14	BLA	ADDED DIMENSIONS

THIS DOCUMENT WAS ORIGINALLY ISSUED
AND SEALED BY DARYL D. KASEMAN,
P.L.S. REGISTRATION NUMBER 3880 ON
6/11/14 AND THE ORIGINAL
DOCUMENTS ARE STORED AT THE
OFFICES OF INTERSTATE ENGINEERING,
INC.

PAD LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5301 44-24 3BR"
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- BERM
- DITCH
- Proposed Contours
- Original Contours



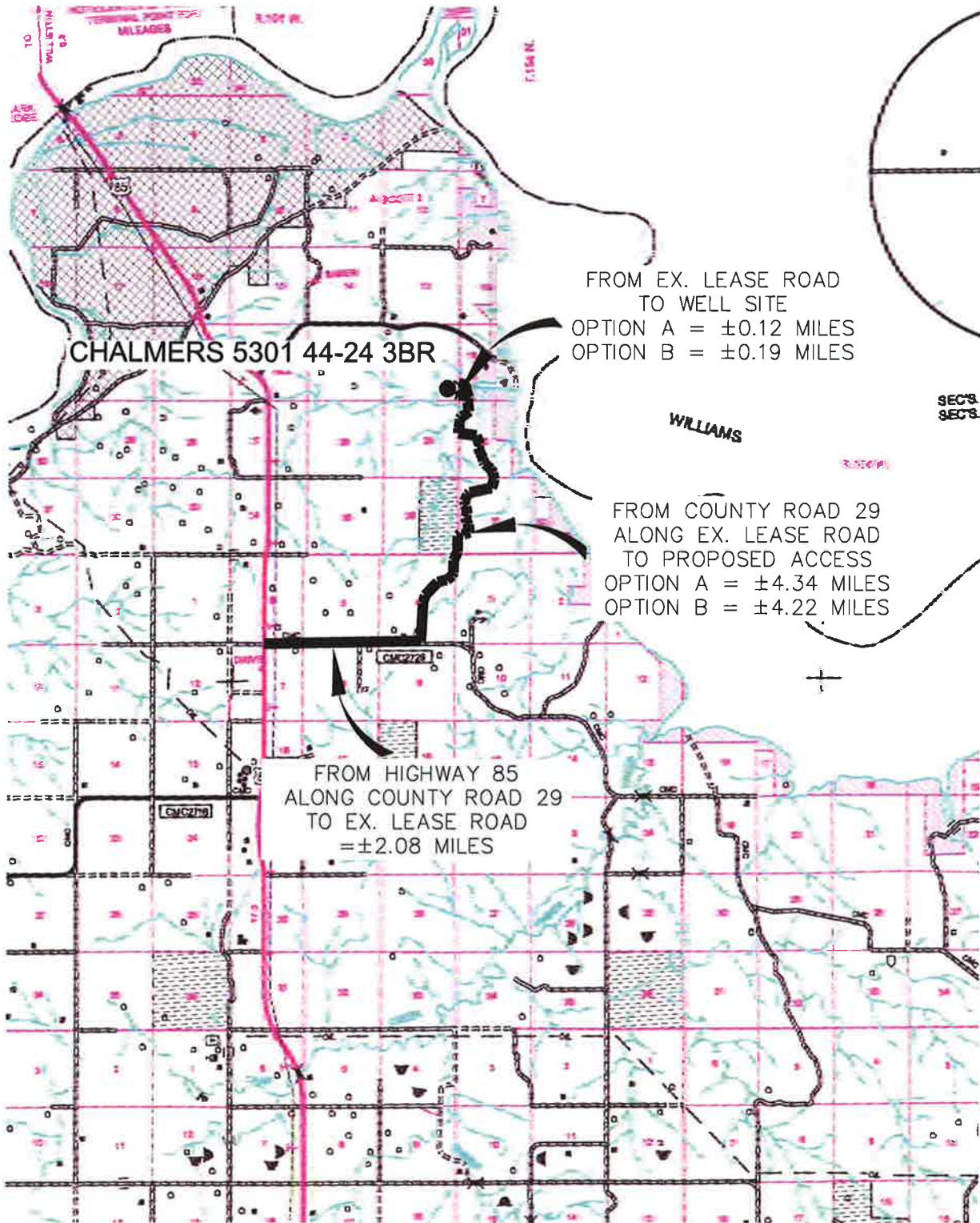
NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5301 44-24 3BR"

926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

6/8



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59720
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

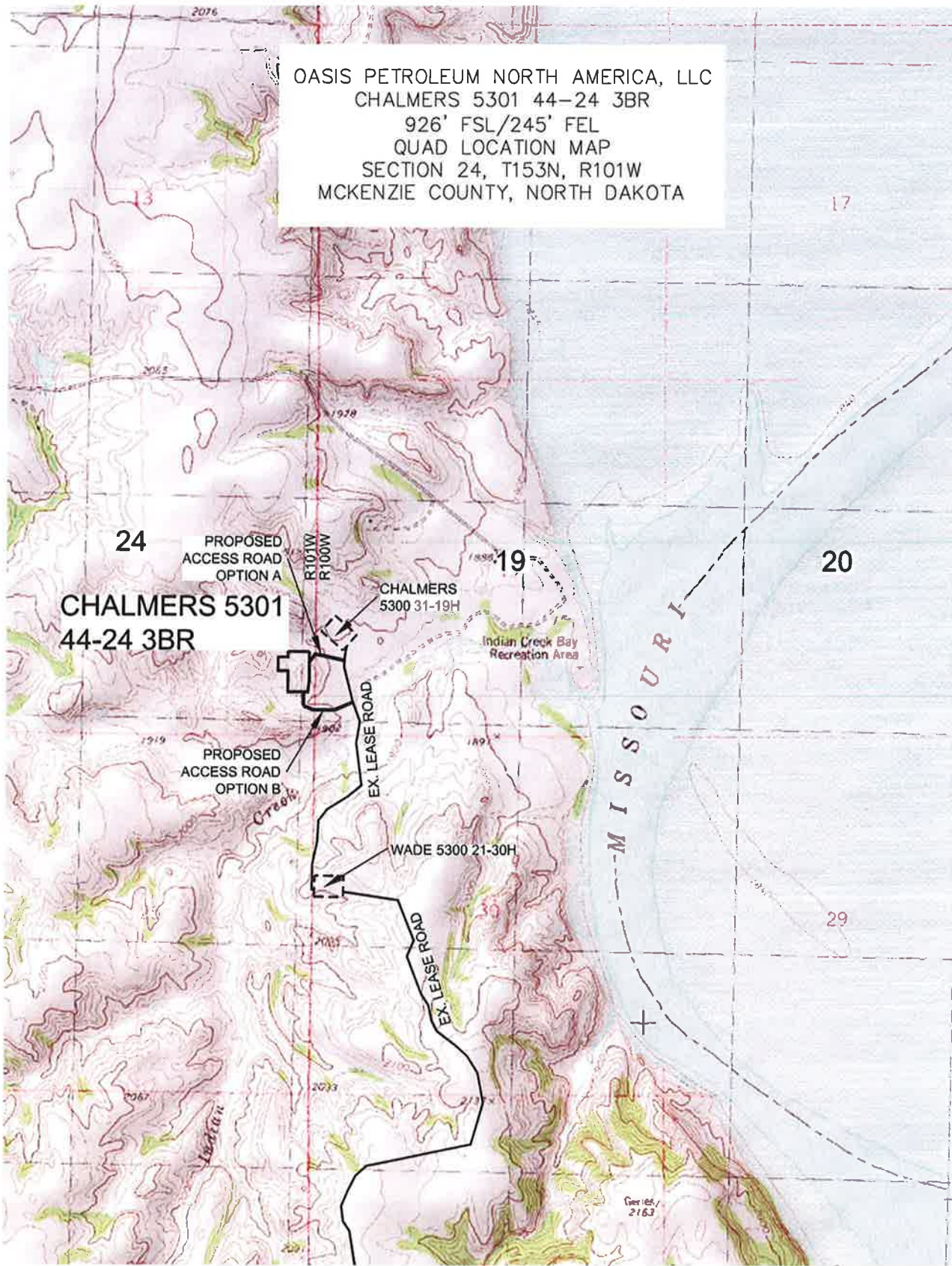
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235.01
Checked By: D.D.K. Date: MAY 2014

Revision No.	Date	By	Description
REV 1	12/2/13	JJS	CHANGED PAD LAYOUT
REV 2	1/28/14	BPH	CHANGED PAD LAYOUT
REV 3	3/28/14	JJS	ADDED WELL TO PAD
REV 4	5/28/14	JJS	CHANGED NAME, MOVED WELL
REV 5	6/11/14	JJS	ADDED DRAINAGE

OASIS PETROLEUM NORTH AMERICA, LLC
 CHALMERS 5301 44-24 3BR
 926' FSL/245' FEL
 QUAD LOCATION MAP
 SECTION 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

5/8



Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 58270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

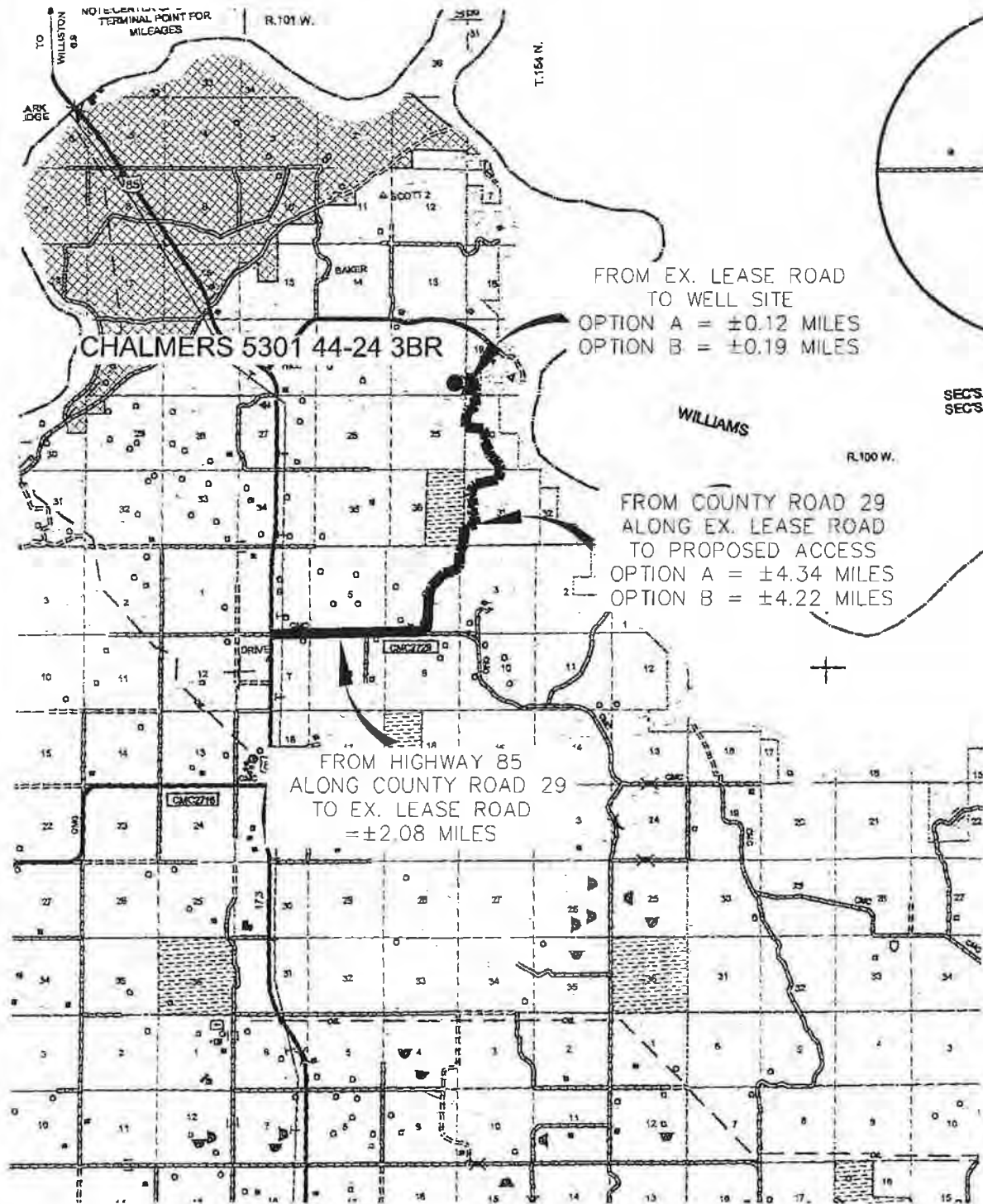
Drawn By: JJS Project No.: 513-09-235.01
 Checked By: DDK Date: MAY 2014

Revision No.	Date	By	Description
REV 2	12/9/13	JJS	CHANGED PAD LAYOUT
REV 3	1/29/14	BH	CHANGED PAD LAYOUT
REV 4	3/29/14	JJS	ADDED WELL TO PAD
REV 5	5/29/14	JJS	CHANGED NAME, DEPTH WELL
REV 6	6/11/14	JMM	ADDED DIMENSIONS

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5301 44-24 3BR"

926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

6/8



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sicily, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

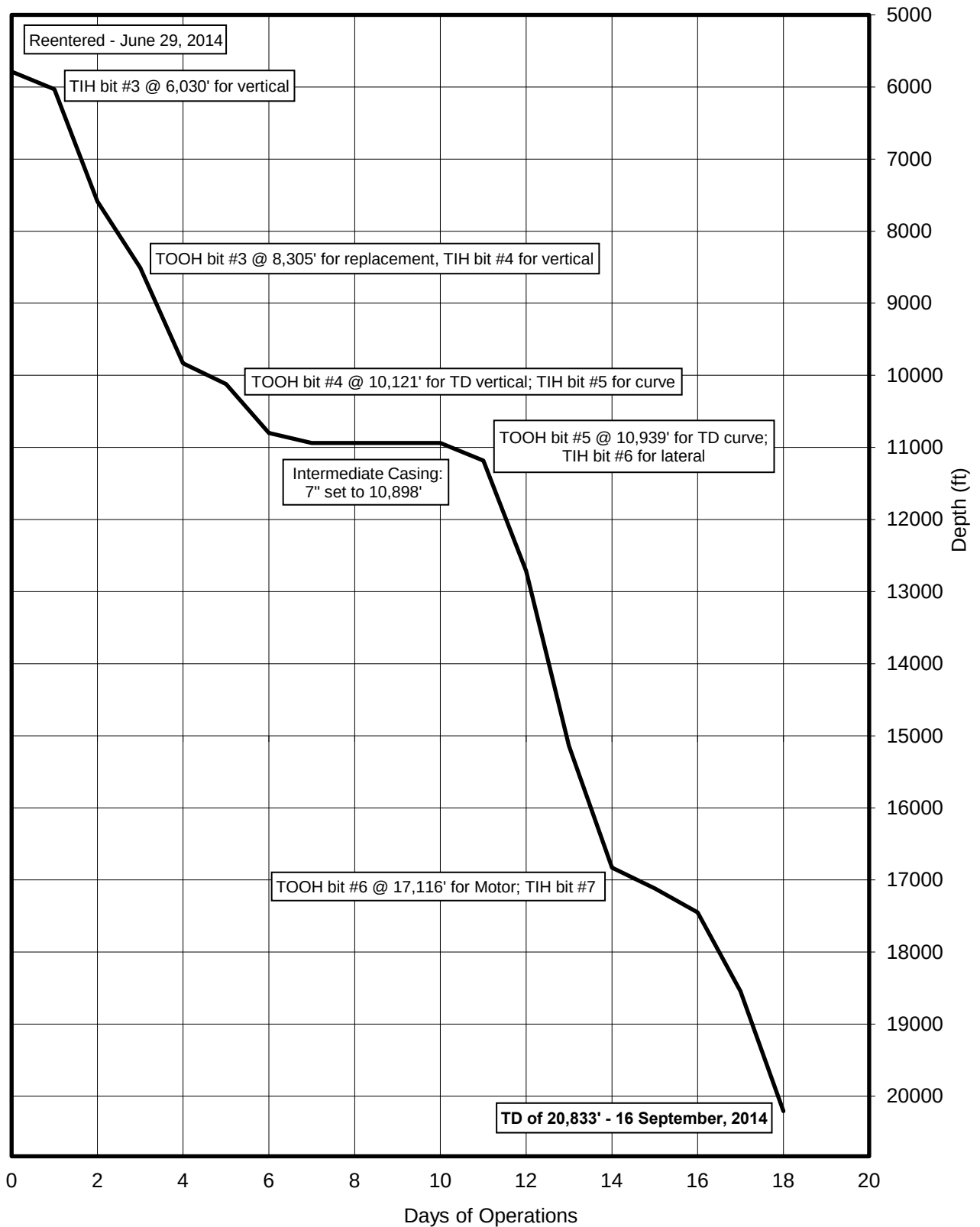
Drawn By: J.J.S. Project No: S13-06-235.01
Checked By: D.D.K. Date: MAY 2014

Revision No.	Date	By	Description
REV 1	10/01/13	JJS	CHANGED WELL NAME
REV 2	12/26/13	JJS	CHANGED PAD LAYOUT
REV 3	1/20/14	BHM	CHANGED PAD LAYOUT
REV 4	4/08/14	JJS	ADDED WELL TO PAD
REV 5	5/25/14	JJS	CHANGED NAME, MOVED WELL

TIME VS. DEPTH

Oasis Petroleum North America, LLC

Chalmers 5301 44-24 3BR



MORNING REPORT SUMMARY

Day	Date 2014	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity Summary		Formation
0	6/29	5,787'	-	-	-	-	-	-	-	-	-	-	Drill F/ 2056 to 4388, rig service, drill F/4388 to 5787		-
1	6/30	6,030'	243	2	25	80	-	-	3400	57	57	401	Drill F/5787 to 6030, circulate and condition, pump pill, BOP drill, service rig, grease C&B, blackjack, TOO H, L/D BHA, install/remove wear bushing, R/U casers, run casing, circulate and condition, R/D casers, R/U cementers, primary cementing, R/D cement		Dakota
2	7/1	7,589'	1559	3	18	50	-	-	2200	70	70	492	P/U BHA, TIH, change rotating head/rubber, TIH, pressure test, drill cement F/5924 to 6040, float at 5977, shoe at 6024, FIT test, drill F/6040 to 6423, service top drive, drill F/6423 to 7589		Amsden
3	7/2	8,506'	917	4	25	50	-	-	3300	70	70	492	BOP drills, drill F/7589 to 8305, TOO H, service rig, TOO H, change rotating head/rubber, remove rotating head/install trip nipple, service rig, level sub, TOO H, change rotating head/rubber, remove trip nipple/install rotating head, TIH, drill F/8305 to 8506		Kibbey
4	7/3	9,832'	1326	4	20	50	-	-	3300	70	70	492	Drill F/8506 to 9284, service top drive, crown, and blocks, function HCR, drill F/9284 to 9832		Mission Canyon
5	7/4	10,121'	289	4	30	50	-	-	3200	70	70	492	Drill F/9832 to 10121, circulate and condition, TOO H, change rotating head/rubber, removed rot head/installed trip nipple, TOO H, case hole logs, R/U loggers, logged, R/D loggers, P/U BHA, TIH		Lodgepole
6	7/5	10,800'	679	5	25	20	28	241	3000	70	70	492	Change rotating head/rubber, install rotating head, TIH, ream F/8405 to 9100,drill F/10121 to 10364, service top drive, function pipe rams, drill F/10364 to 10800		Upper Bakken
7	7/6	10,939'	139	5	20	20	30	241	3300	70	70	492	Drill F/10800 to 10939, circulate and condition, short trip, function HCR, circulate and condition, TOO H, cut and slip, TOO H, checked for flow and worked on flow sensor, TOO H, change rotating head/rubber, installed trip nipple, TOO H, L/D BHA		Middle Bakken
8	7/7	10,939'	0	5	20	20	30	241	3300	70	70	492	Run casing, wash down, run casing, washing casing to bottom, circulate/cement/displace, R/D casing, R/U cementers, waiting on cement		Middle Bakken
9	7/8	10,939'	0	5	20	20	30	241	3300	70	70	492	Waiting on cement, nipple down BOP, working as directed by operator, L/D rotating head, stand back BOP, preparing to set casing slips, install well heads, pull apart B section, clean, put back together, put in slips, install well heads, wait on weldrs, cut casing		Middle Bakken
10	9/8	10,939'	0	5	20	20	30	241	3300	70	70	492	Working as directed by operator on previous well, transport, RB catwalk, flowline, misc. to walk, skid rig, walk 49', soft ground, rig up catwalk, flowline, misc., nipple up BOP		Middle Bakken
11	9/9	11,183'	244	6	18	50	25	147	2800	-	85	299	Nipple up BOP, working as directed by operator, rig up catwalk, flow line, water and air, test BOP, pressure test BOP and casing, P/U BHA, TIH, test tool, service top drive, change out quill, TIH, install/remove wear bushing, pull trip nipple, install rotating head, drill F/10939 to 11183		Middle Bakken
12	9/10	12,714'	1531	6	18	50	35	147	3600	85	-	299	Drill F/11183-11758, service top drive,greasee blocks, crown, swivel, and lower well, drill F/11758 to 12714		Middle Bakken

MORNING REPORT SUMMARY

Day	Date	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity Summary	Formation
13	9/11	15,139'	2425	6	18	50	35	147	3900	84	-	299	Drill F/12714 to 13808, service rig, drill F/13808 to 14732, circulate and condition, circulate gas from wellbore, drill F/14737 to 15139	Middle Bakken
14	9/12	16,829'	1690	6	25	50	30	138	3800	80	-	281	Drill F/15139 to 16047, service rig, drill F/16047 to 16829	Middle Bakken
15	9/13	17,116'	287	6	25	50	35	134	3800	78	-	274	Drill F/16829 to 17116, circulate and condition, TOO, circulate and condition, change rotating head/rubber, remove rotating head, install trip nipple, TOO, L/D BHA, P/U BHA, install/remove wear bushing, install rotating head, TIH, cut drilling line 98', circulate and condition/circulate gas out	Middle Bakken
16	9/14	17,450'	334	7	18	50	25	121	3500	-	70	246	TOOH, L/D BHA, L/D tool and pick up new one, function pipe rams and annular, TIH, change rotating head/rubber, remove trip nipple, install rotating head, TIH, circulate and condition, TIH, circulate and condition/circulate gas out, drill F/17116 to 17450	Middle Bakken
17	9/15	18,542'	1092	7	18	50	25	121	3400	70	-	246	Drill F/17450 to 18012, service rig, drill F/18012 to 18542	Middle Bakken
18	9/16	20,206'	1664	7	18	50	30	121	2755	70	-	246	Drill F/18542 to 19548, service rig, well control BOP drill, drill F/19548 to 20206	Middle Bakken
19	9/17	20,833'	627	7	18	50	30	121	2755	70	-	246	Drill F/20206 to 20833	Middle Bakken

DAILY MUD SUMMARY

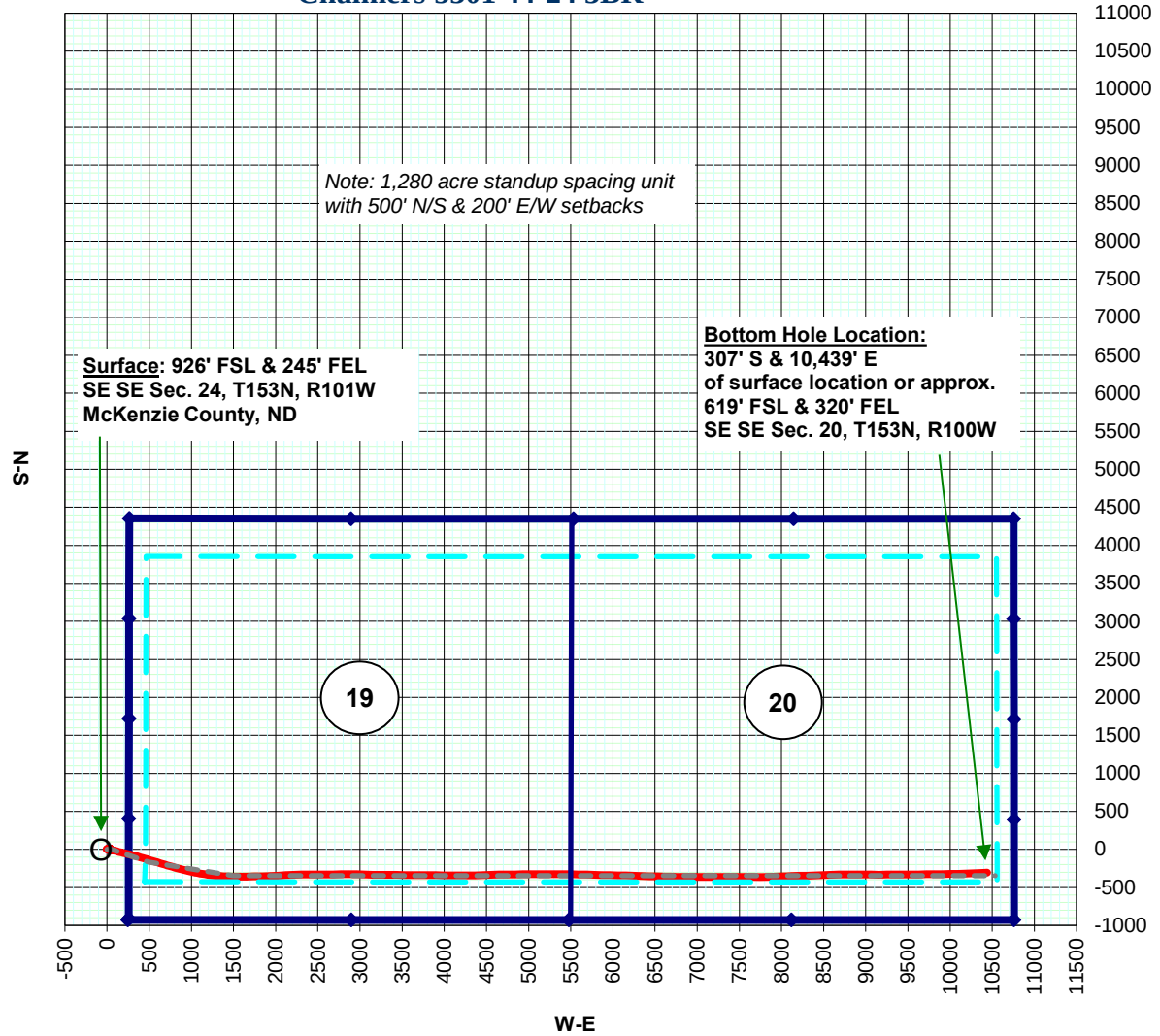
Date 2014	Mud Depth	Mud WT (ppg)	VIS (sec/qt)	PV (cP)	YP (lbs/ 100 ft ²)	Gels (lbs/ 100 ft ²)	600/ 300	NAP/ H ₂ O (ratio)	NAP/ H ₂ O (% by vol)	Cake (API/ HTHP)	Cor. Solids (%)	Oil/ H ₂ O (%)	Alk	pH	Excess Lime (lb/bbl)	Cl ⁻ (mg/L)	LGS/ HGS (%)	Salinity (ppm)	ES	Gain/ Loss (bbls)
06/29	6,030'	11.7	61	26	11	11/16/-	63/37	75.5/24.5	61/19.8	3	17.7	61/19.8	1.7	-	2.2	25k	13.4/4.3	170,813	387	-/107
06/30	6,030'	9.95	81	24	10	9/14/-	58/34	69.7/27	62/27	3	9.5	62/27	2.1	-	2.7	24k	8/1.5	166,498	0	-/188
07/01	8,305'	10	72	18	7	7/11/-	43/25	77.3/22.7	68/20	3	10.7	68/20	2.3	-	3	22k	8.2/2.4	127,856	377	-/70
07/02	8,577'	10.05	48	18	7	7/11/-	43/25	79.3/20.7	69/18	3	11.7	69/18	2.1	-	2.7	22k	7.8/3.9	153,461	370	-/106
07/03	9,975'	9.85	47	17	10	9/14/-	44/27	82.3/17.7	72/15.5	2	11.2	72/15.5	2.3	-	3	22k	7.3/3.9	215,349	676	-/153
07/04	10,122'	10.2	52	15	10	9/14/-	40/25	82.6/17.4	71/15	2	12.5	71/15	2.2	-	2.8	24k	8.5/4.2	240,376	632	-/123
07/05	10,939'	10.4	46	17	11	9/14/-	45/28	83.5/16.5	71/14	2	13.6	71/14	2.2	-	2.8	23k	9.1/4.3	213,059	620	-/146
07/06	10,939'	10.4	46	17	11	9/14/-	45/28	83.5/16.5	71/14	2	13.6	71/14	2.2	-	2.8	23k	9.1/4.3	213,059	620	-/146
07/07	10,939'	10.4	46	17	11	9/14/-	45/29	83.5/16.6	71/15	2	13.6	71/15	3.2	-	3.8	23k	9.1/4.4	213,059	620	-/147
07/08	10,939'																			
Change mud from diesel invert to salt water																				
09/08	10,939'	9.7	26	1	1	-	3/2	-	0/92	-	-	0/92	-	9	-	16k	0/0.1	-	-	-/-
09/09	11,368'	9.7	26	1	1	-	3/2	-	0/92	-	-	0/92	-	9	-	16k	0/0.1	-	-	-/-
09/10	12,714'	9.7	26	1	1	-	3/2	-	0/92	-	-	0/92	-	9	-	16k	0/0.1	-	-	-/-
09/11	15,775'	9.85	27	1	1	-	4/2	-	0.5/88.8	-	-	0.5/88.5	-	8	-	187lk	-/0.8	-	-	-/-
09/12	16,933'	9.85	27	1	1	-	4/2	-	0.5/88.8	-	-	0.5/88.5	-	8	-	187lk	-/0.8	-	-	-/-
09/13	17,116'	9.85	27	1	1	-	4/2	-	0.5/88.8	-	-	0.5/88.5	-	8	-	187lk	-/0.8	-	-	-/-
09/14	17,450'	10.15	28	4	-	-	8/4	-	6.5/79	-	-	6.5/79	-	8	-	8k	-/1.7	-	-	-
09/15	19,061'	10.15	28	4	-	-	8/4	-	6.5/79	-	-	6.5/79	-	8	-	8k	-/1.7	-	-	-
09/16	20,206'	10.15	28	4	-	-	8/4	-	6.5/79	-	-	6.5/79	-	8	-	8k	-/1.7	-	-	-

BOTTOM HOLE ASSEMBLY RECORD

Bit #	Size (in.)	Type	Make	Model	Depth In	Depth Out	Footage	Hours	Σ hrs	Vert. Dev.	Reason For Removal
1	17 1/2	Tri-cone	NOV	-	80'	2,095'	2,015'	14	14	Surface	TD surface
2	12 1/4	PDC	NOV	AA1L	2,040'	6,030'	3,990'	26	40	Vertical	Vertical
3	8 3/4	PDC	Security	MM64D	6,030'	8,305'	2,275'	25	65	Vertical	Casing
4	8 3/4	PDC	Security	MM64D	8,305'	10,121'	1,816'	34.5	99.5	Vertical	TD Vertical
5	8 3/4	PDC	Security	MM55MM	10,121'	10,939'	818'	25.5	125	Curve	TD curve
6	6	PDC	Security	MMD64	10,939'	17,116'	6,177'	76	201	Lateral	Motor Failure
7	6	PDC	Security	MM64	17,116'	20,833'	3,717'	59.5	260.5	Lateral	TD Lateral

PLAN VIEW

Oasis Petroleum North America, LLC
Chalmers 5301 44-24 3BR



Oasis Petroleum North America, LLC

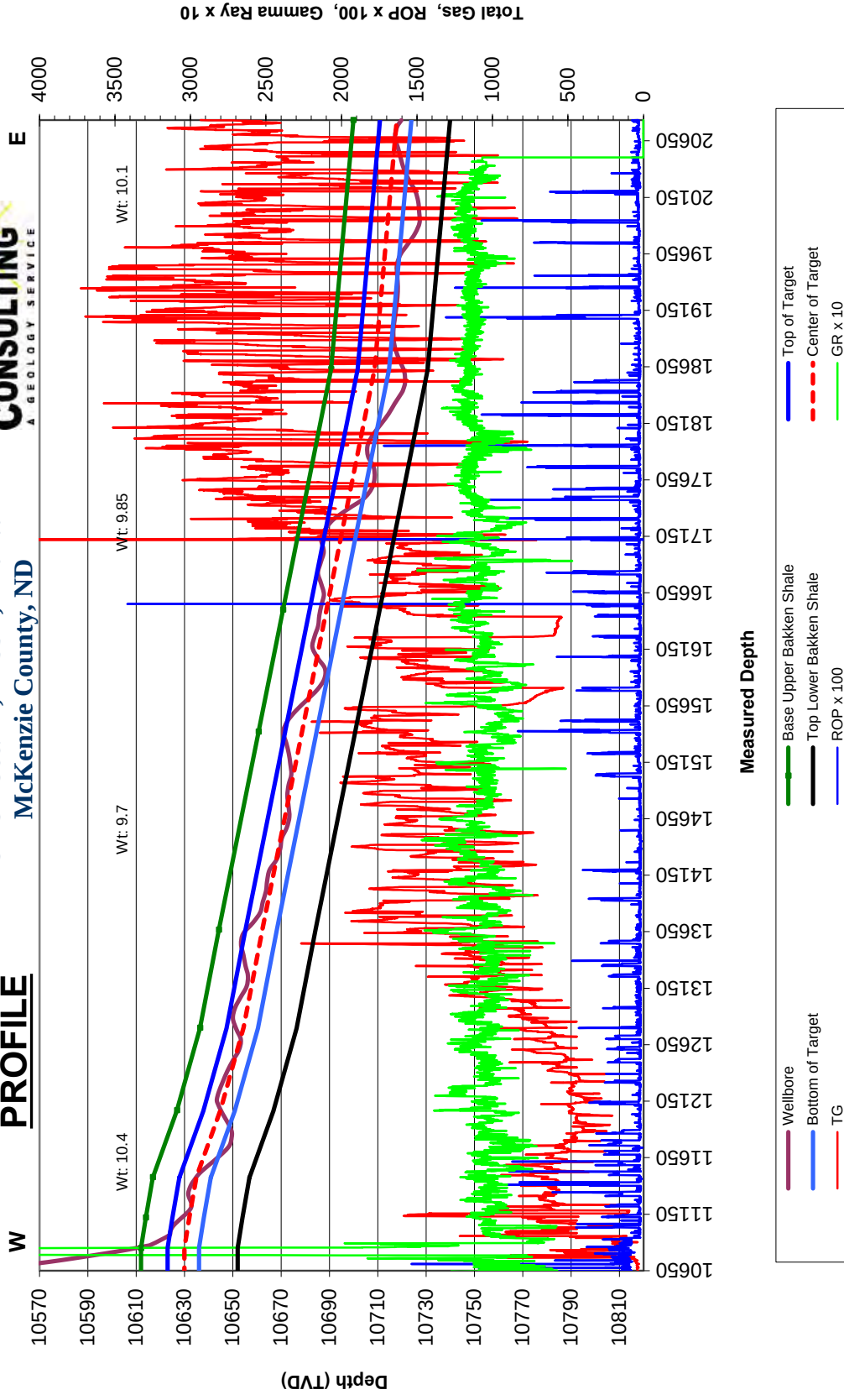
Chalmers 5301 44-24 3BR

SE SE Sec. 24, T153N, R101W

McKenzie County, ND



PROFILE



FORMATION MARKERS & DIP ESTIMATES

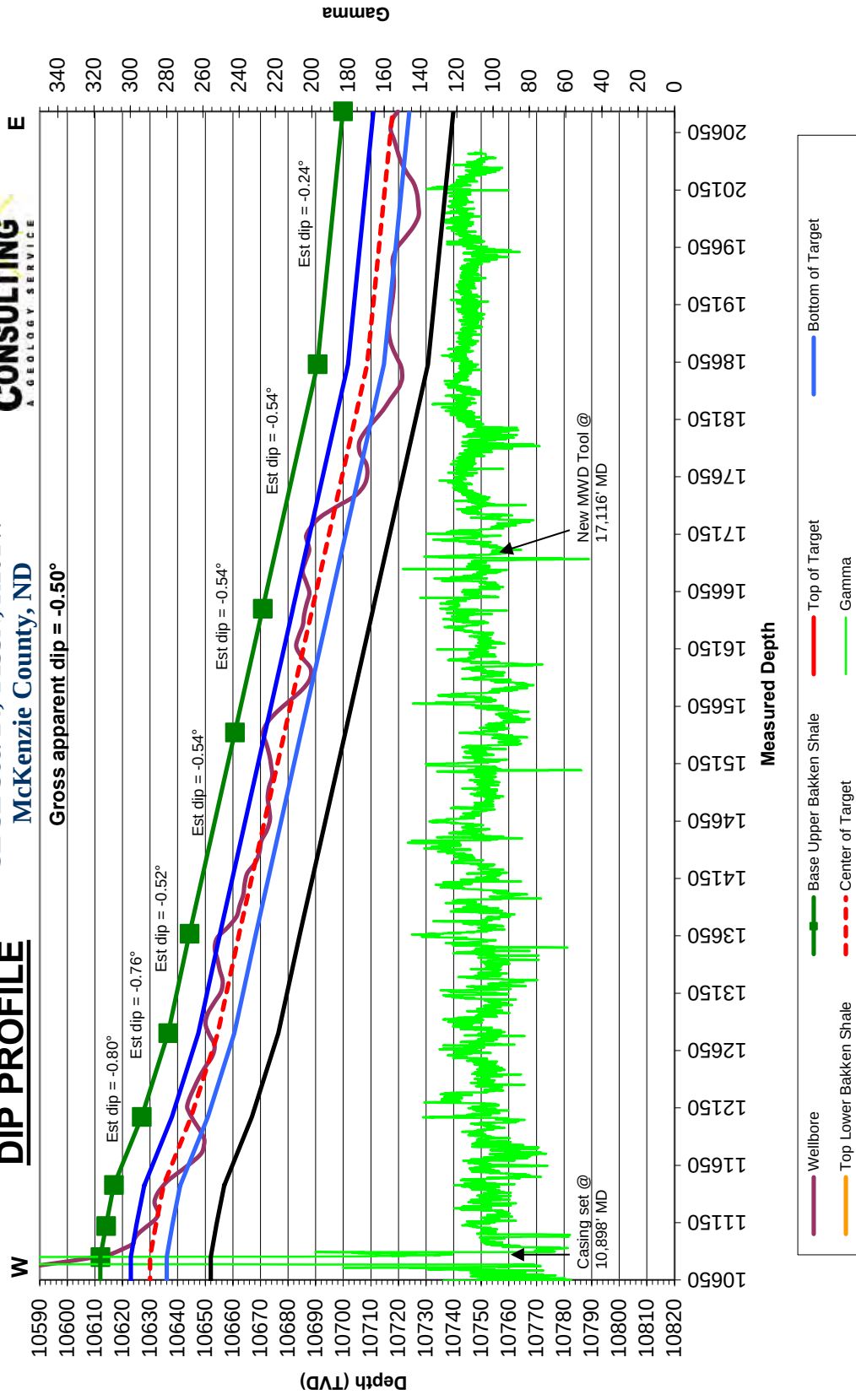
Oasis Petroleum North America, LLC - Chalmers 5301 44-24 3BR

Dip Change Points	MD	TVD	TVD diff.	MD diff.	Dip	Dipping up/down	Type of Marker
Marker							
	10,850'	10,612.00					Gamma
Top of Target	11,120'	10,614.00	2.00	270.00	-0.42	Down	Gamma
Center of Target	11,478'	10,619.00	5.00	358.00	-0.80	Down	Gamma
Center of Target	12,070'	10,626.90	7.90	592.00	-0.76	Down	Gamma
Center of Target	12,800'	10,636.50	9.60	730.00	-0.75	Down	Gamma
Top of Target	13,667'	10,644.30	7.80	867.00	-0.52	Down	Gamma
Top of Target	15,421'	10,660.70	16.40	1754.00	-0.54	Down	Gamma
Center of Target	16,500'	10,670.80	10.10	1079.00	-0.54	Down	Gamma
Nondescript Gamma	18,630'	10,690.70	19.90	2130.00	-0.54	Down	Gamma
TD	20,833'	10,699.90	9.20	2203.00	-0.24	Down	Gamma
Gross Dip							
Initial Target Contact	10,850'	10,612.00					
Projected Final Target Contact	20,833'	10,699.90	87.90	9983.00	-0.50	Down	Projection

Oasis Petroleum North America, LLC
Chalmers 5301 44-24 3BR
SE SE Sec. 24, T153N, R101W
McKenzie County, ND



DIP PROFILE



<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5301 44-24 3BR		
County:	McKenzie	State:	ND
QQ:	SE SE	Section:	24
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	926	FN/SL:	S
	245	FE/WL:	E

Kick-off:	6/30/2014
Finish:	9/16/2014
Directional Supervision:	RPM

Date: 9/23/2014

Time: 16:53

F9 to re-calculate

Proposed dir: 90

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
Tie	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	184.00	0.80	273.10	183.99	0.07	-1.28	-1.28	0.43
2	215.00	0.70	271.20	214.99	0.09	-1.69	-1.69	0.33
3	308.00	0.90	267.70	307.98	0.07	-2.99	-2.99	0.22
4	402.00	0.30	299.60	401.98	0.16	-3.94	-3.94	0.71
5	495.00	0.30	302.10	494.98	0.41	-4.36	-4.36	0.01
6	588.00	0.70	304.50	587.97	0.86	-5.03	-5.03	0.43
7	680.00	0.70	339.40	679.97	1.70	-5.69	-5.69	0.46
8	772.00	1.30	28.50	771.95	3.15	-5.39	-5.39	1.08
9	865.00	1.20	56.40	864.93	4.61	-4.08	-4.08	0.66
10	957.00	1.10	52.90	956.91	5.68	-2.57	-2.57	0.13
11	1049.00	1.00	55.60	1048.90	6.67	-1.20	-1.20	0.12
12	1142.00	1.70	55.50	1141.87	7.91	0.60	0.60	0.75
13	1234.00	1.20	103.10	1233.84	8.46	2.67	2.67	1.37
14	1326.00	0.80	133.60	1325.83	7.80	4.07	4.07	0.71
15	1418.00	0.70	167.60	1417.82	6.81	4.66	4.66	0.49
16	1510.00	0.40	124.20	1509.82	6.08	5.04	5.04	0.54
17	1604.00	0.30	109.00	1603.82	5.81	5.55	5.55	0.14
18	1697.00	0.20	134.70	1696.82	5.62	5.89	5.89	0.16
19	1790.00	0.40	172.40	1789.82	5.18	6.05	6.05	0.29
20	1883.00	0.40	108.80	1882.81	4.76	6.40	6.40	0.45
21	1977.00	0.50	116.60	1976.81	4.47	7.08	7.08	0.12
22	2031.00	0.40	151.40	2030.81	4.20	7.38	7.38	0.53
23	2096.00	0.50	271.90	2095.81	4.01	7.20	7.20	1.20
24	2190.00	0.40	197.90	2189.81	3.71	6.69	6.69	0.58
25	2283.00	0.70	165.90	2282.80	2.85	6.73	6.73	0.45
26	2376.00	1.10	170.60	2375.79	1.42	7.02	7.02	0.44
27	2469.00	1.30	167.40	2468.77	-0.49	7.39	7.39	0.23
28	2563.00	0.90	130.90	2562.75	-2.02	8.18	8.18	0.84
29	2656.00	0.40	132.00	2655.75	-2.71	8.98	8.98	0.54
30	2749.00	0.80	137.60	2748.74	-3.41	9.66	9.66	0.43
31	2843.00	0.70	86.60	2842.73	-3.86	10.67	10.67	0.69
32	2936.00	1.00	99.10	2935.72	-3.95	12.04	12.04	0.38
33	3029.00	0.80	74.90	3028.71	-3.91	13.47	13.47	0.46
34	3122.00	1.10	76.00	3121.70	-3.53	14.96	14.96	0.32

<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5301 44-24 3BR		
County:	McKenzie	State:	ND
QQ:	SE SE	Section:	24
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	926	FN/SL:	S
	245	FE/WL:	E

Kick-off:	6/30/2014
Finish:	9/16/2014
Directional Supervision:	RPM

Date: 9/23/2014

Time: 16:53

F9 to re-calculate

Proposed dir: 90

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
35	3216.00	0.90	83.40	3215.69	-3.23	16.57	16.57	0.25
36	3309.00	0.70	115.00	3308.68	-3.38	17.81	17.81	0.51
37	3402.00	0.60	124.30	3401.67	-3.90	18.73	18.73	0.16
38	3495.00	1.30	50.90	3494.66	-3.50	19.95	19.95	1.36
39	3589.00	1.30	62.60	3588.64	-2.34	21.72	21.72	0.28
40	3682.00	0.70	75.70	3681.62	-1.72	23.21	23.21	0.69
41	3775.00	1.00	93.70	3774.61	-1.63	24.57	24.57	0.43
42	3868.00	1.00	87.90	3867.60	-1.65	26.19	26.19	0.11
43	3962.00	0.80	91.20	3961.59	-1.63	27.67	27.67	0.22
44	4055.00	0.60	112.20	4054.58	-1.83	28.77	28.77	0.35
45	4148.00	0.90	148.50	4147.57	-2.64	29.60	29.60	0.59
46	4242.00	0.70	10.80	4241.57	-2.70	30.09	30.09	1.59
47	4335.00	0.50	1.00	4334.56	-1.74	30.21	30.21	0.24
48	4428.00	0.40	10.40	4427.56	-1.02	30.27	30.27	0.13
49	4521.00	0.30	2.20	4520.56	-0.45	30.34	30.34	0.12
50	4615.00	0.10	305.30	4614.56	-0.16	30.28	30.28	0.28
51	4708.00	0.30	320.50	4707.56	0.08	30.06	30.06	0.22
52	4801.00	0.80	303.50	4800.55	0.62	29.36	29.36	0.56
53	4894.00	0.70	277.90	4893.54	1.06	28.26	28.26	0.37
54	4988.00	0.30	167.50	4987.54	0.90	27.74	27.74	0.91
55	5081.00	0.70	123.70	5080.54	0.34	28.27	28.27	0.57
56	5174.00	1.00	162.10	5173.53	-0.74	28.99	28.99	0.67
57	5268.00	1.10	169.10	5267.51	-2.41	29.41	29.41	0.17
58	5361.00	0.30	157.70	5360.51	-3.51	29.68	29.68	0.87
59	5454.00	0.60	147.50	5453.50	-4.15	30.03	30.03	0.33
60	5547.00	0.70	142.00	5546.50	-5.01	30.64	30.64	0.13
61	5641.00	1.00	165.90	5640.49	-6.25	31.19	31.19	0.49
62	5734.00	1.20	149.70	5733.47	-7.88	31.88	31.88	0.40
63	5827.00	1.00	172.70	5826.45	-9.53	32.48	32.48	0.52
64	5920.00	1.10	185.40	5919.44	-11.22	32.50	32.50	0.27
65	5977.00	1.10	167.90	5976.43	-12.30	32.56	32.56	0.59
66	6062.00	1.20	178.60	6061.41	-13.99	32.75	32.75	0.28
67	6093.00	1.10	184.00	6092.40	-14.61	32.74	32.74	0.48
68	6187.00	0.60	19.80	6186.40	-15.05	32.84	32.84	1.79
69	6280.00	0.70	344.00	6279.39	-14.04	32.85	32.85	0.44

<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5301 44-24 3BR		
County:	McKenzie	State:	ND
QQ:	SE SE	Section:	24
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	926	FN/SL:	S
	245	FE/WL:	E

Kick-off:	6/30/2014
Finish:	9/16/2014
Directional Supervision:	RPM

Date: 9/23/2014

Time: 16:53

F9 to re-calculate

Proposed dir: 90

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
70	6373.00	0.60	322.50	6372.39	-13.11	32.40	32.40	0.28
71	6466.00	0.40	285.40	6465.38	-12.64	31.79	31.79	0.40
72	6560.00	0.90	70.10	6559.38	-12.30	32.17	32.17	1.33
73	6653.00	0.60	65.80	6652.37	-11.85	33.30	33.30	0.33
74	6746.00	0.60	100.50	6745.37	-11.74	34.22	34.22	0.38
75	6839.00	1.10	34.10	6838.36	-11.09	35.20	35.20	1.10
76	6933.00	1.30	16.20	6932.34	-9.32	36.00	36.00	0.45
77	7026.00	1.20	17.60	7025.32	-7.38	36.59	36.59	0.11
78	7119.00	1.20	17.00	7118.30	-5.52	37.17	37.17	0.01
79	7212.00	1.10	6.20	7211.28	-3.70	37.55	37.55	0.26
80	7306.00	0.50	35.80	7305.27	-2.47	37.89	37.89	0.75
81	7399.00	0.40	50.40	7398.27	-1.93	38.38	38.38	0.16
82	7492.00	0.50	41.90	7491.26	-1.42	38.90	38.90	0.13
83	7586.00	0.70	59.10	7585.26	-0.82	39.67	39.67	0.28
84	7679.00	0.90	70.80	7678.25	-0.29	40.84	40.84	0.28
85	7772.00	0.70	65.80	7771.24	0.18	42.05	42.05	0.23
86	7865.00	0.40	9.00	7864.24	0.73	42.62	42.62	0.63
87	7959.00	0.40	307.50	7958.23	1.26	42.41	42.41	0.44
88	8052.00	0.50	318.60	8051.23	1.76	41.89	41.89	0.14
89	8145.00	0.70	315.10	8144.23	2.47	41.22	41.22	0.22
90	8239.00	0.70	312.30	8238.22	3.26	40.39	40.39	0.04
91	8332.00	0.50	328.70	8331.21	3.99	39.75	39.75	0.28
92	8425.00	0.70	329.90	8424.21	4.83	39.26	39.26	0.22
93	8518.00	0.50	336.30	8517.20	5.69	38.81	38.81	0.23
94	8612.00	0.60	358.50	8611.20	6.56	38.63	38.63	0.25
95	8705.00	0.70	330.80	8704.19	7.54	38.34	38.34	0.35
96	8798.00	0.70	322.90	8797.19	8.49	37.72	37.72	0.10
97	8892.00	0.90	317.60	8891.18	9.49	36.88	36.88	0.23
98	8985.00	0.10	318.50	8984.17	10.09	36.33	36.33	0.86
99	9078.00	0.80	149.30	9077.17	9.60	36.61	36.61	0.97
100	9171.00	0.80	154.60	9170.16	8.45	37.22	37.22	0.08
101	9265.00	0.60	154.90	9264.15	7.41	37.71	37.71	0.21
102	9358.00	0.70	158.50	9357.15	6.44	38.13	38.13	0.12
103	9451.00	0.70	168.90	9450.14	5.36	38.44	38.44	0.14
104	9545.00	0.80	168.00	9544.13	4.15	38.69	38.69	0.11

<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5301 44-24 3BR		
County:	McKenzie	State:	ND
QQ:	SE SE	Section:	24
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	926	FN/SL:	S
	245	FE/WL:	E

Kick-off:	6/30/2014
Finish:	9/16/2014
Directional Supervision:	RPM

Date: 9/23/2014

Time: 16:53

F9 to re-calculate

Proposed dir: 90

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
105	9638.00	0.90	178.30	9637.12	2.79	38.85	38.85	0.20
106	9731.00	1.00	194.10	9730.11	1.27	38.67	38.67	0.30
107	9824.00	1.00	210.20	9823.10	-0.22	38.07	38.07	0.30
108	9918.00	0.90	157.50	9917.09	-1.61	37.93	37.93	0.90
109	10011.00	0.80	181.50	10010.08	-2.93	38.20	38.20	0.39
110	10071.00	1.10	207.10	10070.07	-3.87	37.92	37.92	0.85
111	10088.00	1.10	212.70	10087.06	-4.15	37.76	37.76	0.63
112	10119.00	1.50	139.20	10118.06	-4.71	37.87	37.87	5.12
113	10150.00	6.30	110.30	10148.98	-5.60	39.73	39.73	16.25
114	10181.00	11.10	106.30	10179.61	-7.03	44.19	44.19	15.60
115	10212.00	15.40	104.20	10209.78	-8.88	51.05	51.05	13.96
116	10243.00	20.40	103.10	10239.27	-11.12	60.31	60.31	16.17
117	10274.00	24.40	101.80	10267.93	-13.65	71.84	71.84	13.00
118	10305.00	27.20	102.40	10295.83	-16.48	85.03	85.03	9.07
119	10337.00	30.20	105.00	10323.90	-20.14	99.95	99.95	10.15
120	10368.00	33.00	107.10	10350.30	-24.64	115.56	115.56	9.70
121	10399.00	36.70	107.10	10375.74	-29.85	132.48	132.48	11.94
122	10430.00	41.10	106.10	10399.86	-35.40	151.14	151.14	14.34
123	10461.00	45.80	105.00	10422.36	-41.10	171.67	171.67	15.36
124	10492.00	48.80	104.30	10443.38	-46.86	193.71	193.71	9.82
125	10523.00	51.20	104.50	10463.30	-52.77	216.71	216.71	7.76
126	10554.00	51.40	104.70	10482.69	-58.87	240.12	240.12	0.82
127	10585.00	52.10	105.10	10501.88	-65.13	263.65	263.65	2.47
128	10616.00	54.30	106.50	10520.45	-71.89	287.53	287.53	7.96
129	10647.00	57.20	106.50	10537.89	-79.17	312.10	312.10	9.35
130	10678.00	61.50	106.60	10553.69	-86.76	337.65	337.65	13.87
131	10710.00	67.20	106.90	10567.54	-95.07	365.27	365.27	17.83
132	10741.00	71.30	107.60	10578.52	-103.67	392.94	392.94	13.39
133	10772.00	72.30	107.90	10588.20	-112.65	420.99	420.99	3.35
134	10803.00	73.00	108.40	10597.45	-121.86	449.11	449.11	2.73
135	10834.00	74.70	108.60	10606.07	-131.31	477.34	477.34	5.52
136	10865.00	77.80	108.50	10613.43	-140.89	505.89	505.89	10.00
137	10880.00	80.30	108.50	10616.28	-145.56	519.85	519.85	16.67
138	10963.00	89.40	108.60	10623.73	-171.83	598.14	598.14	10.96
139	10994.00	89.00	108.10	10624.16	-181.59	627.56	627.56	2.07

<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5301 44-24 3BR		
County:	McKenzie	State:	ND
QQ:	SE SE	Section:	24
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	926	FN/SL:	S
	245	FE/WL:	E

Kick-off:	6/30/2014
Finish:	9/16/2014
Directional Supervision:	RPM

Date: 9/23/2014

Time: 16:53

F9 to re-calculate

Proposed dir: 90

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
140	11024.00	88.90	107.90	10624.71	-190.86	656.09	656.09	0.75
141	11055.00	87.30	109.20	10625.74	-200.72	685.46	685.46	6.65
142	11086.00	87.20	108.70	10627.22	-210.77	714.75	714.75	1.64
143	11117.00	86.90	108.10	10628.82	-220.55	744.12	744.12	2.16
144	11147.00	87.00	107.10	10630.42	-229.60	772.68	772.68	3.35
145	11178.00	87.70	106.80	10631.85	-238.63	802.30	802.30	2.46
146	11209.00	88.80	107.00	10632.80	-247.64	831.94	831.94	3.61
147	11240.00	90.60	106.80	10632.96	-256.65	861.60	861.60	5.84
148	11271.00	91.20	106.20	10632.47	-265.45	891.32	891.32	2.74
149	11302.00	91.10	106.30	10631.85	-274.13	921.08	921.08	0.46
150	11333.00	89.80	104.50	10631.61	-282.36	950.96	950.96	7.16
151	11364.00	89.00	104.40	10631.93	-290.09	980.98	980.98	2.60
152	11395.00	89.00	104.20	10632.47	-297.75	1011.02	1011.02	0.65
153	11426.00	88.20	102.20	10633.23	-304.83	1041.19	1041.19	6.95
154	11457.00	88.00	102.20	10634.26	-311.37	1071.47	1071.47	0.65
155	11487.00	88.20	101.20	10635.25	-317.45	1100.83	1100.83	3.40
156	11518.00	87.00	99.50	10636.55	-323.02	1131.30	1131.30	6.71
157	11549.00	87.00	99.40	10638.17	-328.10	1161.83	1161.83	0.32
158	11580.00	86.70	99.60	10639.88	-333.21	1192.36	1192.36	1.16
159	11611.00	86.80	97.80	10641.63	-337.89	1222.96	1222.96	5.81
160	11642.00	87.00	97.00	10643.31	-341.88	1253.65	1253.65	2.66
161	11672.00	86.90	97.00	10644.91	-345.53	1283.39	1283.39	0.33
162	11703.00	87.30	94.70	10646.47	-348.68	1314.18	1314.18	7.52
163	11735.00	87.60	94.70	10647.90	-351.30	1346.04	1346.04	0.94
164	11766.00	88.80	94.40	10648.87	-353.76	1376.93	1376.93	3.99
165	11797.00	89.60	92.40	10649.30	-355.60	1407.87	1407.87	6.95
166	11828.00	89.50	92.70	10649.55	-356.98	1438.84	1438.84	1.02
167	11859.00	89.90	92.40	10649.71	-358.36	1469.81	1469.81	1.61
168	11890.00	91.00	90.30	10649.47	-359.09	1500.80	1500.80	7.65
169	11952.00	91.40	89.80	10648.17	-359.14	1562.78	1562.78	1.03
170	12045.00	91.70	89.40	10645.65	-358.49	1655.75	1655.75	0.54
171	12137.00	90.80	88.50	10643.65	-356.81	1747.71	1747.71	1.38
172	12168.00	89.90	87.90	10643.46	-355.83	1778.69	1778.69	3.49
173	12231.00	88.70	87.40	10644.23	-353.25	1841.63	1841.63	2.06
174	12323.00	89.20	88.00	10645.91	-349.56	1933.54	1933.54	0.85

<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5301 44-24 3BR		
County:	McKenzie	State:	ND
QQ:	SE SE	Section:	24
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	926	FN/SL:	S
	245	FE/WL:	E

Kick-off:	6/30/2014
Finish:	9/16/2014
Directional Supervision:	RPM

Date: 9/23/2014

Time: 16:53

F9 to re-calculate

Proposed dir: 90

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
175	12416.00	88.50	85.90	10647.78	-344.61	2026.38	2026.38	2.38
176	12508.00	88.70	87.00	10650.03	-338.92	2118.18	2118.18	1.21
177	12601.00	88.20	86.90	10652.54	-333.97	2211.01	2211.01	0.55
178	12695.00	90.80	88.70	10653.36	-330.36	2304.93	2304.93	3.36
179	12789.00	91.50	90.50	10651.48	-329.71	2398.90	2398.90	2.05
180	12883.00	90.20	89.70	10650.08	-329.87	2492.89	2492.89	1.62
181	12976.00	88.70	89.60	10650.97	-329.30	2585.88	2585.88	1.62
182	13070.00	88.50	89.30	10653.27	-328.40	2679.85	2679.85	0.38
183	13164.00	88.50	89.00	10655.73	-327.01	2773.81	2773.81	0.32
184	13258.00	90.80	90.40	10656.31	-326.52	2867.79	2867.79	2.86
185	13352.00	90.50	91.40	10655.24	-327.99	2961.77	2961.77	1.11
186	13445.00	90.50	91.50	10654.43	-330.34	3054.74	3054.74	0.11
187	13539.00	90.70	91.30	10653.44	-332.64	3148.71	3148.71	0.30
188	13633.00	88.20	91.20	10654.35	-334.69	3242.67	3242.67	2.66
189	13727.00	87.00	89.80	10658.28	-335.51	3336.58	3336.58	1.96
190	13821.00	89.20	90.90	10661.40	-336.09	3430.52	3430.52	2.62
191	13914.00	89.50	89.70	10662.45	-336.57	3523.51	3523.51	1.33
192	14008.00	88.90	90.40	10663.77	-336.66	3617.50	3617.50	0.98
193	14102.00	90.50	90.40	10664.26	-337.31	3711.50	3711.50	1.70
194	14196.00	88.30	90.80	10665.24	-338.30	3805.48	3805.48	2.38
195	14290.00	88.10	90.60	10668.19	-339.44	3899.43	3899.43	0.30
196	14384.00	90.10	91.60	10669.67	-341.25	3993.39	3993.39	2.38
197	14478.00	89.10	90.50	10670.33	-342.97	4087.37	4087.37	1.58
198	14572.00	88.40	89.70	10672.38	-343.13	4181.35	4181.35	1.13
199	14666.00	90.30	89.70	10673.44	-342.64	4275.34	4275.34	2.02
200	14760.00	90.10	89.10	10673.12	-341.66	4369.33	4369.33	0.67
201	14854.00	90.50	88.70	10672.62	-339.85	4463.31	4463.31	0.60
202	14947.00	88.90	86.80	10673.11	-336.20	4556.23	4556.23	2.67
203	15041.00	89.70	88.20	10674.26	-332.10	4650.13	4650.13	1.72
204	15136.00	90.80	88.90	10673.85	-329.70	4745.10	4745.10	1.37
205	15229.00	90.00	89.30	10673.20	-328.24	4838.08	4838.08	0.96
206	15323.00	91.30	89.90	10672.13	-327.58	4932.07	4932.07	1.52
207	15418.00	90.20	89.10	10670.89	-326.75	5027.06	5027.06	1.43
208	15512.00	87.70	90.30	10672.61	-326.26	5121.03	5121.03	2.95
209	15606.00	88.00	90.50	10676.13	-326.92	5214.96	5214.96	0.38

<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5301 44-24 3BR		
County:	McKenzie	State:	ND
QQ:	SE SE	Section:	24
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	926	FN/SL:	S
	245	FE/WL:	E

Kick-off:	6/30/2014
Finish:	9/16/2014
Directional Supervision:	RPM

Date: 9/23/2014

Time: 16:53

F9 to re-calculate

Proposed dir: 90

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
210	15700.00	86.50	89.20	10680.64	-326.67	5308.85	5308.85	2.11
211	15793.00	88.10	90.60	10685.03	-326.51	5401.74	5401.74	2.28
212	15887.00	88.40	91.20	10687.90	-327.99	5495.69	5495.69	0.71
213	15981.00	91.30	91.10	10688.14	-329.87	5589.66	5589.66	3.09
214	16075.00	92.40	91.40	10685.11	-331.92	5683.58	5683.58	1.21
215	16169.00	90.20	91.60	10682.98	-334.38	5777.52	5777.52	2.35
216	16263.00	88.70	91.30	10683.88	-336.76	5871.48	5871.48	1.63
217	16357.00	89.50	91.10	10685.35	-338.73	5965.45	5965.45	0.88
218	16450.00	89.90	91.50	10685.84	-340.84	6058.43	6058.43	0.61
219	16544.00	89.00	91.80	10686.74	-343.54	6152.38	6152.38	1.01
220	16638.00	89.80	92.30	10687.73	-346.91	6246.31	6246.31	1.00
221	16732.00	91.60	92.50	10686.58	-350.84	6340.22	6340.22	1.93
222	16825.00	90.00	91.60	10685.28	-354.17	6433.15	6433.15	1.97
223	16919.00	88.60	91.30	10686.43	-356.55	6527.11	6527.11	1.52
224	17013.00	89.90	91.20	10687.66	-358.60	6621.08	6621.08	1.39
225	17108.00	91.30	91.20	10686.67	-360.59	6716.05	6716.05	1.47
226	17139.00	89.00	90.30	10686.58	-360.99	6747.04	6747.04	7.97
227	17201.00	88.50	90.60	10687.94	-361.48	6809.03	6809.03	0.94
228	17295.00	87.20	90.10	10691.46	-362.05	6902.96	6902.96	1.48
229	17389.00	85.40	89.60	10697.53	-361.81	6996.76	6996.76	1.99
230	17483.00	86.50	89.80	10704.17	-361.32	7090.52	7090.52	1.19
231	17576.00	89.30	90.80	10707.58	-361.80	7183.44	7183.44	3.20
232	17670.00	89.30	90.00	10708.72	-362.46	7277.43	7277.43	0.85
233	17764.00	91.20	90.60	10708.31	-362.95	7371.43	7371.43	2.12
234	17857.00	91.60	90.10	10706.04	-363.52	7464.40	7464.40	0.69
235	17951.00	88.70	90.40	10705.80	-363.93	7558.39	7558.39	3.10
236	18045.00	88.70	90.40	10707.93	-364.59	7652.36	7652.36	0.00
237	18138.00	87.30	89.70	10711.17	-364.67	7745.30	7745.30	1.68
238	18231.00	88.70	88.60	10714.42	-363.29	7838.23	7838.23	1.91
239	18325.00	88.10	87.20	10717.04	-359.85	7932.13	7932.13	1.62
240	18418.00	88.40	86.90	10719.88	-355.06	8024.96	8024.96	0.46
241	18450.00	89.10	87.40	10720.58	-353.47	8056.91	8056.91	2.69
242	18512.00	89.50	88.10	10721.34	-351.04	8118.86	8118.86	1.30
243	18606.00	90.90	88.20	10721.01	-348.00	8212.81	8212.81	1.49
244	18699.00	91.30	88.20	10719.23	-345.08	8305.74	8305.74	0.43

<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Chalmers 5301 44-24 3BR		
County:	McKenzie	State:	ND
QQ:	SE SE	Section:	24
Township:	153	N/S:	N
Range:	101	E/W:	W
Footages:	926	FN/SL:	S
	245	FE/WL:	E

Kick-off:	6/30/2014
Finish:	9/16/2014
Directional Supervision:	RPM

Date: 9/23/2014

Time: 16:53

F9 to re-calculate

Proposed dir: 90

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
245	18793.00	90.80	88.10	10717.50	-342.05	8399.68	8399.68	0.54
246	18887.00	90.30	87.50	10716.60	-338.44	8493.60	8493.60	0.83
247	18982.00	89.80	87.40	10716.52	-334.21	8588.51	8588.51	0.54
248	19076.00	89.50	88.20	10717.09	-330.61	8682.44	8682.44	0.91
249	19171.00	89.80	90.80	10717.67	-329.78	8777.42	8777.42	2.75
250	19266.00	89.60	90.30	10718.17	-330.69	8872.42	8872.42	0.57
251	19361.00	90.20	90.50	10718.34	-331.35	8967.42	8967.42	0.67
252	19455.00	90.10	90.60	10718.09	-332.25	9061.41	9061.41	0.15
253	19549.00	90.10	89.90	10717.93	-332.66	9155.41	9155.41	0.74
254	19644.00	88.10	89.00	10719.42	-331.75	9250.39	9250.39	2.31
255	19739.00	88.10	89.60	10722.57	-330.59	9345.33	9345.33	0.63
256	19833.00	88.30	89.90	10725.52	-330.18	9439.28	9439.28	0.38
257	19927.00	89.50	89.70	10727.33	-329.85	9533.26	9533.26	1.29
258	20020.00	90.70	89.40	10727.16	-329.12	9626.26	9626.26	1.33
259	20114.00	89.80	89.10	10726.75	-327.89	9720.25	9720.25	1.01
260	20208.00	91.70	89.20	10725.52	-326.50	9814.22	9814.22	2.02
261	20302.00	91.40	88.70	10722.98	-324.78	9908.17	9908.17	0.62
262	20396.00	91.10	89.00	10720.93	-322.89	10002.13	10002.13	0.45
263	20489.00	90.50	88.50	10719.63	-320.86	10095.10	10095.10	0.84
264	20584.00	90.90	88.60	10718.47	-318.46	10190.06	10190.06	0.43
265	20678.00	90.70	87.80	10717.16	-315.51	10284.01	10284.01	0.88
266	20772.00	88.30	86.70	10717.98	-311.00	10377.88	10377.88	2.81
267	20833.00	88.30	86.70	10719.79	-307.49	10438.76	10438.76	0.00

FORMATION TOPS & STRUCTURAL RELATIONSHIPS

Operator: Well Name: Location:	Subject Well:										Offset Wells:		
	Oasis Petroleum North America, LLC Chalmers 5301 44-24 3BR 926' FSL & 245' FSL SE SE Section 24, T153N, R101W GL: 1,943' Sub: 25' KB: 1,968'												
Elevation:													
Formation/ Marker	Prog. Top	Prog. Datum (MSL)	Driller's Depth Top (MD)	Driller's Depth Top (TVD)	Datum (MSL)	Interval Thickness	Thickness to Target	Dip To Prog.	Dip To Chalmers 5301 44-24 2TR	Dip To Chalmers 5300 31-19H	Dip To Lynn 1		
Kibbey Lime	8,269'	-6,301'	8,269'	8,268'	-6,300'	171'	2,353'	1'	-4'	11'	-47'		
Charles	8,433'	-6,465'	8,440'	8,439'	-6,471'	548'	2,182'	-6'	-10'	4'	-54'		
UB	9,010'	-7,042'	8,988'	8,987'	-7,019'	84'	1,634'	23'	-5'	30'	-28'		
Base Last Salt	9,091'	-7,123'	9,072'	9,071'	-7,103'	46'	1,550'	20'	-9'	27'	-31'		
Ratcliffe	9,141'	-7,173'	9,118'	9,117'	-7,149'	174'	1,504'	24'	-8'	30'	-27'		
Mission Canyon	9,311'	-7,343'	9,292'	9,291'	-7,323'	558'	1,330'	20'	-5'	26'	-27'		
Lodgepole	9,876'	-7,908'	9,850'	9,849'	-7,881'	99'	772'	27'	-4'	34'	-35'		
Lodgepole A	9,953'	-7,985'	9,949'	9,948'	-7,980'	105'	673'	5'	5'	14'	-56'		
Lodgepole B	10,109'	-8,141'	10,054'	10,053'	-8,085'	38'	568'	56'	-2'	63'	-70'		
Lodgepole C	10,180'	-8,212'	10,092'	10,091'	-8,123'	187'	530'	89'	0'	96'	-27'		
Lodgepole D	10,283'	-8,315'	10,286'	10,278'	-8,310'	151'	343'	5'	-3'	12'	-86'		
Lodgepole E	10,429'	-8,461'	10,470'	10,429'	-8,461'	68'	192'	0'	-8'	7'	-57'		
Lodgepole F	10,502'	-8,534'	10,583'	10,497'	-8,529'	87'	124'	5'	-5'	11'	-44'		
False Bakken	10,588'	-8,620'	10,754'	10,584'	-8,616'	8'	37'	4'	-7'	11'	-54'		
Upper Bakken Shale	10,599'	-8,631'	10,785'	10,592'	-8,624'	20'	29'	7'	-7'	12'	-52'		
Middle Bakken	10,616'	-8,648'	10,853'	10,612'	-8,644'	9'	9'	4'	-10'	10'	-57'		
Landing Target	10,630'	-8,662'	10,939'	10,621'	-8,653'	-	-	9'	-12'	11'	-59'		

CONTROL DATA

Operator: Well Name: Location:	Oasis Petroleum North America Chalmers 5301 44-24 2TR SE Sec. 24, T153N, R101W McKenzie Co., ND shares pad with subject well KB: 1,968'					Oasis Petroleum North America Chalmers 5300 31-19H NW SW Sec. 19, T153N, R100W McKenzie Co., ND 0.16 miles northeast of Chalmers 5301 44-24 3BR KB: 1,929'					Missouri Basin Well Lynn 1 SW NW Sec. 23, T153N, R101W McKenzie Co., ND 83 miles northwest of Chalmers 5301 44-24 3BR KB: 2,142'				
Elevation: Formation/ Zone	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target		E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target		E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target	
Kibbey Lime	8,264'	-6,296'	165'	2,352'		8,240'	-6,311'	164'	2,358'		8,395'	-6,253'	164'	2,346'	
Charles	8,429'	-6,461'	553'	2,187'		8,404'	-6,475'	574'	2,194'		8,559'	-6,417'	574'	2,182'	
UB	8,982'	-7,014'	80'	1,634'		8,978'	-7,049'		1,620'		9,133'	-6,991'		1,608'	
Base Last Salt	9,062'	-7,094'	47'	1,554'		9,059'	-7,130'	81'	1,539'		9,214'	-7,072'	81'	1,527'	
Ratcliffe	9,109'	-7,141'	177'	1,507'		9,108'	-7,179'	49'	1,490'		9,264'	-7,122'	50'	1,477'	
Mission Canyon	9,286'	-7,318'	559'	1,330'		9,278'	-7,349'	566'	1,320'		9,438'	-7,296'	550'	1,303'	
Lodgepole	9,845'	-7,877'	108'	771'		9,844'	-7,915'	79'	754'		9,988'	-7,846'	78'	753'	
Lodgepole A	9,953'	-7,985'	98'	663'		9,923'	-7,994'	154'	675'		10,066'	-7,924'	91'	675'	
Lodgepole B	10,051'	-8,083'	40'	565'		10,077'	-8,148'	71'	521'		10,157'	-8,015'	81'	584'	
Lodgepole C	10,091'	-8,123'	184'	525'		10,148'	-8,219'	103'	450'		10,238'	-8,096'	128'	503'	
Lodgepole D	10,275'	-8,307'	146'	341'		10,251'	-8,322'	146'	347'		10,366'	-8,224'	180'	375'	
Lodgepole E	10,421'	-8,453'	71'	195'		10,397'	-8,468'	72'	201'		10,546'	-8,404'	81'	195'	
Lodgepole F	10,492'	-8,524'	85'	124'		10,469'	-8,540'	87'	129'		10,627'	-8,485'	77'	114'	
False Bakken	10,577'	-8,609'	8'	39'		10,556'	-8,627'	9'	42'		10,704'	-8,562'	10'	37'	
Upper Bakken Shale	10,585'	-8,617'	17'	31'		10,565'	-8,636'	18'	33'		10,714'	-8,572'	15'	27'	
Middle Bakken	10,602'	-8,634'	7'	14'		10,583'	-8,654'	10'	15'		10,729'	-8,587'	7'	12'	
Top of Target	10,609'	-8,641'	7'	7'		10,593'	-8,664'	5'	5'		10,736'	-8,594'	5'	5'	
Landing Target	10,616'	-8,648'	5'	0'		10,598'	-8,669'	3'	0'		10,741'	-8,599'	4'	0'	
Base of Target	10,621'	-8,653'	23'	-		10,601'	-8,672'	13'	-		10,745'	-8,603'	19'	-	
Lower Bakken Shale	10,644'	-8,676'	-	-		10,614'	-8,685'	-	-		10,764'	-8,622'	-	-	

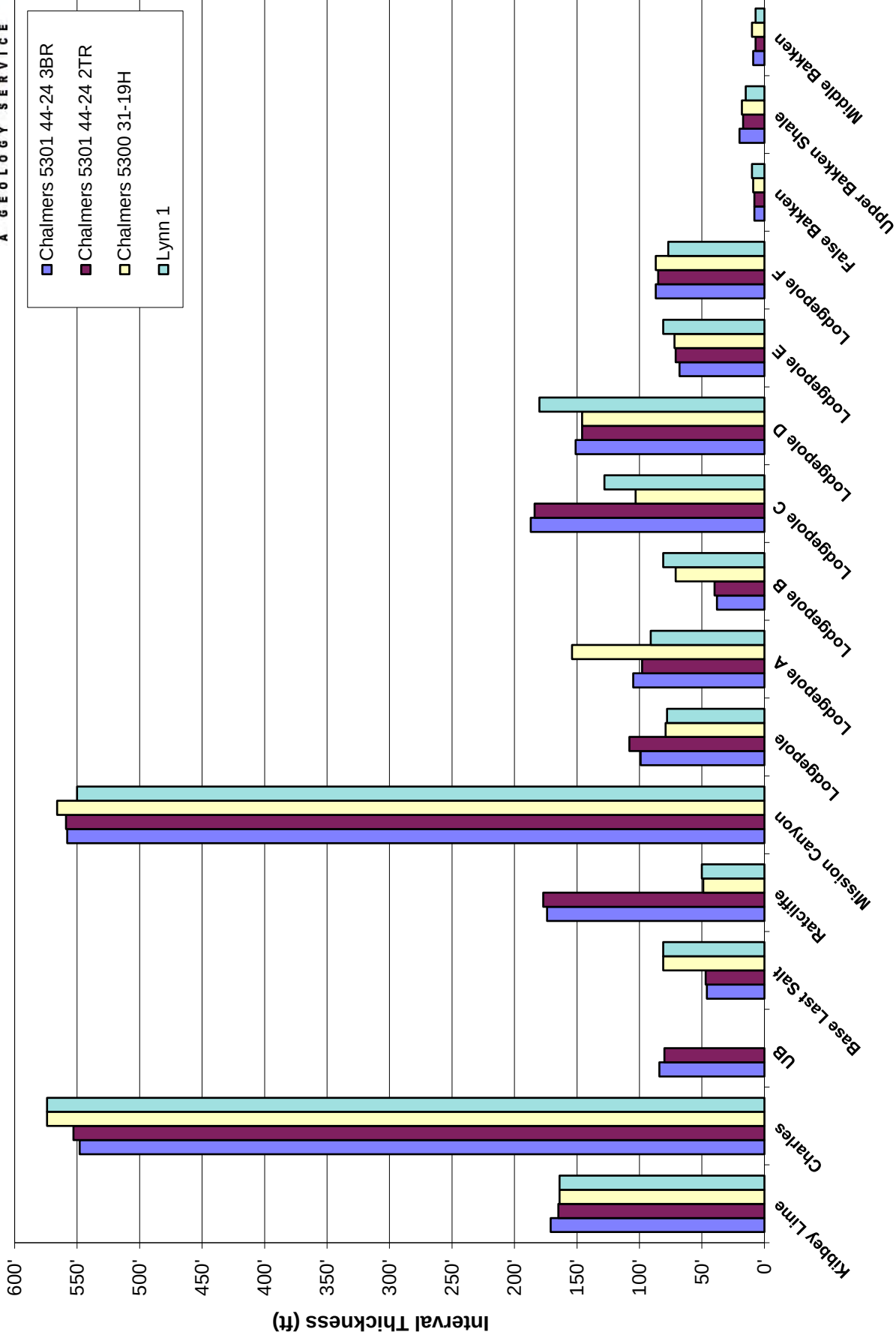
LANDING PROJECTION

Formation/ Zone:	Proposed Top of Target From:		
	Chalmers 5301 44-24 2TR	Chalmers 5300 31-19H	Lynn 1
			Average of Offset Wells
Kibbey Lime	10,620'	10,626'	10,614'
Charles	10,626'	10,633'	10,621'
UB	10,621'	10,691'	10,679'
Base Last Salt	10,625'	10,656'	10,644'
Ratcliffe	10,624'	10,611'	10,594'
Mission Canyon	10,621'	10,603'	10,602'
Lodgepole	10,620'	10,623'	10,623'
Lodgepole A	10,611'	10,574'	10,637'
Lodgepole B	10,618'	10,541'	10,594'
Lodgepole C	10,616'	10,625'	10,653'
Lodgepole D	10,619'	10,630'	10,624'
Lodgepole E	10,624'	10,626'	10,611'
Lodgepole F	10,621'	10,626'	10,621'
False Bakken	10,623'	10,625'	10,619'
Upper Bakken Shale	10,623'	10,627'	10,624'
Middle Bakken	10,626'		
Landing Target	10,628'	10,621'	10,621'

Current Landing Target (9' below the base of the UBS): **10,621'**

INTERVAL THICKNESS

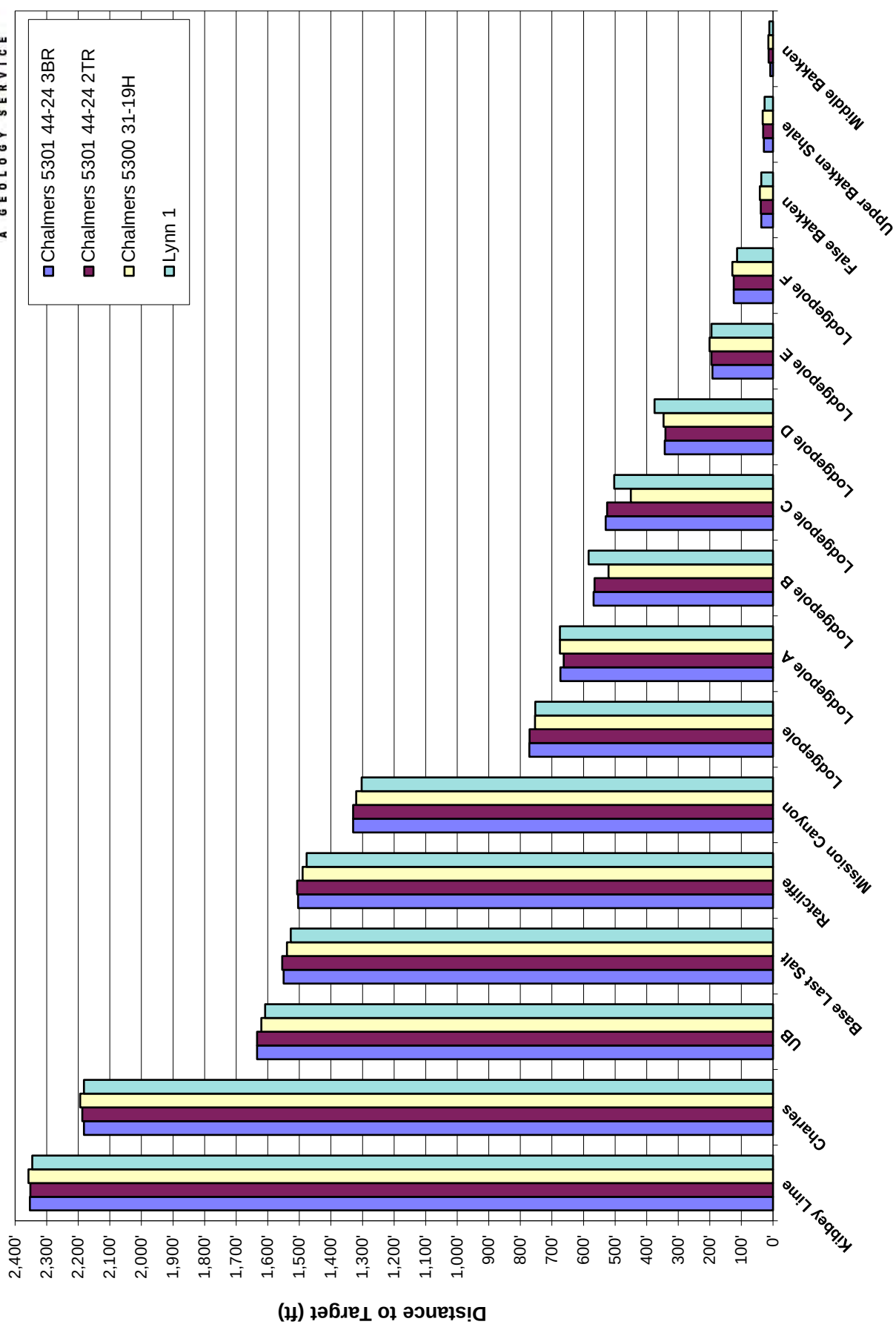
Oasis Petroleum North America, LLC - Chalmers 5301 44-24 3BR





ISOPACH TO TARGET

Oasis Petroleum North America, LLC - Chalmers 5301 44-24 3BR



LITHOLOGY

Rig crews caught lagged samples in 30' intervals under the supervision of Sunburst geologists. A detailed list of sampling intervals is included in the well data summary page. Sample or gamma ray marker tops have been inserted in the sample descriptions below for reference. Samples were examined wet and dry under a trinocular microscope. The drilling fluid was diesel-based invert from surface casing to intermediate casing, while salt water drilling fluid was used throughout the lateral. Sample collection began at 8,200' MD.

Kibbey

8200-8230 SILTSTONE: red orange, tan, pink, soft to firm, sub-blocky, calcite cement, poorly cemented; rare SILTY SANDSTONE: tan to off white, very fine grained, sub round, moderately sorted, calcite cement, poorly cemented; trace ANHYDRITE: off white, soft, massive, amorphous texture

8230-8260 SILTSTONE: dark orange to light brown, tan, pink, rare gray, soft to firm, sub-blocky, calcite cement, poorly cemented; rare ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: tan to off white, very fine grained, sub round, moderately sorted, calcite cement, poorly cemented

Kibbey "Lime" [8,269' MD, 8,268' TVD (-6,300')]

8260-8290 SILTSTONE: orange to light brown, tan, gray, soft, sub-blocky, calcite cement, poorly cemented; rare ANHYDRITE: off white, light pink, soft, amorphous texture; trace SILTY SANDSTONE: tan to off white, very fine grained, sub round, moderately sorted, calcite cement, poorly cemented

8290-8320 SILTSTONE: dark orange to light brown, tan, pink, soft, sub-blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

8320-8350 SILTSTONE: dark orange to light brown, tan, pink, soft, sub-blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

8350-8380 SILTSTONE: dark orange to light brown, tan, pink, soft, sub-blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

8380-8410 SILTSTONE: orange to light brown, tan, pink, soft, sub-blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

8410-8440 SILTSTONE: dark orange to light brown, tan to gray, trace pink, soft, sub-blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

The First Charles Salt [8,440' MD, 8,439' TVD (-6,471')]

8440-8470 SALT: translucent, microcrystalline, anhedral, crystalline texture, hard, frosted; trace LIMESTONE: mudstone, light to medium gray to gray brown, microcrystalline, firm to hard, argillaceous in part, dense, chalky texture, no visible porosity

8470-8500 SALT: translucent, trace frosted, microcrystalline, anhedral, crystalline texture, hard; rare LIMESTONE: mudstone, light to medium gray to gray brown, microcrystalline, firm to hard, argillaceous in part, dense, chalky texture, no visible porosity

8500-8530 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8530-8560 SALT: translucent, trace frosted, microcrystalline, anhedral, crystalline texture, hard; trace LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8560-8590 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; trace LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8590-8620 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, occasional gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8620-8650 ANHYDRITE: off white, soft, amorphous; abundant SALT: clear to milky, off white, crystalline, hard, euhedral; trace LIMESTONE: as above

8650-8680 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, occasional gray brown, microcrystalline, firm to hard, dense, earthy texture, no visible porosity; trace ANHYDRITE: off white, soft, massive, amorphous

8680-8710 SALT: translucent, microcrystalline, anhedral, crystalline texture, hard, frosted; common LIMESTONE: mudstone, light to medium gray to gray brown, microcrystalline, firm to hard, argillaceous in part, dense, chalky texture; rare ANHYDRITE: off white, soft, amorphous, no visible porosity

8710-8740 SALT: translucent, rare milky, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, medium gray, occasional gray brown, trace light gray, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8740-8770 SALT: translucent, occasional milky, microcrystalline, anhedral, crystalline texture, hard; rare LIMESTONE: mudstone, light gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8770-8800 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light gray, trace gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8800-8830 LIMESTONE: mudstone, light gray, trace gray brown, rare off white, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8830-8860 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: off white, soft, massive, amorphous, no visible porosity

8860-8890 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional SALT: clear to milky, off white, crystalline, hard, euhedral; trace ANHYDRITE: as above

8890-8920 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity, trace SALT: as above

8920-8950 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity, trace SALT: as above

8950-8980 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, trace ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity

8980-9010 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white, trace light tan, soft, massive, amorphous, no visible porosity

9010-9040 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: as above; rare SALT: translucent, rare milky, crystalline, hard, euhedral

9040-9070 SALT: translucent, rare milky, crystalline, hard, euhedral; occasional ARGILLACEOUS LIMESTONE: mudstone, cream to tan, rare light to medium gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity

Base Last Salt [9,072' MD, 9,071' TVD (-7,103')]

9070-9100 ARGILLACEOUS LIMESTONE: mudstone, cream to tan, rare light to medium gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity; trace ANHYDRITE: off white, soft, massive, amorphous, no visible porosity

Ratcliffe [9,118' MD, 9,117' TVD (-7,149')]

9100-9130 ARGILLACEOUS LIMESTONE: mudstone, cream to tan, rare brown, trace gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity; trace ANHYDRITE: off white, soft, massive, amorphous, no visible porosity

9130-9160 ARGILLACEOUS LIMESTONE: mudstone, cream to tan, rare brown, trace gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity; trace ANHYDRITE: off white, soft, massive, amorphous, no visible porosity

9160-9190 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9190-9220 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9220-9250 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9250-9280 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

Mission Canyon [9,292' MD, 9,291' TVD (-7,323')]

9280-9310 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9310-9340 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9340-9370 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9370-9400 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9400-9430 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9430-9460 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9460-9490 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9490-9520 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9520-9550 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9550-9580 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9580-9610 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9610-9640 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9640-9670 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9670-9700 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9700-9730 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9730-9760 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9760-9790 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9790-9820 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, microcrystalline, firm to friable, rare algal material; trace ANHYDRITE: off white, massive, soft, amorphous

9820-9850 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, microcrystalline, firm to friable, earthy texture, rare algal material

Lodgepole [9,850' MD, 9,849' TVD (-7,881')]

9850-9880 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, microcrystalline, firm to friable, earthy texture, rare algal material

9880-9910 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture, trace algal material

9910-9940 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture

9940-9970 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture

9970-10000 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture; trace ANHYDRITE: off white, massive, soft, amorphous

10000-10030 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture, trace algal material; trace ANHYDRITE: off white, massive, soft, amorphous

10030-10060 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture, trace algal material; trace ANHYDRITE: off white, massive, soft, amorphous

10060-10090 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, light tan, trace cream, slightly mottled, microcrystalline, firm to friable, earthy texture, trace algal material; trace ANHYDRITE: off white, massive, soft, amorphous

10090-10120 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, light tan, trace cream, slightly mottled, microcrystalline, firm to friable, earthy texture, trace algal material; trace ANHYDRITE: off white, massive, soft, amorphous

10120-10150 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, microcrystalline, firm to friable, rare algal material; rare ANHYDRITE: off white, massive, soft, amorphous

10150-10180 ARGILLACEOUS LIMESTONE: mudstone, occasional gray brown, rare tan, microcrystalline, firm to friable, trace algal material; trace ANHYDRITE: off white, massive, soft, amorphous

10180-10210 ARGILLACEOUS LIMESTONE: mudstone, occasional gray brown, rare tan, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, massive, soft, amorphous

10210-10240 ARGILLACEOUS LIMESTONE: mudstone, occasional gray brown, rare tan, rare cream, microcrystalline, firm to friable, earthy, trace algal material

10240-10270 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, rare cream, microcrystalline, firm to friable, earthy, trace algal material

10270-10300 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10300-10330 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10330-10360 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10360-10390 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10390-10420 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10420-10450 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10450-10480 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10480-10510 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10510-10540 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10540-10570 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10570-10600 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10600-10630 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10630-10660 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10660-10690 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10690-10720 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy, trace algal material; trace ANHYDRITE: off white, soft, massive, amorphous

10720-10750 ARGILLACEOUS LIMESTONE: mudstone, gray brown, rare tan, trace cream, microcrystalline, firm to friable, earthy

False Bakken [10,754' MD, 10,584' TVD (-8,616')]

10750-10780 LIMESTONE: mudstone, light brown, rare tan, trace cream, microcrystalline, firm to friable, earthy

Upper Bakken [10,785' MD, 10,592' TVD (-8,624')]

10780-10810 SHALE: black to dark brown, firm, blocky, earthy, disseminated pyrite, petroliferous, carbonaceous, rare fracture porosity

10810-10820 SHALE: black to dark brown, firm, blocky, earthy, disseminated pyrite, petroliferous, carbonaceous, rare fracture porosity

10820-10830 SHALE: black to dark brown, firm, blocky, earthy, disseminated pyrite, petroliferous, carbonaceous, rare fracture porosity

10830-10840 SHALE: black, firm, blocky, earthy, disseminated pyrite, petroliferous, carbonaceous, rare fracture porosity

10840-10850 SHALE: black, firm, blocky, earthy, disseminated pyrite, petroliferous, carbonaceous, rare fracture porosity

Middle Bakken [10,853' MD, 10,612' TVD (-8,644')]

10850-10860 SHALE: black to dark brown, friable to firm, blocky, earthy, disseminated pyrite, petroliferous, carbonaceous, rare fracture porosity

10860-10870 SILTSTONE: light gray to light gray brown, sub-blocky, calcite cemented, trace disseminated pyrite, trace intergranular porosity

10870-10880 SILTSTONE: light gray to light gray brown, sub-blocky, calcite cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10880-10890 SILTSTONE: light gray to light gray brown, sub-blocky, calcite cemented, trace disseminated pyrite, trace intergranular porosity; trace SILTY SANDSTONE: medium to light gray, fine grained, sub round, firm, moderate sorted, calcite cement, moderate cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain

10890-10910 SILTSTONE: as above; occasional SILTY SANDSTONE: medium to light gray, trace tan, fine grained, sub round, firm, moderate sorted, calcite cement, moderate cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

10910-10920 SILTY SANDSTONE: medium to light gray, trace tan, fine grained, sub round, firm, moderate sorted, calcite cement, moderate cemented, trace disseminated pyrite, trace intergranular porosity trace spotty light brown oil stain; trace SILTSTONE: as above

10920-10930 SILTY SANDSTONE: medium to light gray, trace tan, fine grained, sub round, firm, moderate sorted, calcite cement, moderate cemented, trace disseminated pyrite, trace intergranular porosity, trace spotty light brown oil stain

11330-11360 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast cloudy pale yellow cut florescence

11780-11810 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast diffuse pale yellow cut florescence

12230-12260 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast diffuse pale yellow cut florescence

12680-12710 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast diffuse pale yellow cut florescence

13130-13160 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast diffuse pale yellow cut florescence

13580-13610 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast diffuse pale yellow cut florescence

14030-14060 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast diffuse pale yellow cut florescence

14480-14510 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast diffuse pale yellow cut florescence

14930-14960 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast cloudy pale yellow cut florescence

15380-15410 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, moderately contaminated with lube

15830-15860 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavy contaminated with lube

16280-16310 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavy contaminated with lube

16730-16760 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavy contaminated with lube

17180-17210 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast cloudy pale yellow cut florescence

17630-17660 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast cloudy pale yellow cut florescence

18080-18110 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, fast cloudy pale yellow cut florescence

18530-18560 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavily contaminated with lube

18980-19010 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavily contaminated with lube

19430-19460 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavily contaminated with lube

19880-19910 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavily contaminated with lube

20330-20360 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavily contaminated with lube

20780-20810 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavily contaminated with lube

20810-20833 SILTY SANDSTONE: light to medium gray to brown, fine grained, sub round, firm, moderately sorted, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace spotty light brown oil stain, heavily contaminated with lube

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28599

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed September 23, 2014
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☒ Reclamation
☐ Other	<u>Reserve pit reclamation</u>

Well Name and Number Chalmers 5301 44-24 3BR					
Footages		Qtr-Qtr	Section	Township	Range
926 F S L 245 F E L		SESE	24	153 N	101 W
Field	Pool	County			
Baker	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Neu Construction			
Address 602 W 9th Street	City Fairview	State MT	Zip Code 59221

DETAILS OF WORK

Oasis Petroleum North America LLC plans to reclaim the reserve pit for the above referenced wells as follows:

The NDIC field inspector, Rick Dunn (NDIC) was notified on 09/23/2014

The surface owners, Wes Lindvig was contacted on 09/23/2014
Wes Lindvig 140758 41st Street NW Alexander, ND 58831

Spread material out in pit, cut top edge of liner and fold over cuttings, cover entire pit with line, back fill with clay slope and contour well site to ensure proper drainage

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9591	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Chelsea Covington		
Title Regulatory Assistant	Date September 24, 2014		
Email Address ccovington@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date 10-6-14	
By 	
Title 	



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Thursday, September 18, 2014

State of North Dakota

Subject: **Surveys**

Re: **Oasis**
Chalmers 5301 44-24 3BR
McKenzie, ND
A.P.I. No: 33-053-06010

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc.. Other information required by your office is as follows:

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of</i>	<i>TD Straight Line Projection</i>
Ogden, Daniel	MWD Operator	O.H.	0'	10880'	06/24/14	07/05/14	MWD	10880'
Ogden, Daniel	MWD Operator	O.H.	18880'	20772'	09/08/14	09/15/14	MWD	20883'

A certified plat on which the bottom hole location is oriented both to the surface location and to the lease lines (or unit lines in case of pooling) is attached to the survey report. If any other information is required please contact the undersigned at the letterhead address or phone number.

Hudson, Douglas
Well Planner



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Directional Services, Inc.
19510 Oil Center Blvd.
Houston, Texas 77073
Bus: 281.443.1414
Fax: 281.443.1676

Saturday, July 05, 2014

State of North Dakota
County of McKenzie Co.

Subject: **Survey Certification Letter**

Survey Company: Ryan Directional Services, Inc.
Job Number: 7798
Survey Job Type: Ryan MWD
Customer: Oasis Petroleum N.A. LLC
Well Name: Chalmers 5301 44-24 3BR
Rig Name: Nabors B-25

Surface: 48°03'20.49", -103°36'18.55"
A.P.I. No: 33-053-06010
Location: McKenzie Co., North Dakota
RKB Height: 25'
Distance to Bit: 59'

Surveyor Name	Surveyor Title	Borehole Number	Start Depth	End Depth	Start Date	End Date	Type of	TD Straight
								Line Projection
Daniel Ogden	MWD Supervisor	OH	100'	10880'	06/24/14	07/05/14	MWD	10939'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

Daniel Ogden
MWD Supervisor
Ryan Directional Services, Inc.



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Directional Services, Inc.
19510 Oil Center Blvd.
Houston, Texas 77073
Bus: 281.443.1414
Fax: 281.443.1676

Tuesday, September 16, 2014

State of North Dakota
County of McKenzie

Subject: **Survey Certification Letter**

Survey Company: Ryan Directional Services, Inc.
Job Number: 7893
Survey Job Type: Ryan MWD
Customer: Oasis Petroleum N.A. LLC
Well Name: Chalmers 5301 44-24 3BR
Rig Name: Nabors B-25

Surface: 48°03'20.49", -103°36'18.55"
A.P.I. No: 33-053-06010
Location: McKenzie, ND
RKB Height: 25'
Distance to Bit: 61'

Surveyor Name	Surveyor Title	Borehole Number	Start Depth	End Depth	Start Date	End Date	Type of	TD Straight Line Projection
Daniel Ogden	MWD Supervisor	OH	10939'	20772'	09/08/14	09/16/14	MWD	20833'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.


Daniel Ogden
MWD Supervisor
Ryan Directional Services, Inc.

Report #: **1**
Date: **24-Jun-14**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **7798**

SURVEY REPORT

Customer: **Oasis Petroleum N.A. LLC**
Well Name: **Chalmers 5301 44-24 3BR**
Rig #: **Nabors B-25**
API #: **33-053-06010**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **90**
Total Correction: **8.27**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Conductor Well Depth									
Tie In	0	0.00	0.00		0.00	0.00	0.00	0.00	0.00
1	184	0.80	273.10	1.00	183.99	-1.28	0.07	-1.28	0.43
2	215	0.70	271.20	71.00	214.99	-1.69	0.09	-1.69	0.33
3	308	0.90	267.70	71.00	307.98	-2.99	0.07	-2.99	0.22
4	402	0.30	299.60	71.00	401.98	-3.94	0.16	-3.94	0.71
5	495	0.30	302.10	73.00	494.98	-4.36	0.41	-4.36	0.01
6	588	0.70	304.50	75.00	587.97	-5.03	0.86	-5.03	0.43
7	680	0.70	339.40	75.00	679.97	-5.69	1.70	-5.69	0.46
8	772	1.30	28.50	78.00	771.95	-5.39	3.15	-5.39	1.08
9	865	1.20	56.40	80.00	864.93	-4.08	4.61	-4.08	0.66
10	957	1.10	52.90	82.00	956.91	-2.57	5.68	-2.57	0.13
11	1049	1.00	55.60	86.00	1048.90	-1.20	6.67	-1.20	0.12
12	1142	1.70	55.50	87.00	1141.87	0.60	7.91	0.60	0.75
13	1234	1.20	103.10	91.00	1233.84	2.67	8.46	2.67	1.37
14	1326	0.80	133.60	93.00	1325.83	4.07	7.80	4.07	0.71
15	1418	0.70	167.60	98.00	1417.82	4.66	6.81	4.66	0.49
16	1510	0.40	124.20	98.00	1509.82	5.04	6.08	5.04	0.54
17	1604	0.30	109.00	100.00	1603.82	5.55	5.81	5.55	0.14
18	1697	0.20	134.70	104.00	1696.82	5.89	5.62	5.89	0.16
19	1790	0.40	172.40	104.00	1789.82	6.05	5.18	6.05	0.29
20	1883	0.40	108.80	105.00	1882.81	6.40	4.76	6.40	0.45
21	1977	0.50	116.60	107.00	1976.81	7.08	4.47	7.08	0.12
22	2031	0.40	151.40	102.00	2030.81	7.38	4.20	7.38	0.53
23	2096	0.50	271.90	96.00	2095.81	7.20	4.01	7.20	1.20
24	2190	0.40	197.90	100.00	2189.81	6.69	3.71	6.69	0.58
25	2283	0.70	165.90	104.00	2282.80	6.73	2.85	6.73	0.45
26	2376	1.10	170.60	107.00	2375.79	7.02	1.42	7.02	0.44
27	2469	1.30	167.40	109.00	2468.77	7.39	-0.49	7.39	0.23
28	2563	0.90	130.90	111.00	2562.75	8.18	-2.02	8.18	0.84
29	2656	0.40	132.00	114.00	2655.75	8.98	-2.71	8.98	0.54
30	2749	0.80	137.60	116.00	2748.74	9.66	-3.41	9.66	0.43
31	2843	0.70	86.60	120.00	2842.73	10.67	-3.86	10.67	0.69
32	2936	1.00	99.10	122.00	2935.72	12.04	-3.95	12.04	0.38
33	3029	0.80	74.90	125.00	3028.71	13.47	-3.91	13.47	0.46
34	3122	1.10	76.00	127.00	3121.70	14.96	-3.53	14.96	0.32
35	3216	0.90	83.40	127.00	3215.69	16.57	-3.23	16.57	0.25
36	3309	0.70	115.00	127.00	3308.68	17.81	-3.38	17.81	0.51
37	3402	0.60	124.30	127.00	3401.67	18.73	-3.90	18.73	0.16
38	3495	1.30	50.90	127.00	3494.66	19.95	-3.50	19.95	1.36
39	3589	1.30	62.60	129.00	3588.64	21.72	-2.34	21.72	0.28
40	3682	0.70	75.70	131.00	3681.62	23.21	-1.72	23.21	0.69
41	3775	1.00	93.70	132.00	3774.61	24.57	-1.63	24.57	0.43
42	3868	1.00	87.90	132.00	3867.60	26.19	-1.65	26.19	0.11
43	3962	0.80	91.20	134.00	3961.59	27.67	-1.63	27.67	0.22
44	4055	0.60	112.20	136.00	4054.58	28.77	-1.83	28.77	0.35
45	4148	0.90	148.50	134.00	4147.57	29.60	-2.64	29.60	0.59
46	4242	0.70	10.80	136.00	4241.57	30.09	-2.70	30.09	1.59
47	4335	0.50	1.00	140.00	4334.56	30.21	-1.74	30.21	0.24
48	4428	0.40	10.40	141.00	4427.56	30.27	-1.02	30.27	0.13
49	4521	0.30	2.20	143.00	4520.56	30.34	-0.45	30.34	0.12
50	4615	0.10	305.30	143.00	4614.56	30.28	-0.16	30.28	0.28
51	4708	0.30	320.50	145.00	4707.56	30.06	0.08	30.06	0.22
52	4801	0.80	303.50	147.00	4800.55	29.36	0.62	29.36	0.56
53	4894	0.70	277.90	147.00	4893.54	28.26	1.06	28.26	0.37
54	4988	0.30	167.50	149.00	4987.54	27.74	0.90	27.74	0.91
55	5081	0.70	123.70	150.00	5080.54	28.27	0.34	28.27	0.57
56	5174	1.00	162.10	150.00	5173.53	28.99	-0.74	28.99	0.67
57	5268	1.10	169.10	152.00	5267.51	29.41	-2.41	29.41	0.17
58	5361	0.30	157.70	152.00	5360.51	29.68	-3.51	29.68	0.87
59	5454	0.60	147.50	154.00	5453.50	30.03	-4.15	30.03	0.33
60	5547	0.70	142.00	156.00	5546.50	30.64	-5.01	30.64	0.13

Report #: **1**
Date: **24-Jun-14**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **7798**

SURVEY REPORT

Customer: **Oasis Petroleum N.A. LLC**
Well Name: **Chalmers 5301 44-24 3BR**
Rig #: **Nabors B-25**
API #: **33-053-06010**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **90**
Total Correction: **8.27**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
61	5641	1.00	165.90	158.00	5640.49	31.19	-6.25	31.19	0.49
62	5734	1.20	149.70	158.00	5733.47	31.88	-7.88	31.88	0.40
63	5827	1.00	172.70	158.00	5826.45	32.48	-9.53	32.48	0.52
64	5920	1.10	185.40	158.00	5919.44	32.50	-11.22	32.50	0.27
65	5977	1.10	167.90	158.00	5976.43	32.56	-12.30	32.56	0.59
66	6062	1.20	178.60	122.00	6061.41	32.75	-13.99	32.75	0.28
67	6093	1.10	184.00	122.00	6092.40	32.74	-14.61	32.74	0.48
68	6187	0.60	19.80	125.00	6186.40	32.84	-15.05	32.84	1.79
69	6280	0.70	344.00	129.00	6279.39	32.85	-14.04	32.85	0.44
70	6373	0.60	322.50	132.00	6372.39	32.40	-13.11	32.40	0.28
71	6466	0.40	285.40	136.00	6465.38	31.79	-12.64	31.79	0.40
72	6560	0.90	70.10	140.00	6559.38	32.17	-12.30	32.17	1.33
73	6653	0.60	65.80	145.00	6652.37	33.30	-11.85	33.30	0.33
74	6746	0.60	100.50	147.00	6745.37	34.22	-11.74	34.22	0.38
75	6839	1.10	34.10	150.00	6838.36	35.20	-11.09	35.20	1.10
76	6933	1.30	16.20	152.00	6932.34	36.00	-9.32	36.00	0.45
77	7026	1.20	17.60	156.00	7025.32	36.59	-7.38	36.59	0.11
78	7119	1.20	17.00	158.00	7118.30	37.17	-5.52	37.17	0.01
79	7212	1.10	6.20	159.00	7211.28	37.55	-3.70	37.55	0.26
80	7306	0.50	35.80	161.00	7305.27	37.89	-2.47	37.89	0.75
81	7399	0.40	50.40	165.00	7398.27	38.38	-1.93	38.38	0.16
82	7492	0.50	41.90	167.00	7491.26	38.90	-1.42	38.90	0.13
83	7586	0.70	59.10	168.00	7585.26	39.67	-0.82	39.67	0.28
84	7679	0.90	70.80	170.00	7678.25	40.84	-0.29	40.84	0.28
85	7772	0.70	65.80	172.00	7771.24	42.05	0.18	42.05	0.23
86	7865	0.40	9.00	172.00	7864.24	42.62	0.73	42.62	0.63
87	7959	0.40	307.50	174.00	7958.23	42.41	1.26	42.41	0.44
88	8052	0.50	318.60	176.00	8051.23	41.89	1.76	41.89	0.14
89	8145	0.70	315.10	177.00	8144.23	41.22	2.47	41.22	0.22
90	8239	0.70	312.30	179.00	8238.22	40.39	3.26	40.39	0.04
91	8332	0.50	328.70	167.00	8331.21	39.75	3.99	39.75	0.28
92	8425	0.70	329.90	170.00	8424.21	39.26	4.83	39.26	0.22
93	8518	0.50	336.30	174.00	8517.20	38.81	5.69	38.81	0.23
94	8612	0.60	358.50	177.00	8611.20	38.63	6.56	38.63	0.25
95	8705	0.70	330.80	179.00	8704.19	38.34	7.54	38.34	0.35
96	8798	0.70	322.90	183.00	8797.19	37.72	8.49	37.72	0.10
97	8892	0.90	317.60	185.00	8891.18	36.88	9.49	36.88	0.23
98	8985	0.10	318.50	186.00	8984.17	36.33	10.09	36.33	0.86
99	9078	0.80	149.30	188.00	9077.17	36.61	9.60	36.61	0.97
100	9171	0.80	154.60	192.00	9170.16	37.22	8.45	37.22	0.08
101	9265	0.60	154.90	194.00	9264.15	37.71	7.41	37.71	0.21
102	9358	0.70	158.50	194.00	9357.15	38.13	6.44	38.13	0.12
103	9451	0.70	168.90	197.00	9450.14	38.44	5.36	38.44	0.14
104	9545	0.80	168.00	199.00	9544.13	38.69	4.15	38.69	0.11
105	9638	0.90	178.30	199.00	9637.12	38.85	2.79	38.85	0.20
106	9731	1.00	194.10	201.00	9730.11	38.67	1.27	38.67	0.30
107	9824	1.00	210.20	203.00	9823.10	38.07	-0.22	38.07	0.30
108	9918	0.90	157.50	201.00	9917.09	37.93	-1.61	37.93	0.90
109	10011	0.80	181.50	203.00	10010.08	38.20	-2.93	38.20	0.39
110	10071	1.10	207.10	203.00	10070.07	37.92	-3.87	37.92	0.85
111	10088	1.10	212.70	185.00	10087.06	37.76	-4.15	37.76	0.63
112	10119	1.50	139.20	185.00	10118.06	37.87	-4.71	37.87	5.12
113	10150	6.30	110.30	188.00	10148.98	39.73	-5.60	39.73	16.25
114	10181	11.10	106.30	188.00	10179.61	44.19	-7.03	44.19	15.60
115	10212	15.40	104.20	188.00	10209.78	51.05	-8.88	51.05	13.96
116	10243	20.40	103.10	192.00	10239.27	60.31	-11.12	60.31	16.17
117	10274	24.40	101.80	192.00	10267.93	71.84	-13.65	71.84	13.00
118	10305	27.20	102.40	195.00	10295.83	85.03	-16.48	85.03	9.07
119	10337	30.20	105.00	195.00	10323.90	99.95	-20.14	99.95	10.15
120	10368	33.00	107.10	195.00	10350.30	115.56	-24.64	115.56	9.70

Report #: **1**
Date: **24-Jun-14**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **7798**

SURVEY REPORT

Customer: **Oasis Petroleum N.A. LLC**
Well Name: **Chalmers 5301 44-24 3BR**
Rig #: **Nabors B-25**
API #: **33-053-06010**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **90**
Total Correction: **8.27**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
121	10399	36.70	107.10	195.00	10375.74	132.48	-29.85	132.48	11.94
122	10430	41.10	106.10	199.00	10399.86	151.14	-35.40	151.14	14.34
123	10461	45.80	105.00	199.00	10422.36	171.67	-41.10	171.67	15.36
124	10492	48.80	104.30	199.00	10443.38	193.71	-46.86	193.71	9.82
125	10523	51.20	104.50	203.00	10463.30	216.71	-52.77	216.71	7.76
126	10554	51.40	104.70	203.00	10482.69	240.12	-58.87	240.12	0.82
127	10585	52.10	105.10	203.00	10501.88	263.65	-65.13	263.65	2.47
128	10616	54.30	106.50	203.00	10520.45	287.53	-71.89	287.53	7.96
129	10647	57.20	106.50	203.00	10537.89	312.10	-79.17	312.10	9.35
130	10678	61.50	106.60	203.00	10553.69	337.65	-86.76	337.65	13.87
131	10710	67.20	106.90	203.00	10567.54	365.27	-95.07	365.27	17.83
132	10741	71.30	107.60	203.00	10578.52	392.94	-103.67	392.94	13.39
133	10772	72.30	107.90	203.00	10588.20	420.99	-112.65	420.99	3.35
134	10803	73.00	108.40	203.00	10597.45	449.11	-121.86	449.11	2.73
135	10834	74.70	108.60	203.00	10606.07	477.34	-131.31	477.34	5.52
136	10865	77.80	108.50	203.00	10613.43	505.89	-140.89	505.89	10.00
137	10880	80.30	108.50	203.00	10616.28	519.85	-145.56	519.85	16.67
138	10963	89.40	108.60	221.00	10623.73	598.14	-171.83	598.14	10.96
139	10994	89.00	108.10	213.00	10624.16	627.56	-181.59	627.56	2.07
140	11024	88.90	107.90	213.00	10624.71	656.09	-190.86	656.09	0.75
141	11055	87.30	109.20	215.00	10625.74	685.46	-200.72	685.46	6.65
142	11086	87.20	108.70	212.00	10627.22	714.75	-210.77	714.75	1.64
143	11117	86.90	108.10	212.00	10628.82	744.12	-220.55	744.12	2.16
144	11147	87.00	107.10	213.00	10630.42	772.68	-229.60	772.68	3.35
145	11178	87.70	106.80	210.00	10631.85	802.30	-238.63	802.30	2.46
146	11209	88.80	107.00	212.00	10632.80	831.94	-247.64	831.94	3.61
147	11240	90.60	106.80	213.00	10632.96	861.60	-256.65	861.60	5.84
148	11271	91.20	106.20	210.00	10632.47	891.32	-265.45	891.32	2.74
149	11302	91.10	106.30	210.00	10631.85	921.08	-274.13	921.08	0.46
150	11333	89.80	104.50	212.00	10631.61	950.96	-282.36	950.96	7.16
151	11364	89.00	104.40	210.00	10631.93	980.98	-290.09	980.98	2.60
152	11395	89.00	104.20	210.00	10632.47	1011.02	-297.75	1011.02	0.65
153	11426	88.20	102.20	213.00	10633.23	1041.19	-304.83	1041.19	6.95
154	11457	88.00	102.20	210.00	10634.26	1071.47	-311.37	1071.47	0.65
155	11487	88.20	101.20	212.00	10635.25	1100.83	-317.45	1100.83	3.40
156	11518	87.00	99.50	215.00	10636.55	1131.30	-323.02	1131.30	6.71
157	11549	87.00	99.40	213.00	10638.17	1161.83	-328.10	1161.83	0.32
158	11580	86.70	99.60	213.00	10639.88	1192.36	-333.21	1192.36	1.16
159	11611	86.80	97.80	215.00	10641.63	1222.96	-337.89	1222.96	5.81
160	11642	87.00	97.00	213.00	10643.31	1253.65	-341.88	1253.65	2.66
161	11672	86.90	97.00	215.00	10644.91	1283.39	-345.53	1283.39	0.33
162	11703	87.30	94.70	213.00	10646.47	1314.18	-348.68	1314.18	7.52
163	11735	87.60	94.70	213.00	10647.90	1346.04	-351.30	1346.04	0.94
164	11766	88.80	94.40	215.00	10648.87	1376.93	-353.76	1376.93	3.99
165	11797	89.60	92.40	217.00	10649.30	1407.87	-355.60	1407.87	6.95
166	11828	89.50	92.70	215.00	10649.55	1438.84	-356.98	1438.84	1.02
167	11859	89.90	92.40	217.00	10649.71	1469.81	-358.36	1469.81	1.61
168	11890	91.00	90.30	219.00	10649.47	1500.80	-359.09	1500.80	7.65
169	11952	91.40	89.80	221.00	10648.17	1562.78	-359.14	1562.78	1.03
170	12045	91.70	89.40	222.00	10645.65	1655.75	-358.49	1655.75	0.54
171	12137	90.80	88.50	222.00	10643.65	1747.71	-356.81	1747.71	1.38
172	12168	89.90	87.90	222.00	10643.46	1778.69	-355.83	1778.69	3.49
173	12231	88.70	87.40	226.00	10644.23	1841.63	-353.25	1841.63	2.06
174	12323	89.20	88.00	224.00	10645.91	1933.54	-349.56	1933.54	0.85
175	12416	88.50	85.90	228.00	10647.78	2026.38	-344.61	2026.38	2.38
176	12508	88.70	87.00	226.00	10650.03	2118.18	-338.92	2118.18	1.21
177	12601	88.20	86.90	226.00	10652.54	2211.01	-333.97	2211.01	0.55
178	12695	90.80	88.70	226.00	10653.36	2304.93	-330.36	2304.93	3.36
179	12789	91.50	90.50	228.00	10651.48	2398.90	-329.71	2398.90	2.05
180	12883	90.20	89.70	231.00	10650.08	2492.89	-329.87	2492.89	1.62

Report #: **1**
Date: **24-Jun-14**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **7798**

SURVEY REPORT

Customer: **Oasis Petroleum N.A. LLC**
Well Name: **Chalmers 5301 44-24 3BR**
Rig #: **Nabors B-25**
API #: **33-053-06010**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **90**
Total Correction: **8.27**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
181	12976	88.70	89.60	231.00	10650.97	2585.88	-329.30	2585.88	1.62
182	13070	88.50	89.30	235.00	10653.27	2679.85	-328.40	2679.85	0.38
183	13164	88.50	89.00	237.00	10655.73	2773.81	-327.01	2773.81	0.32
184	13258	90.80	90.40	235.00	10656.31	2867.79	-326.52	2867.79	2.86
185	13352	90.50	91.40	235.00	10655.24	2961.77	-327.99	2961.77	1.11
186	13445	90.50	91.50	239.00	10654.43	3054.74	-330.34	3054.74	0.11
187	13539	90.70	91.30	235.00	10653.44	3148.71	-332.64	3148.71	0.30
188	13633	88.20	91.20	239.00	10654.35	3242.67	-334.69	3242.67	2.66
189	13727	87.00	89.80	240.00	10658.28	3336.58	-335.51	3336.58	1.96
190	13821	89.20	90.90	242.00	10661.40	3430.52	-336.09	3430.52	2.62
191	13914	89.50	89.70	239.00	10662.45	3523.51	-336.57	3523.51	1.33
192	14008	88.90	90.40	242.00	10663.77	3617.50	-336.66	3617.50	0.98
193	14102	90.50	90.40	244.00	10664.26	3711.50	-337.31	3711.50	1.70
194	14196	88.30	90.80	242.00	10665.24	3805.48	-338.30	3805.48	2.38
195	14290	88.10	90.60	246.00	10668.19	3899.43	-339.44	3899.43	0.30
196	14384	90.10	91.60	246.00	10669.67	3993.39	-341.25	3993.39	2.38
197	14478	89.10	90.50	248.00	10670.33	4087.37	-342.97	4087.37	1.58
198	14572	88.40	89.70	249.00	10672.38	4181.35	-343.13	4181.35	1.13
199	14666	90.30	89.70	248.00	10673.44	4275.34	-342.64	4275.34	2.02
200	14760	90.10	89.10	249.00	10673.12	4369.33	-341.66	4369.33	0.67
201	14854	90.50	88.70	251.00	10672.62	4463.31	-339.85	4463.31	0.60
202	14947	88.90	86.80	253.00	10673.11	4556.23	-336.20	4556.23	2.67
203	15041	89.70	88.20	249.00	10674.26	4650.13	-332.10	4650.13	1.72
204	15136	90.80	88.90	253.00	10673.85	4745.10	-329.70	4745.10	1.37
205	15229	90.00	89.30	251.00	10673.20	4838.08	-328.24	4838.08	0.96
206	15323	91.30	89.90	253.00	10672.13	4932.07	-327.58	4932.07	1.52
207	15418	90.20	89.10	249.00	10670.89	5027.06	-326.75	5027.06	1.43
208	15512	87.70	90.30	251.00	10672.61	5121.03	-326.26	5121.03	2.95
209	15606	88.00	90.50	255.00	10676.13	5214.96	-326.92	5214.96	0.38
210	15700	86.50	89.20	255.00	10680.64	5308.85	-326.67	5308.85	2.11
211	15793	88.10	90.60	255.00	10685.03	5401.74	-326.51	5401.74	2.28
212	15887	88.40	91.20	255.00	10687.90	5495.69	-327.99	5495.69	0.71
213	15981	91.30	91.10	255.00	10688.14	5589.66	-329.87	5589.66	3.09
214	16075	92.40	91.40	255.00	10685.11	5683.58	-331.92	5683.58	1.21
215	16169	90.20	91.60	255.00	10682.98	5777.52	-334.38	5777.52	2.35
216	16263	88.70	91.30	255.00	10683.88	5871.48	-336.76	5871.48	1.63
217	16357	89.50	91.10	258.00	10685.35	5965.45	-338.73	5965.45	0.88
218	16450	89.90	91.50	257.00	10685.84	6058.43	-340.84	6058.43	0.61
219	16544	89.00	91.80	257.00	10686.74	6152.38	-343.54	6152.38	1.01
220	16638	89.80	92.30	255.00	10687.73	6246.31	-346.91	6246.31	1.00
221	16732	91.60	92.50	258.00	10686.58	6340.22	-350.84	6340.22	1.93
222	16825	90.00	91.60	257.00	10685.28	6433.15	-354.17	6433.15	1.97
223	16919	88.60	91.30	260.00	10686.43	6527.11	-356.55	6527.11	1.52
224	17013	89.90	91.20	260.00	10687.66	6621.08	-358.60	6621.08	1.39
225	17108	91.30	91.20	257.00	10686.67	6716.05	-360.59	6716.05	1.47
226	17139	89.00	90.30	258.00	10686.58	6747.04	-360.99	6747.04	7.97
227	17201	88.50	90.60	260.00	10687.94	6809.03	-361.48	6809.03	0.94
228	17295	87.20	90.10	258.00	10691.46	6902.96	-362.05	6902.96	1.48
229	17389	85.40	89.60	262.00	10697.53	6996.76	-361.81	6996.76	1.99
230	17483	86.50	89.80	260.00	10704.17	7090.52	-361.32	7090.52	1.19
231	17576	89.30	90.80	260.00	10707.58	7183.44	-361.80	7183.44	3.20
232	17670	89.30	90.00	262.00	10708.72	7277.43	-362.46	7277.43	0.85
233	17764	91.20	90.60	260.00	10708.31	7371.43	-362.95	7371.43	2.12
234	17857	91.60	90.10	264.00	10706.04	7464.40	-363.52	7464.40	0.69
235	17951	88.70	90.40	262.00	10705.80	7558.39	-363.93	7558.39	3.10
236	18045	88.70	90.40	264.00	10707.93	7652.36	-364.59	7652.36	0.00
237	18138	87.30	89.70	266.00	10711.17	7745.30	-364.67	7745.30	1.68
238	18231	88.70	88.60	262.00	10714.42	7838.23	-363.29	7838.23	1.91
239	18325	88.10	87.20	262.00	10717.04	7932.13	-359.85	7932.13	1.62
240	18418	88.40	86.90	262.00	10719.88	8024.96	-355.06	8024.96	0.46

Report #: **1**
 Date: **24-Jun-14**



Ryan Job # **7798**

SURVEY REPORT

Customer: **Oasis Petroleum N.A. LLC**
 Well Name: **Chalmers 5301 44-24 3BR**
 Rig #: **Nabors B-25**
 API #: **33-053-06010**
 Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**
 Directional Drillers: **RPM**
 Survey Corrected To: **True North**
 Vertical Section Direction: **90**
 Total Correction: **8.27**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
241	18450	89.10	87.40	264.00	10720.58	8056.91	-353.47	8056.91	2.69
242	18512	89.50	88.10	262.00	10721.34	8118.86	-351.04	8118.86	1.30
243	18606	90.90	88.20	266.00	10721.01	8212.81	-348.00	8212.81	1.49
244	18699	91.30	88.20	267.00	10719.23	8305.74	-345.08	8305.74	0.43
245	18793	90.80	88.10	267.00	10717.50	8399.68	-342.05	8399.68	0.54
246	18887	90.30	87.50	267.00	10716.60	8493.60	-338.44	8493.60	0.83
247	18982	89.80	87.40	269.00	10716.52	8588.51	-334.21	8588.51	0.54
248	19076	89.50	88.20	269.00	10717.09	8682.44	-330.61	8682.44	0.91
249	19171	89.80	90.80	267.00	10717.67	8777.42	-329.78	8777.42	2.75
250	19266	89.60	90.30	267.00	10718.17	8872.42	-330.69	8872.42	0.57
251	19361	90.20	90.50	266.00	10718.34	8967.42	-331.35	8967.42	0.67
252	19455	90.10	90.60	266.00	10718.09	9061.41	-332.25	9061.41	0.15
253	19549	90.10	89.90	267.00	10717.93	9155.41	-332.66	9155.41	0.74
254	19644	88.10	89.00	269.00	10719.42	9250.39	-331.75	9250.39	2.31
255	19739	88.10	89.60	269.00	10722.57	9345.33	-330.59	9345.33	0.63
256	19833	88.30	89.90	267.00	10725.52	9439.28	-330.18	9439.28	0.38
257	19927	89.50	89.70	266.00	10727.33	9533.26	-329.85	9533.26	1.29
258	20020	90.70	89.40	267.00	10727.16	9626.26	-329.12	9626.26	1.33
259	20114	89.80	89.10	269.00	10726.75	9720.25	-327.89	9720.25	1.01
260	20208	91.70	89.20	266.00	10725.52	9814.22	-326.50	9814.22	2.02
261	20302	91.40	88.70	269.00	10722.98	9908.17	-324.78	9908.17	0.62
262	20396	91.10	89.00	267.00	10720.93	10002.13	-322.89	10002.13	0.45
263	20489	90.50	88.50	269.00	10719.63	10095.10	-320.86	10095.10	0.84
264	20584	90.90	88.60	269.00	10718.47	10190.06	-318.46	10190.06	0.43
265	20678	90.70	87.80	262.00	10717.16	10284.01	-315.51	10284.01	0.88
266	20772	88.30	86.70	262.00	10717.98	10377.88	-311.00	10377.88	2.81
Projection	20833	88.30	86.70	262.00	10719.79	10438.76	-307.49	10438.76	0.00



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28599

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: 6/17/2014

RE: CORES AND SAMPLES

Well Name: **CHALMERS 5301 44-24 3BR** Well File No.: **28599**
Location: **SESE 24-153-101** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **BAKKEN**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

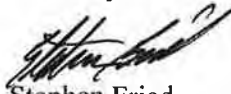
- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
 - Sample cuttings shall be collected at:
 - o 30' maximum intervals through all vertical and build sections.
 - o 100' maximum intervals through any horizontal sections.
 - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
 - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely


Stephen Fried
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28599

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date June 1, 2014	☒ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other Waiver to rule Rule 43-02-03-31	

Well Name and Number Chalmers 5301 44-24 3BR					
Footages 926 F S L 245 F E L		Qtr-Qtr SESE	Section 24	Township 153 N	Range 101 W
Field		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address

City

State

Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

Oasis Petroleum/ Chalmers 5300 31-19H (NDIC 20407) located within one mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD GR.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Heather McCowan</i>		Printed Name Heather McCowan	
Title Regulatory Assistant		Date May 28, 2014	
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 6-11-2014	
By <i>Stephen Fried</i>	
Title Stephen Fried Geologist	



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
28599

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date June 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Suspension of Drilling

Well Name and Number Chalmers 5301 44-24 3BR							
Footages		Qtr-Qtr	Section	Township	Range		
926 F S L 245 F E L		SESE	24	153 N	101 W		
Field		Pool		County			
		Bakken		McKENZIE			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services			
Address	City	State	Zip Code

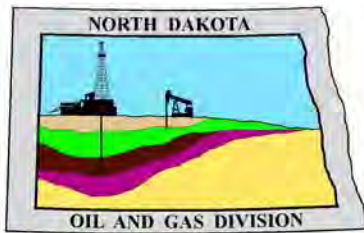
DETAILS OF WORK

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Notify NDIC inspector Richard Dunn at 701-770-3554 with spud and TD info.

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Heather McCowan</i>	Printed Name Heather McCowan		
Title Regulatory Assistant	Date May 28, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date	6/11/14
By	Nathaniel Erbele
Title	Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

June 11, 2014

Heather McCowan
Regulatory Assistant
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

**RE: HORIZONTAL WELL
CHALMERS 5301 44-24 3BR
SESE Section 24-153N-101W
McKenzie County
Well File # 28599**

Dear Heather:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500' setback** from the north & south boundaries and **200' setback** from the east & west boundaries within the 1280 acre spacing unit consisting of Sections 19 & 20 T153N R100W.

PERMIT STIPULATIONS: DUE TO STREAM ADJACENT TO THE WELL SITE, A DIKE IS REQUIRED SURROUNDING THE ENTIRE LOCATION. Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. In cases where a spacing unit is accessed from an off-site drill pad, an affidavit must be provided affirming that the surface owner of the multi-well pad agrees to accept burial on their property of the cuttings generated from drilling the well(s) into an offsite spacing/drilling unit. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to setting conductor.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 445' east. Also, based on the azimuth of the proposed lateral the maximum legal coordinates from the well head are: 426' south and 10558' east.

Heather McCowan

June 11, 2014

Page 2

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Nathaniel Erbele

Petroleum Resource Specialist



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 6 / 1 / 2013	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9563
Address 1001 Fannin Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name CHALMERS				Well Number 5301 44-24 3BR			
Surface Footages 926 F S L 245 F E L		Qtr-Qtr SESE	Section 24	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Footages 760 F S L 250 F W L		Qtr-Qtr SWSW	Section 19	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 166 S From WH 495 E From WH		Azimuth 110.4 °	Longstring Total Depth 10899 Feet MD 10629 Feet TVD				
Bottom Hole Footages From Nearest Section Line 580 F S L 224 F E L		Qtr-Qtr SESE	Section 20	Township 153 N	Range 101 W	County Williams	
Bottom Hole Coordinates From Well Head 346 S From WH 10534 E From WH		KOP Lateral 1 10151 Feet MD	Azimuth Lateral 1 90 °		Estimated Total Depth Lateral 1 20960 Feet MD 10699 Feet TVD		
Latitude of Well Head 48 ° 03 ' 20.5 "	Longitude of Well Head -103 ° 36 ' 18.55 "	NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Sections 19 & 20 T153N R100W				
Ground Elevation 1970 Feet Above S.L.	Acres in Spacing/Drilling Unit 1280	Spacing/Drilling Unit Setback Requirement 500 Feet N/S 200 Feet E/W			Industrial Commission Order 23752		
North Line of Spacing/Drilling Unit 10489 Feet	South Line of Spacing/Drilling Unit 10513 Feet	East Line of Spacing/Drilling Unit 5280 Feet			West Line of Spacing/Drilling Unit 5263 Feet		
Objective Horizons Bakken						Pierre Shale Top 1918	
Proposed Surface Casing	Size 13 - 3/8 "	Weight 54 Lb./Ft.	Depth 2018 Feet	Cement Volume 967 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 32 Lb./Ft.	Longstring Total Depth 10899 Feet MD 10629 Feet TVD		Cement Volume 821 Sacks	Cement Top 3841 Feet	Top Dakota Sand 5341 Feet
Base Last Charles Salt (If Applicable) 9091 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD	Feet TVD	KOP Coordinates From Well Head From WH	From WH
Formation Entry Point Coordinates From Well Head From WH	From WH	Bottom Hole Coordinates From Well Head From WH	From WH		
KOP Footages From Nearest Section Line F L F L	Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L F L	Qtr-Qtr	Section	Township N	Range W	County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD	Feet TVD	KOP Coordinates From Well Head From WH	From WH
Formation Entry Point Coordinates From Well Head From WH	From WH	Bottom Hole Coordinates From Well Head From WH	From WH		
KOP Footages From Nearest Section Line F L F L	Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L F L	Qtr-Qtr	Section	Township N	Range W	County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD	Feet TVD	KOP Coordinates From Well Head From WH	From WH
Formation Entry Point Coordinates From Well Head From WH	From WH	Bottom Hole Coordinates From Well Head From WH	From WH		
KOP Footages From Nearest Section Line F L F L	Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L F L	Qtr-Qtr	Section	Township N	Range W	County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD	Feet TVD	KOP Coordinates From Well Head From WH	From WH
Formation Entry Point Coordinates From Well Head From WH	From WH	Bottom Hole Coordinates From Well Head From WH	From WH		
KOP Footages From Nearest Section Line F L F L	Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L F L	Qtr-Qtr	Section	Township N	Range W	County

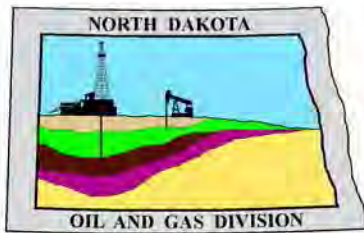
I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 5 / 28 / 2014
ePermit	Printed Name Heather McCowan	Title Regulatory Assistant

FOR STATE USE ONLY

Permit and File Number 28599	API Number 33 - 053 - 06010
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 6 / 11 / 2014
By Nathaniel Erbele
Title Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

**RE: Filter Socks and Other Filter Media
Leakproof Container Required
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

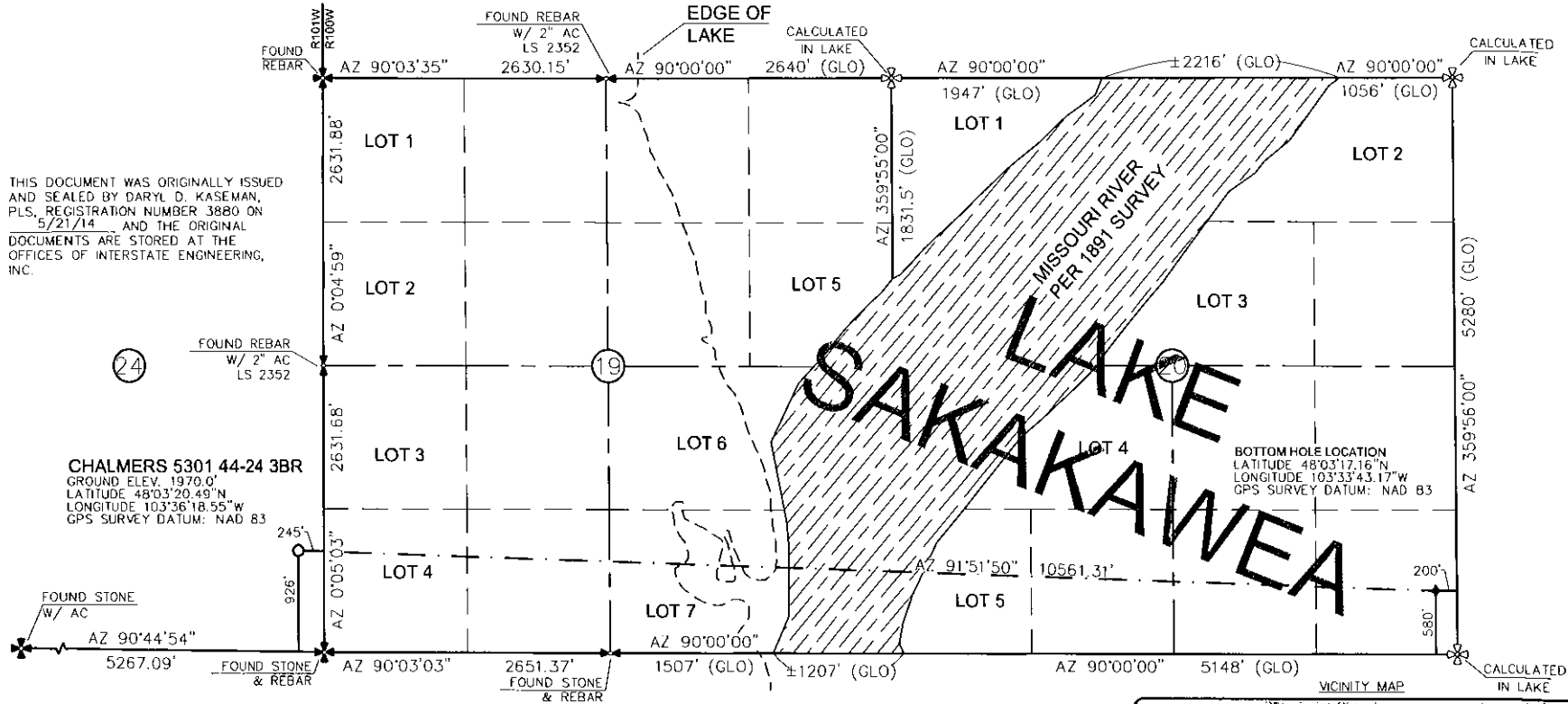
Thank you for your cooperation.

Sincerely,

Bruce E. Hicks

Assistant Director

WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5301 44-24 3BR"
 926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
 SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

Revision	Date	By	Description
REV 1	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 2	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 3	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 4	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 5	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 6	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 7	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 8	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 9	5/21/14	JK	CHALMERS 5301 44-24 3BR
REV 10	5/21/14	JK	CHALMERS 5301 44-24 3BR

OASIS PETROLEUM NORTH AMERICA, LLC WELL LOCATION PLAT SECTION 24, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA	Drawn By: JK Checked By: DDK Date: 5/21/14
---	--

Interstate Engineering, Inc.
 P.O. Box 848
 425 East Main Street
 Suite 200
 Fargo, ND 58103
 Ph: (701) 433-5617
 Fax: (701) 433-5618
 www.interstate-engineering.com
 One of the nation's leading civil engineering firms

INTERSTATE ENGINEERING

1/8
 SHEET NO.

DRILLING PLAN				COUNTY/STATE		McKenzie Co., ND	
OPERATOR		Oasis Petroleum		RIG		Nabors B25	
WELL NAME		Chalmers 5301 44-24 38R					
WELL TYPE		Horizontal Middle Bakken					
LOCATION		SESE 24-153N-101W		Surface Location (survey plat): 926' FSL		245' FEL	
EST. T.D.		20,960'		GROUND ELEV:		1943 Finished Pad Elev.	
TOTAL LATERA		10,061'		KB ELEV:		1968	
PROGNOSIS:		Based on 1,968' KB(est)		LOGS:			
MARKER		DEPTH (Surf Loc)		DATUM (Surf Loc)		Type Interval	
Pierre		NDIC MAP	1,918	50		OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota	
Greenhorn			4,510	(2,542)		CBL/GR: Above top of cement/GR to base of casing	
Mowry			4,914	(2,946)		MWD GR: KOP to lateral TD	
Dakota			5,341	(3,373)		Logging Waiver Oasis Chalmers 5300 31-19H 894' to NE sec 19 153N 100W	
Rierdon			6,356	(4,388)		DEVIATION:	
Dunham Salt			6,683	(4,715)		Surf: 3 deg. max., 1 deg / 100'; srvy every 500'	
Dunham Salt Base			6,800	(4,832)		Prod: 5 deg. max., 1 deg / 100'; srvy every 100'	
Spearfish			6,887	(4,919)		DST'S:	
Pine Salt			7,135	(5,167)		None planned	
Pine Salt Base			7,168	(5,200)		CORES:	
Opeche Salt			7,235	(5,267)		None planned	
Opeche Salt Base			7,269	(5,301)		MUDLOGGING:	
Broom Creek (Top of Minnelusa Gp.)			7,469	(5,501)		Two-Man: 8,233'	
Amsden			7,553	(5,585)		~200' above the Charles (Kibbey) to Casing point; Casing point to TD	
Tyler			7,720	(5,752)		30' samples at direction of wellsite geologist; 10' through target @ curve land	
Otter (Base of Minnelusa Gp.)			7,920	(5,952)		BOP:	
Kibbey Lime			8,269	(6,301)		11" 5000 psi blind, pipe & annular	
Charles Salt			8,433	(6,465)			
UB			9,010	(7,042)			
Base Last Salt			9,091	(7,123)			
Ratcliffe			9,141	(7,173)			
Mission Canyon			9,311	(7,343)			
Lodgepole			9,876	(7,908)			
Lodgepole Fracture Zone			10,109	(8,141)			
False Bakken			10,588	(8,620)			
Upper Bakken			10,599	(8,631)			
Middle Bakken			10,616	(8,648)			
Middle Bakken Sand Target			10,625	(8,657)			
Base Middle Bakken Sand Target			10,634	(8,666)			
Lower Bakken			10,648	(8,680)			
Three Forks			10,676	(8,708)			
Dip Rate:		-0.40° or .7' /100' down					
Max. Anticipated BHP:		4614		Surface Formation: Glacial till			
MUD:		Interval		Type		WT Vis WL Remarks	
Surface:		0' -		2,018' FW/Gel - Lime Sweeps		8.4-9.0 28-32 NC Circ Mud Tanks	
Intermediate:		2,018' -		10,899' Invert		9.5-10.4 40-50 30+HTHp Circ Mud Tanks	
Lateral:		10,899' -		20,960' Salt Water		9.8-10.2 28-32 NC Circ Mud Tanks	
CASING:		Size		Wt ppf Hole Depth		Cement WOC Remarks	
Surface:		13-3/8"		54.5# 17-1/2" 2,018'		Surface 12 100' into Pierre	
Intermediate (Dakota):		9-5/8"		40# 12-1/4" 6,000'		Surface 24 Set Casing across Dakota	
Intermediate:		7"		32# 8-3/4" 10,899'		3,841' 24 1500' above Dakota	
Production Liner:		4.5"		13.5# 6" 20,960'		TOL @ 10,101' 24 50' above KOP	
PROBABLE PLUGS, IF REQ'D:							
OTHER:		MD		TVD		FNL/FSL FEL/FWL S-T-R AZI	
Surface:		2,018		2,018		926' FSL 245' FEL Sec. 24 153N 101W	
KOP:		10,151'		10,151'		926' FSL 195' FEL Sec. 24 153N 101W	
EOC:		10,899'		10,629'		761' FSL 249' FWL Sec. 19 153N 100W	
Casing Point:		10,899'		10,630'		761' FSL 249' FWL Sec. 19 153N 100W	
Middle Bakken Lateral TD:		20,960'		10,700'		580' FSL 200' FEL Sec. 20 153N 100W	
Comments:							
Request a Sundry for an Open Hole Log Waiver							
Exception well: Oasis Petroleum's J O Anderson 5200 31-28T (33053039510000); 152N/100W/S28; located 0.30 miles S of the proposed location							
No frac string planned							
35 packers and 2 sleeves							
**Note Hardlines							
Pit location as depicted on plats. Constructed on native ground, not fill							
OASIS PETROLEUM							
Geology: C. Hargett 9/30/2013		Engineering: M. Brown 5-28-14					

**Oasis Petroleum
Well Summary
Chalmers 5301 44-24 3BR
Sec. 24 T153N R101W
McKenzie County, North Dakota**

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
13-3/8"	0' to 2,018'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,018'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 2.13	2730 / 2.89	514 / 2.63	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2018' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2018' setting depth).
- c) Based on string weight in 9 ppg fluid at 2018' TVD plus 100k# overpull. (Buoyed weight equals 95k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **593 sks** (307 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCL₂, 4% D079 Extender, and 2% D053 Expanding Agent.

Tail Slurry: **374 sks** (77 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25 lb/sk Lost Circulation Control Agent, and .25% CaCL₂.

**Oasis Petroleum
Well Summary
Chalmers 5301 44-24 3BR
Sec. 24 T153N R101W
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' - 6000'	40	HCL-80	LTC	8.835"	8.75"***	5,450	7,270	9,090

**Special Drift

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 6000'	9-5/8", 40#, HCL-80, LTC, 8rd	3090 / 3.96*	5750 / 1.23	837 / 2.75

API Rating & Safety Factor

- a) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- b) Collapse pressure based on 11.5ppg fluid on backside and 9ppg fluid inside of casing.
- c) Yield based on string weight in 10 ppg fluid, (217k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 9-5/8" casing set in an 12-1/4" hole with **10%** excess in OH and **0%** excess inside surface casing. TOC at surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **540 sks** (280 bbls) Conventional system with 75 lb/sk cement, 0.5lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl₂, 0.2% anti foam, and 0.4% fluid loss

Tail Slurry: **373 sks** (77 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti foam, and 0.1% retarder

**Oasis Petroleum
Well Summary
Chalmers 5301 44-24 3BR
Sec. 24 T153N R101W
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' - 10899'	32	HCP-110	LTC	6.094"	6.000"	6,730	8,970	11210

**Special Drift

Interval	Description	Collapse (psi) a	Burst (psi) b	Tension (1000 lbs) c
0' - 10899'	7", 32#, HCP-110, LTC, 8rd	11890 / 2.15*	12450 / 1.28	897 / 2.26
6600' - 9091'	7", 32#, HCP-110, LTC, 8rd	11890 / 1.85**	12450 / 1.30	

API Rating & Safety Factor

- a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,630' TVD.
- c. Based on string weight in 10 ppg fluid, (295k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls** Saltwater
 40 bbls Weighted MudPush Express

Lead Slurry: **207 sks** (82 bbls) 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 Extender, 3.0% KCl, 3.0% D154 Extender, 0.3% D208 Viscosifier, 0.07% Retarder, 0.2% Anti Foam, 0.5lb/sk D130 LCM

Tail Slurry: **614 sks** (168 bbls) 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% Retarder, 0.2% Fluid Loss, 0.2% Anti Foam, 0.5 lb/sk LCM

**Oasis Petroleum
Well Summary
Chalmers 5301 44-24 3BR
Sec. 24 T153N R101W
McKenzie County, North Dakota**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Torque
4-1/2"	10101' - 20960'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
10101' - 20960'	10,859'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10680 / 2.02	12410 / 1.28	422 / 1.87

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10700' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10700' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 125k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)

Oasis

Indian Hills

153N-100W-19/20_Altered

Chalmers 5301 44-24 3BR

Chalmers 5301 44-24 3BR

Plan: Design #2

Standard Planning Report

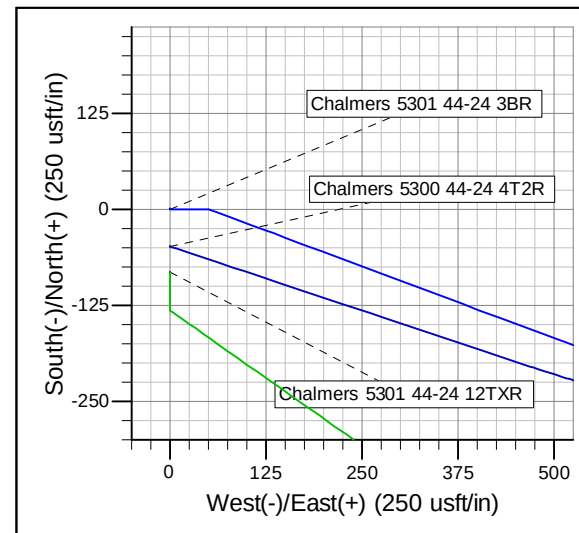
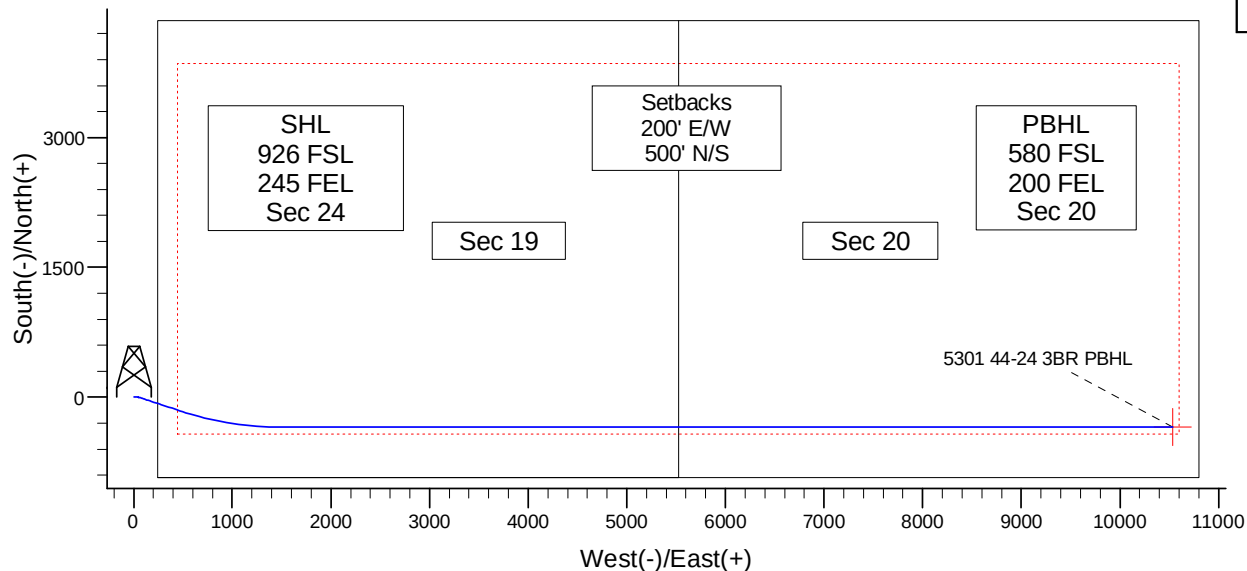
27 May, 2014

Project: Indian Hills
 Site: 153N-100W-19/20_Altered
 Well: Chalmers 5301 44-24 3BR
 Wellbore: Chalmers 5301 44-24 3BR
 Design: Design #2



WELL DETAILS: Chalmers 5301 44-24 3BR

Ground Level: 1942.0
 Northing 400390.13 Easting 1209328.09 Latitude 48° 3' 20.490 N Longitude 103° 36' 18.550 W



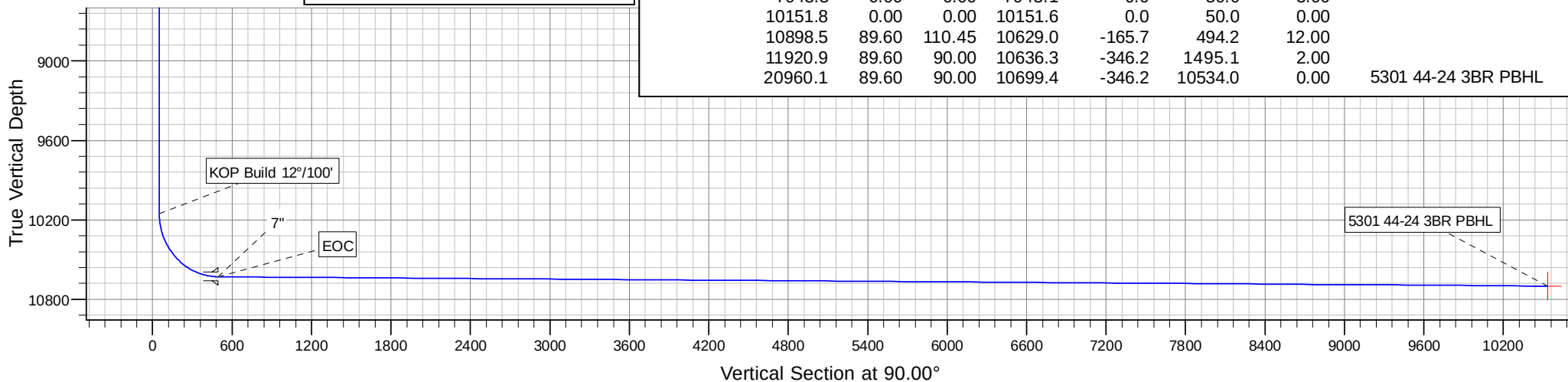
CASING DETAILS			
TVD	MD	Name	Size
10629.0	10899.0	7"	7"



Azimuths to True North
 Magnetic North: 8.27°
 Magnetic Field
 Strength: 56433.4snT
 Dip Angle: 72.98°
 Date: 5/22/2014
 Model: IGRF2010

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2200.0	0.00	0.00	2200.0	0.0	0.0	0.00	
2216.7	0.50	90.00	2216.7	0.0	0.1	3.00	
7926.7	0.50	90.00	7926.4	0.0	49.9	0.00	
7943.3	0.00	0.00	7943.1	0.0	50.0	3.00	
10151.8	0.00	0.00	10151.6	0.0	50.0	0.00	
10898.5	89.60	110.45	10629.0	-165.7	494.2	12.00	
11920.9	89.60	90.00	10636.3	-346.2	1495.1	2.00	
20960.1	89.60	90.00	10699.4	-346.2	10534.0	0.00	5301 44-24 3BR PBHL



Ryan Directional Services

Planning Report

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Company:	Oasis	TVD Reference:	Well @ 1967.0usft
Project:	Indian Hills	MD Reference:	Well @ 1967.0usft
Site:	153N-100W-19/20_Altered	North Reference:	True
Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Wellbore:	Chalmers 5301 44-24 3BR		
Design:	Design #2		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site		153N-100W-19/20_Altered			
Site Position:		Northing:	400,357.73 usft	Latitude:	48° 3' 20.170 N
From:	Lat/Long	Easting:	1,209,326.78 usft	Longitude:	103° 36' 18.550 W
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "	Grid Convergence:	-2.31 °

Well		Chalmers 5301 44-24 3BR				
Well Position	+N/-S	32.4 usft	Northing:	400,390.13 usft	Latitude:	48° 3' 20.490 N
	+E/-W	0.0 usft	Easting:	1,209,328.09 usft	Longitude:	103° 36' 18.550 W
Position Uncertainty		0.0 usft	Wellhead Elevation:	0.0 usft	Ground Level:	1,942.0 usft

Wellbore	Chalmers 5301 44-24 3BR				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	5/22/2014	8.27	72.98	56,433

Design	Design #2			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	90.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,216.7	0.50	90.00	2,216.7	0.0	0.1	3.00	3.00	0.00	90.00	
7,926.7	0.50	90.00	7,926.4	0.0	49.9	0.00	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	0.0	50.0	3.00	-3.00	0.00	180.00	
10,151.8	0.00	0.00	10,151.6	0.0	50.0	0.00	0.00	0.00	0.00	
10,898.5	89.60	110.45	10,629.0	-165.7	494.2	12.00	12.00	0.00	110.45	
11,920.9	89.60	90.00	10,636.3	-346.2	1,495.1	2.00	0.00	-2.00	-90.07	
20,960.1	89.60	90.00	10,699.4	-346.2	10,534.0	0.00	0.00	0.00	0.00	5301 44-24 3BR PBH

Ryan Directional Services

Planning Report

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Company:	Oasis	TVD Reference:	Well @ 1967.0usft
Project:	Indian Hills	MD Reference:	Well @ 1967.0usft
Site:	153N-100W-19/20_Altered	North Reference:	True
Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Wellbore:	Chalmers 5301 44-24 3BR		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,216.7	0.50	90.00	2,216.7	0.0	0.1	0.1	3.00	3.00	0.00
2,300.0	0.50	90.00	2,300.0	0.0	0.8	0.8	0.00	0.00	0.00
2,400.0	0.50	90.00	2,400.0	0.0	1.7	1.7	0.00	0.00	0.00
2,500.0	0.50	90.00	2,500.0	0.0	2.5	2.5	0.00	0.00	0.00
2,600.0	0.50	90.00	2,600.0	0.0	3.4	3.4	0.00	0.00	0.00
2,700.0	0.50	90.00	2,700.0	0.0	4.3	4.3	0.00	0.00	0.00
2,800.0	0.50	90.00	2,800.0	0.0	5.2	5.2	0.00	0.00	0.00
2,900.0	0.50	90.00	2,900.0	0.0	6.0	6.0	0.00	0.00	0.00
3,000.0	0.50	90.00	3,000.0	0.0	6.9	6.9	0.00	0.00	0.00
3,100.0	0.50	90.00	3,100.0	0.0	7.8	7.8	0.00	0.00	0.00
3,200.0	0.50	90.00	3,200.0	0.0	8.7	8.7	0.00	0.00	0.00
3,300.0	0.50	90.00	3,300.0	0.0	9.5	9.5	0.00	0.00	0.00
3,400.0	0.50	90.00	3,400.0	0.0	10.4	10.4	0.00	0.00	0.00
3,500.0	0.50	90.00	3,500.0	0.0	11.3	11.3	0.00	0.00	0.00
3,600.0	0.50	90.00	3,599.9	0.0	12.1	12.1	0.00	0.00	0.00
3,700.0	0.50	90.00	3,699.9	0.0	13.0	13.0	0.00	0.00	0.00
3,800.0	0.50	90.00	3,799.9	0.0	13.9	13.9	0.00	0.00	0.00
3,900.0	0.50	90.00	3,899.9	0.0	14.8	14.8	0.00	0.00	0.00
4,000.0	0.50	90.00	3,999.9	0.0	15.6	15.6	0.00	0.00	0.00
4,100.0	0.50	90.00	4,099.9	0.0	16.5	16.5	0.00	0.00	0.00
4,200.0	0.50	90.00	4,199.9	0.0	17.4	17.4	0.00	0.00	0.00
4,300.0	0.50	90.00	4,299.9	0.0	18.3	18.3	0.00	0.00	0.00
4,400.0	0.50	90.00	4,399.9	0.0	19.1	19.1	0.00	0.00	0.00
4,500.0	0.50	90.00	4,499.9	0.0	20.0	20.0	0.00	0.00	0.00
4,510.1	0.50	90.00	4,510.0	0.0	20.1	20.1	0.00	0.00	0.00
Greenhorn									
4,600.0	0.50	90.00	4,599.9	0.0	20.9	20.9	0.00	0.00	0.00
4,700.0	0.50	90.00	4,699.9	0.0	21.7	21.7	0.00	0.00	0.00
4,800.0	0.50	90.00	4,799.9	0.0	22.6	22.6	0.00	0.00	0.00
4,900.0	0.50	90.00	4,899.9	0.0	23.5	23.5	0.00	0.00	0.00
4,914.1	0.50	90.00	4,914.0	0.0	23.6	23.6	0.00	0.00	0.00
Mowry									
5,000.0	0.50	90.00	4,999.9	0.0	24.4	24.4	0.00	0.00	0.00
5,100.0	0.50	90.00	5,099.9	0.0	25.2	25.2	0.00	0.00	0.00
5,200.0	0.50	90.00	5,199.9	0.0	26.1	26.1	0.00	0.00	0.00
5,300.0	0.50	90.00	5,299.9	0.0	27.0	27.0	0.00	0.00	0.00
5,341.1	0.50	90.00	5,341.0	0.0	27.3	27.3	0.00	0.00	0.00
Dakota									
5,400.0	0.50	90.00	5,399.9	0.0	27.9	27.9	0.00	0.00	0.00
5,500.0	0.50	90.00	5,499.9	0.0	28.7	28.7	0.00	0.00	0.00
5,600.0	0.50	90.00	5,599.9	0.0	29.6	29.6	0.00	0.00	0.00
5,700.0	0.50	90.00	5,699.9	0.0	30.5	30.5	0.00	0.00	0.00
5,800.0	0.50	90.00	5,799.9	0.0	31.3	31.3	0.00	0.00	0.00
5,900.0	0.50	90.00	5,899.9	0.0	32.2	32.2	0.00	0.00	0.00
6,000.0	0.50	90.00	5,999.9	0.0	33.1	33.1	0.00	0.00	0.00
6,100.0	0.50	90.00	6,099.9	0.0	34.0	34.0	0.00	0.00	0.00
6,200.0	0.50	90.00	6,199.8	0.0	34.8	34.8	0.00	0.00	0.00
6,300.0	0.50	90.00	6,299.8	0.0	35.7	35.7	0.00	0.00	0.00
6,356.2	0.50	90.00	6,356.0	0.0	36.2	36.2	0.00	0.00	0.00
Rierdon									
6,400.0	0.50	90.00	6,399.8	0.0	36.6	36.6	0.00	0.00	0.00
6,500.0	0.50	90.00	6,499.8	0.0	37.5	37.5	0.00	0.00	0.00

Ryan Directional Services

Planning Report

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Company:	Oasis	TVD Reference:	Well @ 1967.0usft
Project:	Indian Hills	MD Reference:	Well @ 1967.0usft
Site:	153N-100W-19/20_Altered	North Reference:	True
Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Wellbore:	Chalmers 5301 44-24 3BR		
Design:	Design #2		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
6,600.0	0.50	90.00	6,599.8	0.0	38.3	38.3	0.00	0.00	0.00	
6,683.2	0.50	90.00	6,683.0	0.0	39.0	39.0	0.00	0.00	0.00	
Dunham Salt										
6,700.0	0.50	90.00	6,699.8	0.0	39.2	39.2	0.00	0.00	0.00	
6,800.0	0.50	90.00	6,799.8	0.0	40.1	40.1	0.00	0.00	0.00	
6,800.2	0.50	90.00	6,800.0	0.0	40.1	40.1	0.00	0.00	0.00	
Dunham Salt Base										
6,887.2	0.50	90.00	6,887.0	0.0	40.8	40.8	0.00	0.00	0.00	
Spearfish										
6,900.0	0.50	90.00	6,899.8	0.0	40.9	40.9	0.00	0.00	0.00	
7,000.0	0.50	90.00	6,999.8	0.0	41.8	41.8	0.00	0.00	0.00	
7,100.0	0.50	90.00	7,099.8	0.0	42.7	42.7	0.00	0.00	0.00	
7,135.2	0.50	90.00	7,135.0	0.0	43.0	43.0	0.00	0.00	0.00	
Pine Salt										
7,168.2	0.50	90.00	7,168.0	0.0	43.3	43.3	0.00	0.00	0.00	
Pine Salt Base										
7,200.0	0.50	90.00	7,199.8	0.0	43.6	43.6	0.00	0.00	0.00	
7,235.2	0.50	90.00	7,235.0	0.0	43.9	43.9	0.00	0.00	0.00	
Opeche Salt										
7,269.2	0.50	90.00	7,269.0	0.0	44.2	44.2	0.00	0.00	0.00	
Opeche Salt Base										
7,300.0	0.50	90.00	7,299.8	0.0	44.4	44.4	0.00	0.00	0.00	
7,400.0	0.50	90.00	7,399.8	0.0	45.3	45.3	0.00	0.00	0.00	
7,469.2	0.50	90.00	7,469.0	0.0	45.9	45.9	0.00	0.00	0.00	
Broom Creek (Top of Minnelusa Gp.)										
7,500.0	0.50	90.00	7,499.8	0.0	46.2	46.2	0.00	0.00	0.00	
7,553.2	0.50	90.00	7,553.0	0.0	46.6	46.6	0.00	0.00	0.00	
Amsden										
7,600.0	0.50	90.00	7,599.8	0.0	47.1	47.1	0.00	0.00	0.00	
7,700.0	0.50	90.00	7,699.8	0.0	47.9	47.9	0.00	0.00	0.00	
7,720.2	0.50	90.00	7,720.0	0.0	48.1	48.1	0.00	0.00	0.00	
Tyler										
7,800.0	0.50	90.00	7,799.8	0.0	48.8	48.8	0.00	0.00	0.00	
7,900.0	0.50	90.00	7,899.8	0.0	49.7	49.7	0.00	0.00	0.00	
7,920.2	0.50	90.00	7,920.0	0.0	49.8	49.8	0.00	0.00	0.00	
Otter (Base of Minnelusa Gp.)										
7,926.7	0.50	90.00	7,926.4	0.0	49.9	49.9	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	0.0	50.0	50.0	3.00	-3.00	0.00	
8,000.0	0.00	0.00	7,999.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,100.0	0.00	0.00	8,099.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,200.0	0.00	0.00	8,199.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,269.2	0.00	0.00	8,269.0	0.0	50.0	50.0	0.00	0.00	0.00	
Kibbey Lime										
8,300.0	0.00	0.00	8,299.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,400.0	0.00	0.00	8,399.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,433.2	0.00	0.00	8,433.0	0.0	50.0	50.0	0.00	0.00	0.00	
Charles Salt										
8,500.0	0.00	0.00	8,499.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,600.0	0.00	0.00	8,599.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,700.0	0.00	0.00	8,699.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,800.0	0.00	0.00	8,799.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,900.0	0.00	0.00	8,899.8	0.0	50.0	50.0	0.00	0.00	0.00	

Ryan Directional Services

Planning Report

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Company:	Oasis	TVD Reference:	Well @ 1967.0usft
Project:	Indian Hills	MD Reference:	Well @ 1967.0usft
Site:	153N-100W-19/20_Altered	North Reference:	True
Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Wellbore:	Chalmers 5301 44-24 3BR		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
9,000.0	0.00	0.00	8,999.8	0.0	50.0	50.0	0.00	0.00	0.00
9,010.2	0.00	0.00	9,010.0	0.0	50.0	50.0	0.00	0.00	0.00
UB									
9,091.2	0.00	0.00	9,091.0	0.0	50.0	50.0	0.00	0.00	0.00
Base Last Salt									
9,100.0	0.00	0.00	9,099.8	0.0	50.0	50.0	0.00	0.00	0.00
9,141.2	0.00	0.00	9,141.0	0.0	50.0	50.0	0.00	0.00	0.00
Ratcliffe									
9,200.0	0.00	0.00	9,199.8	0.0	50.0	50.0	0.00	0.00	0.00
9,300.0	0.00	0.00	9,299.8	0.0	50.0	50.0	0.00	0.00	0.00
9,311.2	0.00	0.00	9,311.0	0.0	50.0	50.0	0.00	0.00	0.00
Mission Canyon									
9,400.0	0.00	0.00	9,399.8	0.0	50.0	50.0	0.00	0.00	0.00
9,500.0	0.00	0.00	9,499.8	0.0	50.0	50.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	0.0	50.0	50.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	0.0	50.0	50.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	0.0	50.0	50.0	0.00	0.00	0.00
9,876.2	0.00	0.00	9,876.0	0.0	50.0	50.0	0.00	0.00	0.00
Lodgepole									
9,900.0	0.00	0.00	9,899.8	0.0	50.0	50.0	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	0.0	50.0	50.0	0.00	0.00	0.00
10,100.0	0.00	0.00	10,099.8	0.0	50.0	50.0	0.00	0.00	0.00
10,109.2	0.00	0.00	10,109.0	0.0	50.0	50.0	0.00	0.00	0.00
Lodgepole Fracture Zone									
10,151.8	0.00	0.00	10,151.6	0.0	50.0	50.0	0.00	0.00	0.00
KOP Build 12°/100'									
10,175.0	2.78	110.45	10,174.8	-0.2	50.5	50.5	12.00	12.00	0.00
10,200.0	5.78	110.45	10,199.7	-0.8	52.3	52.3	12.00	12.00	0.00
10,225.0	8.78	110.45	10,224.5	-2.0	55.2	55.2	12.00	12.00	0.00
10,250.0	11.78	110.45	10,249.1	-3.5	59.4	59.4	12.00	12.00	0.00
10,275.0	14.78	110.45	10,273.4	-5.5	64.8	64.8	12.00	12.00	0.00
10,300.0	17.78	110.45	10,297.4	-8.0	71.4	71.4	12.00	12.00	0.00
10,325.0	20.78	110.45	10,321.0	-10.9	79.1	79.1	12.00	12.00	0.00
10,350.0	23.78	110.45	10,344.1	-14.2	88.0	88.0	12.00	12.00	0.00
10,375.0	26.78	110.45	10,366.7	-17.9	98.0	98.0	12.00	12.00	0.00
10,400.0	29.78	110.45	10,388.8	-22.0	109.1	109.1	12.00	12.00	0.00
10,425.0	32.78	110.45	10,410.1	-26.6	121.2	121.2	12.00	12.00	0.00
10,450.0	35.78	110.45	10,430.8	-31.5	134.4	134.4	12.00	12.00	0.00
10,475.0	38.78	110.45	10,450.7	-36.8	148.6	148.6	12.00	12.00	0.00
10,500.0	41.78	110.45	10,469.7	-42.4	163.8	163.8	12.00	12.00	0.00
10,525.0	44.78	110.45	10,487.9	-48.4	179.8	179.8	12.00	12.00	0.00
10,550.0	47.78	110.45	10,505.2	-54.7	196.7	196.7	12.00	12.00	0.00
10,575.0	50.78	110.45	10,521.5	-61.3	214.5	214.5	12.00	12.00	0.00
10,600.0	53.78	110.45	10,536.8	-68.3	233.0	233.0	12.00	12.00	0.00
10,625.0	56.78	110.45	10,551.0	-75.4	252.3	252.3	12.00	12.00	0.00
10,650.0	59.78	110.45	10,564.2	-82.9	272.2	272.2	12.00	12.00	0.00
10,675.0	62.78	110.45	10,576.2	-90.5	292.7	292.7	12.00	12.00	0.00
10,700.0	65.78	110.45	10,587.0	-98.4	313.8	313.8	12.00	12.00	0.00
10,702.4	66.07	110.45	10,588.0	-99.2	315.9	315.9	12.00	12.00	0.00
False Bakken									
10,725.0	68.78	110.45	10,596.7	-106.5	335.4	335.4	12.00	12.00	0.00
10,731.5	69.57	110.45	10,599.0	-108.6	341.2	341.2	12.00	12.00	0.00
Upper Bakken									

Ryan Directional Services

Planning Report

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Company:	Oasis	TVD Reference:	Well @ 1967.0usft
Project:	Indian Hills	MD Reference:	Well @ 1967.0usft
Site:	153N-100W-19/20_Altered	North Reference:	True
Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Wellbore:	Chalmers 5301 44-24 3BR		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,750.0	71.78	110.45	10,605.1	-114.7	357.5	357.5	12.00	12.00	0.00
10,775.0	74.78	110.45	10,612.3	-123.0	379.9	379.9	12.00	12.00	0.00
10,789.9	76.57	110.45	10,616.0	-128.1	393.5	393.5	12.00	12.00	0.00
Middle Bakken									
10,800.0	77.78	110.45	10,618.2	-131.5	402.7	402.7	12.00	12.00	0.00
10,825.0	80.78	110.45	10,622.9	-140.1	425.7	425.7	12.00	12.00	0.00
10,839.6	82.53	110.45	10,625.0	-145.1	439.2	439.2	12.00	12.00	0.00
Middle Bakken Sand Target									
10,850.0	83.78	110.45	10,626.2	-148.8	448.9	448.9	12.00	12.00	0.00
10,875.0	86.78	110.45	10,628.3	-157.5	472.2	472.2	12.00	12.00	0.00
10,898.5	89.60	110.45	10,629.0	-165.7	494.3	494.3	11.98	11.98	0.00
EOC									
10,899.0	89.60	110.44	10,629.0	-165.8	494.7	494.7	2.13	0.00	-2.13
7"									
10,900.0	89.60	110.42	10,629.0	-166.2	495.7	495.7	2.00	0.00	-2.00
11,000.0	89.60	108.42	10,629.7	-199.4	590.0	590.0	2.00	0.00	-2.00
11,100.0	89.60	106.42	10,630.4	-229.4	685.4	685.4	2.00	0.00	-2.00
11,200.0	89.59	104.42	10,631.2	-256.0	781.8	781.8	2.00	0.00	-2.00
11,300.0	89.59	102.42	10,631.9	-279.2	879.0	879.0	2.00	0.00	-2.00
11,400.0	89.59	100.42	10,632.6	-299.0	977.0	977.0	2.00	0.00	-2.00
11,500.0	89.59	98.42	10,633.3	-315.3	1,075.7	1,075.7	2.00	0.00	-2.00
11,600.0	89.59	96.42	10,634.0	-328.2	1,174.8	1,174.8	2.00	0.00	-2.00
11,601.2	89.59	96.42	10,634.0	-328.4	1,176.0	1,176.0	0.00	0.00	0.00
Base Middle Bakken Sand Target									
11,700.0	89.60	94.42	10,634.7	-337.7	1,274.4	1,274.4	2.02	0.00	-2.02
11,800.0	89.60	92.42	10,635.4	-343.6	1,374.2	1,374.2	2.00	0.00	-2.00
11,900.0	89.60	90.42	10,636.1	-346.1	1,474.2	1,474.2	2.00	0.00	-2.00
11,920.9	89.60	90.00	10,636.3	-346.2	1,495.1	1,495.1	2.00	0.00	-2.00
12,000.0	89.60	90.00	10,636.8	-346.2	1,574.2	1,574.2	0.00	0.00	0.00
12,100.0	89.60	90.00	10,637.5	-346.2	1,674.2	1,674.2	0.00	0.00	0.00
12,200.0	89.60	90.00	10,638.2	-346.2	1,774.2	1,774.2	0.00	0.00	0.00
12,300.0	89.60	90.00	10,638.9	-346.2	1,874.2	1,874.2	0.00	0.00	0.00
12,400.0	89.60	90.00	10,639.6	-346.2	1,974.1	1,974.1	0.00	0.00	0.00
12,500.0	89.60	90.00	10,640.3	-346.2	2,074.1	2,074.1	0.00	0.00	0.00
12,600.0	89.60	90.00	10,641.0	-346.2	2,174.1	2,174.1	0.00	0.00	0.00
12,700.0	89.60	90.00	10,641.7	-346.2	2,274.1	2,274.1	0.00	0.00	0.00
12,800.0	89.60	90.00	10,642.4	-346.2	2,374.1	2,374.1	0.00	0.00	0.00
12,900.0	89.60	90.00	10,643.1	-346.2	2,474.1	2,474.1	0.00	0.00	0.00
13,000.0	89.60	90.00	10,643.8	-346.2	2,574.1	2,574.1	0.00	0.00	0.00
13,100.0	89.60	90.00	10,644.5	-346.2	2,674.1	2,674.1	0.00	0.00	0.00
13,200.0	89.60	90.00	10,645.2	-346.2	2,774.1	2,774.1	0.00	0.00	0.00
13,300.0	89.60	90.00	10,645.9	-346.2	2,874.1	2,874.1	0.00	0.00	0.00
13,400.0	89.60	90.00	10,646.6	-346.2	2,974.1	2,974.1	0.00	0.00	0.00
13,500.0	89.60	90.00	10,647.3	-346.2	3,074.1	3,074.1	0.00	0.00	0.00
13,600.0	89.60	90.00	10,648.0	-346.2	3,174.1	3,174.1	0.00	0.00	0.00
13,604.0	89.60	90.00	10,648.0	-346.2	3,178.1	3,178.1	0.00	0.00	0.00
Lower Bakken									
13,700.0	89.60	90.00	10,648.7	-346.2	3,274.1	3,274.1	0.00	0.00	0.00
13,800.0	89.60	90.00	10,649.4	-346.2	3,374.1	3,374.1	0.00	0.00	0.00
13,900.0	89.60	90.00	10,650.1	-346.2	3,474.1	3,474.1	0.00	0.00	0.00
14,000.0	89.60	90.00	10,650.8	-346.2	3,574.1	3,574.1	0.00	0.00	0.00
14,100.0	89.60	90.00	10,651.5	-346.2	3,674.1	3,674.1	0.00	0.00	0.00
14,200.0	89.60	90.00	10,652.2	-346.2	3,774.1	3,774.1	0.00	0.00	0.00

Ryan Directional Services

Planning Report

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Company:	Oasis	TVD Reference:	Well @ 1967.0usft
Project:	Indian Hills	MD Reference:	Well @ 1967.0usft
Site:	153N-100W-19/20_Altered	North Reference:	True
Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Wellbore:	Chalmers 5301 44-24 3BR		
Design:	Design #2		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
14,300.0	89.60	90.00	10,652.9	-346.2	3,874.1	3,874.1	0.00	0.00	0.00	
14,400.0	89.60	90.00	10,653.6	-346.2	3,974.1	3,974.1	0.00	0.00	0.00	
14,500.0	89.60	90.00	10,654.3	-346.2	4,074.1	4,074.1	0.00	0.00	0.00	
14,600.0	89.60	90.00	10,655.0	-346.2	4,174.1	4,174.1	0.00	0.00	0.00	
14,700.0	89.60	90.00	10,655.7	-346.2	4,274.1	4,274.1	0.00	0.00	0.00	
14,800.0	89.60	90.00	10,656.3	-346.2	4,374.1	4,374.1	0.00	0.00	0.00	
14,900.0	89.60	90.00	10,657.0	-346.2	4,474.1	4,474.1	0.00	0.00	0.00	
15,000.0	89.60	90.00	10,657.7	-346.2	4,574.1	4,574.1	0.00	0.00	0.00	
15,100.0	89.60	90.00	10,658.4	-346.2	4,674.1	4,674.1	0.00	0.00	0.00	
15,200.0	89.60	90.00	10,659.1	-346.2	4,774.1	4,774.1	0.00	0.00	0.00	
15,300.0	89.60	90.00	10,659.8	-346.2	4,874.1	4,874.1	0.00	0.00	0.00	
15,400.0	89.60	90.00	10,660.5	-346.2	4,974.1	4,974.1	0.00	0.00	0.00	
15,500.0	89.60	90.00	10,661.2	-346.2	5,074.1	5,074.1	0.00	0.00	0.00	
15,600.0	89.60	90.00	10,661.9	-346.2	5,174.1	5,174.1	0.00	0.00	0.00	
15,700.0	89.60	90.00	10,662.6	-346.2	5,274.1	5,274.1	0.00	0.00	0.00	
15,800.0	89.60	90.00	10,663.3	-346.2	5,374.1	5,374.1	0.00	0.00	0.00	
15,900.0	89.60	90.00	10,664.0	-346.2	5,474.1	5,474.1	0.00	0.00	0.00	
16,000.0	89.60	90.00	10,664.7	-346.2	5,574.1	5,574.1	0.00	0.00	0.00	
16,100.0	89.60	90.00	10,665.4	-346.2	5,674.1	5,674.1	0.00	0.00	0.00	
16,200.0	89.60	90.00	10,666.1	-346.2	5,774.1	5,774.1	0.00	0.00	0.00	
16,300.0	89.60	90.00	10,666.8	-346.2	5,874.1	5,874.1	0.00	0.00	0.00	
16,400.0	89.60	90.00	10,667.5	-346.2	5,974.1	5,974.1	0.00	0.00	0.00	
16,500.0	89.60	90.00	10,668.2	-346.2	6,074.0	6,074.0	0.00	0.00	0.00	
16,600.0	89.60	90.00	10,668.9	-346.2	6,174.0	6,174.0	0.00	0.00	0.00	
16,700.0	89.60	90.00	10,669.6	-346.2	6,274.0	6,274.0	0.00	0.00	0.00	
16,800.0	89.60	90.00	10,670.3	-346.2	6,374.0	6,374.0	0.00	0.00	0.00	
16,900.0	89.60	90.00	10,671.0	-346.2	6,474.0	6,474.0	0.00	0.00	0.00	
17,000.0	89.60	90.00	10,671.7	-346.2	6,574.0	6,574.0	0.00	0.00	0.00	
17,100.0	89.60	90.00	10,672.4	-346.2	6,674.0	6,674.0	0.00	0.00	0.00	
17,200.0	89.60	90.00	10,673.1	-346.2	6,774.0	6,774.0	0.00	0.00	0.00	
17,300.0	89.60	90.00	10,673.8	-346.2	6,874.0	6,874.0	0.00	0.00	0.00	
17,400.0	89.60	90.00	10,674.5	-346.2	6,974.0	6,974.0	0.00	0.00	0.00	
17,500.0	89.60	90.00	10,675.2	-346.2	7,074.0	7,074.0	0.00	0.00	0.00	
17,600.0	89.60	90.00	10,675.9	-346.2	7,174.0	7,174.0	0.00	0.00	0.00	
17,614.7	89.60	90.00	10,676.0	-346.2	7,188.7	7,188.7	0.00	0.00	0.00	
Three Forks										
17,700.0	89.60	90.00	10,676.6	-346.2	7,274.0	7,274.0	0.00	0.00	0.00	
17,800.0	89.60	90.00	10,677.3	-346.2	7,374.0	7,374.0	0.00	0.00	0.00	
17,900.0	89.60	90.00	10,678.0	-346.2	7,474.0	7,474.0	0.00	0.00	0.00	
18,000.0	89.60	90.00	10,678.7	-346.2	7,574.0	7,574.0	0.00	0.00	0.00	
18,100.0	89.60	90.00	10,679.4	-346.2	7,674.0	7,674.0	0.00	0.00	0.00	
18,200.0	89.60	90.00	10,680.1	-346.2	7,774.0	7,774.0	0.00	0.00	0.00	
18,300.0	89.60	90.00	10,680.8	-346.2	7,874.0	7,874.0	0.00	0.00	0.00	
18,400.0	89.60	90.00	10,681.5	-346.2	7,974.0	7,974.0	0.00	0.00	0.00	
18,500.0	89.60	90.00	10,682.2	-346.2	8,074.0	8,074.0	0.00	0.00	0.00	
18,600.0	89.60	90.00	10,682.9	-346.2	8,174.0	8,174.0	0.00	0.00	0.00	
18,700.0	89.60	90.00	10,683.6	-346.2	8,274.0	8,274.0	0.00	0.00	0.00	
18,800.0	89.60	90.00	10,684.3	-346.2	8,374.0	8,374.0	0.00	0.00	0.00	
18,900.0	89.60	90.00	10,685.0	-346.2	8,474.0	8,474.0	0.00	0.00	0.00	
19,000.0	89.60	90.00	10,685.7	-346.2	8,574.0	8,574.0	0.00	0.00	0.00	
19,100.0	89.60	90.00	10,686.4	-346.2	8,674.0	8,674.0	0.00	0.00	0.00	
19,200.0	89.60	90.00	10,687.1	-346.2	8,774.0	8,774.0	0.00	0.00	0.00	
19,300.0	89.60	90.00	10,687.8	-346.2	8,874.0	8,874.0	0.00	0.00	0.00	
19,400.0	89.60	90.00	10,688.5	-346.2	8,974.0	8,974.0	0.00	0.00	0.00	

Ryan Directional Services

Planning Report

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Company:	Oasis	TVD Reference:	Well @ 1967.0usft
Project:	Indian Hills	MD Reference:	Well @ 1967.0usft
Site:	153N-100W-19/20_Altered	North Reference:	True
Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Wellbore:	Chalmers 5301 44-24 3BR		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
19,500.0	89.60	90.00	10,689.2	-346.2	9,074.0	9,074.0	0.00	0.00	0.00
19,600.0	89.60	90.00	10,689.9	-346.2	9,174.0	9,174.0	0.00	0.00	0.00
19,700.0	89.60	90.00	10,690.6	-346.2	9,274.0	9,274.0	0.00	0.00	0.00
19,800.0	89.60	90.00	10,691.3	-346.2	9,374.0	9,374.0	0.00	0.00	0.00
19,900.0	89.60	90.00	10,692.0	-346.2	9,474.0	9,474.0	0.00	0.00	0.00
20,000.0	89.60	90.00	10,692.7	-346.2	9,574.0	9,574.0	0.00	0.00	0.00
20,100.0	89.60	90.00	10,693.4	-346.2	9,674.0	9,674.0	0.00	0.00	0.00
20,200.0	89.60	90.00	10,694.0	-346.2	9,774.0	9,774.0	0.00	0.00	0.00
20,300.0	89.60	90.00	10,694.7	-346.2	9,874.0	9,874.0	0.00	0.00	0.00
20,400.0	89.60	90.00	10,695.4	-346.2	9,974.0	9,974.0	0.00	0.00	0.00
20,500.0	89.60	90.00	10,696.1	-346.2	10,074.0	10,074.0	0.00	0.00	0.00
20,600.0	89.60	90.00	10,696.8	-346.2	10,173.9	10,173.9	0.00	0.00	0.00
20,700.0	89.60	90.00	10,697.5	-346.2	10,273.9	10,273.9	0.00	0.00	0.00
20,800.0	89.60	90.00	10,698.2	-346.2	10,373.9	10,373.9	0.00	0.00	0.00
20,900.0	89.60	90.00	10,698.9	-346.2	10,473.9	10,473.9	0.00	0.00	0.00
20,960.1	89.60	90.00	10,699.4	-346.2	10,534.0	10,534.0	0.00	0.00	0.00
5301 44-24 3BR PBHL									

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
5301 44-24 3BR PBHL	0.00	0.00	10,699.2	-346.0	10,534.0	399,619.70	1,219,839.57	48° 3' 17.046 N	103° 33' 43.485 W
- hit/miss target									
- Shape									
- plan misses target center by 0.2usft at 20960.1usft MD (10699.4 TVD, -346.2 N, 10534.0 E)									
- Point									

Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")	
10,899.0	10,629.0	7"	7	8-3/4	

Ryan Directional Services

Planning Report

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Company:	Oasis	TVD Reference:	Well @ 1967.0usft
Project:	Indian Hills	MD Reference:	Well @ 1967.0usft
Site:	153N-100W-19/20_Altered	North Reference:	True
Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Wellbore:	Chalmers 5301 44-24 3BR		
Design:	Design #2		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,918.0	1,918.0	Pierre				
4,510.1	4,510.0	Greenhorn				
4,914.1	4,914.0	Mowry				
5,341.1	5,341.0	Dakota				
6,356.2	6,356.0	Rierdon				
6,683.2	6,683.0	Dunham Salt				
6,800.2	6,800.0	Dunham Salt Base				
6,887.2	6,887.0	Spearfish				
7,135.2	7,135.0	Pine Salt				
7,168.2	7,168.0	Pine Salt Base				
7,235.2	7,235.0	Opeche Salt				
7,269.2	7,269.0	Opeche Salt Base				
7,469.2	7,469.0	Broom Creek (Top of Minnelusa Gp.)				
7,553.2	7,553.0	Amsden				
7,720.2	7,720.0	Tyler				
7,920.2	7,920.0	Otter (Base of Minnelusa Gp.)				
8,269.2	8,269.0	Kibbey Lime				
8,433.2	8,433.0	Charles Salt				
9,010.2	9,010.0	UB				
9,091.2	9,091.0	Base Last Salt				
9,141.2	9,141.0	Ratcliffe				
9,311.2	9,311.0	Mission Canyon				
9,876.2	9,876.0	Lodgepole				
10,109.2	10,109.0	Lodgepole Fracture Zone				
10,702.4	10,588.0	False Bakken				
10,731.5	10,599.0	Upper Bakken				
10,789.9	10,616.0	Middle Bakken				
10,839.6	10,625.0	Middle Bakken Sand Target				
11,601.2	10,634.0	Base Middle Bakken Sand Target				
13,604.0	10,648.0	Lower Bakken				
17,614.7	10,676.0	Three Forks				

Plan Annotations					
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates			
		+N/-S (usft)	+E/-W (usft)	Comment	
10,151.8	10,151.6	0.0	50.0	KOP Build 12°/100'	
10,898.5	10,629.0	-165.7	494.3	EOC	

Oasis

Indian Hills

153N-100W-19/20_Altered

Chalmers 5301 44-24 3BR

Chalmers 5301 44-24 3BR

Design #2

Anticollision Report

27 May, 2014

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Reference	Design #2		
Filter type:	NO GLOBAL FILTER: Using user defined selection & filtering criteria		
Interpolation Method:	Stations	Error Model:	ISCWSA
Depth Range:	Unlimited	Scan Method:	Closest Approach 3D
Results Limited by:	Maximum center-center distance of 10,000.0 usft	Error Surface:	Elliptical Conic
Warning Levels Evaluated at:	2.00 Sigma	Casing Method:	Not applied

Survey Tool Program	Date 5/27/2014			
From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description
0.0	20,960.1	Design #2 (Chalmers 5301 44-24 3BR)	MWD	MWD - Standard

Summary						
Site Name	Reference Measured Depth (usft)	Offset Measured Depth (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Separation Factor	Warning
Offset Well - Wellbore - Design						
153N-100W-19/20_Altered						
Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R	2,200.1	2,200.1	48.6	38.6	4.859	CC
Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R	20,960.1	21,105.8	108.2	-47.9	0.693	Level 1, ES, SF
Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12T	2,200.0	2,200.0	82.1	72.1	8.201	CC
Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12T	20,960.1	21,225.9	585.3	-22.5	0.963	Level 1, ES, SF

Offset Design 153N-100W-19/20_Altered - Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R - Design #1											
Survey Program: 0-MWD											
Reference											
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Semi Major Axis		Highside Toolface (°)	Offset Wellbore Centre		Distance		Warning
				Reference (usft)	Offset (usft)		+N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)
											Factor
0.0	0.0	0.0	0.0	0.0	2.0	-180.00	-48.6	0.0	48.6		
100.0	100.0	100.0	100.0	0.1	2.0	-180.00	-48.6	0.0	48.6	46.6	2.09
200.0	200.0	200.0	200.0	0.3	2.0	-180.00	-48.6	0.0	48.6	46.3	2.33
300.0	300.0	300.0	300.0	0.5	2.1	-180.00	-48.6	0.0	48.6	46.0	2.60
400.0	400.0	400.0	400.0	0.8	2.1	-180.00	-48.6	0.0	48.6	45.7	2.90
500.0	500.0	500.0	500.0	1.0	2.2	-180.00	-48.6	0.0	48.6	45.4	3.21
600.0	600.0	600.0	600.0	1.2	2.3	-180.00	-48.6	0.0	48.6	45.1	3.54
700.0	700.0	700.0	700.0	1.4	2.5	-180.00	-48.6	0.0	48.6	44.7	3.89
800.0	800.0	800.0	800.0	1.7	2.6	-180.00	-48.6	0.0	48.6	44.4	4.26
900.0	900.0	900.0	900.0	1.9	2.7	-180.00	-48.6	0.0	48.6	44.0	4.63
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.9	-180.00	-48.6	0.0	48.6	43.6	5.01
1,100.0	1,100.0	1,100.0	1,100.0	2.3	3.1	-180.00	-48.6	0.0	48.6	43.2	5.40
1,200.0	1,200.0	1,200.0	1,200.0	2.6	3.2	-180.00	-48.6	0.0	48.6	42.8	5.80
1,300.0	1,300.0	1,300.0	1,300.0	2.8	3.4	-180.00	-48.6	0.0	48.6	42.4	6.21
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.6	-180.00	-48.6	0.0	48.6	42.0	6.62
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.8	-180.00	-48.6	0.0	48.6	41.6	7.03
1,600.0	1,600.0	1,600.0	1,600.0	3.5	4.0	-180.00	-48.6	0.0	48.6	41.2	7.45
1,700.0	1,700.0	1,700.0	1,700.0	3.7	4.2	-180.00	-48.6	0.0	48.6	40.8	7.87
1,800.0	1,800.0	1,800.0	1,800.0	3.9	4.4	-180.00	-48.6	0.0	48.6	40.3	8.29
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.6	-180.00	-48.6	0.0	48.6	39.9	8.72
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.8	-180.00	-48.6	0.0	48.6	39.5	9.15
2,100.0	2,100.0	2,100.0	2,100.0	4.6	5.0	-180.00	-48.6	0.0	48.6	39.1	9.58
2,200.0	2,200.0	2,200.0	2,200.0	4.8	5.2	-180.00	-48.6	0.0	48.6	38.6	10.01

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
2,200.1	2,200.1	2,200.1	2,200.1	4.8	5.2	-180.00	-48.6	0.0	48.6	38.6	10.01	4.859	CC	
2,216.7	2,216.7	2,216.7	2,216.7	4.8	5.2	90.09	-48.6	0.0	48.6	38.6	10.08	4.826		
2,300.0	2,300.0	2,300.0	2,300.0	5.0	5.4	90.94	-48.6	0.0	48.6	38.2	10.42	4.670		
2,400.0	2,400.0	2,400.0	2,400.0	5.2	5.6	91.97	-48.6	0.0	48.7	37.8	10.82	4.496		
2,500.0	2,500.0	2,500.0	2,500.0	5.4	5.8	93.00	-48.6	0.0	48.7	37.5	11.24	4.334		
2,600.0	2,600.0	2,600.0	2,600.0	5.6	6.0	94.02	-48.6	0.0	48.8	37.1	11.65	4.185		
2,700.0	2,700.0	2,700.0	2,700.0	5.8	6.3	95.04	-48.6	0.0	48.8	36.8	12.07	4.046		
2,800.0	2,800.0	2,800.0	2,800.0	6.0	6.5	96.06	-48.6	0.0	48.9	36.4	12.49	3.916		
2,900.0	2,900.0	2,900.0	2,900.0	6.2	6.7	97.07	-48.6	0.0	49.0	36.1	12.91	3.796		
3,000.0	3,000.0	3,000.0	3,000.0	6.4	6.9	98.08	-48.6	0.0	49.1	35.8	13.33	3.684		
3,100.0	3,100.0	3,100.0	3,100.0	6.6	7.1	99.09	-48.6	0.0	49.3	35.5	13.76	3.580		
3,200.0	3,200.0	3,200.0	3,200.0	6.9	7.3	100.09	-48.6	0.0	49.4	35.2	14.19	3.482		
3,300.0	3,300.0	3,300.0	3,300.0	7.1	7.5	101.08	-48.6	0.0	49.6	34.9	14.61	3.391		
3,400.0	3,400.0	3,400.0	3,400.0	7.3	7.8	102.07	-48.6	0.0	49.7	34.7	15.04	3.306		
3,500.0	3,500.0	3,500.0	3,500.0	7.5	8.0	103.05	-48.6	0.0	49.9	34.5	15.48	3.226		
3,600.0	3,599.9	3,599.9	3,599.9	7.7	8.2	104.02	-48.6	0.0	50.1	34.2	15.91	3.151		
3,700.0	3,699.9	3,699.9	3,699.9	7.9	8.4	104.98	-48.6	0.0	50.3	34.0	16.34	3.081		
3,800.0	3,799.9	3,799.9	3,799.9	8.1	8.6	105.94	-48.6	0.0	50.6	33.8	16.77	3.016		
3,900.0	3,899.9	3,899.9	3,899.9	8.4	8.9	106.88	-48.6	0.0	50.8	33.6	17.21	2.954		
4,000.0	3,999.9	3,999.9	3,999.9	8.6	9.1	107.82	-48.6	0.0	51.1	33.4	17.64	2.896		
4,100.0	4,099.9	4,099.9	4,099.9	8.8	9.3	108.75	-48.6	0.0	51.4	33.3	18.08	2.841		
4,200.0	4,199.9	4,199.9	4,199.9	9.0	9.5	109.66	-48.6	0.0	51.6	33.1	18.52	2.789		
4,300.0	4,299.9	4,299.9	4,299.9	9.2	9.7	110.57	-48.6	0.0	51.9	33.0	18.95	2.741		
4,400.0	4,399.9	4,399.9	4,399.9	9.4	10.0	111.47	-48.6	0.0	52.3	32.9	19.39	2.695		
4,500.0	4,499.9	4,499.9	4,499.9	9.7	10.2	112.35	-48.6	0.0	52.6	32.8	19.83	2.652		
4,600.0	4,599.9	4,599.9	4,599.9	9.9	10.4	113.22	-48.6	0.0	52.9	32.7	20.27	2.611		
4,700.0	4,699.9	4,699.9	4,699.9	10.1	10.6	114.09	-48.6	0.0	53.3	32.6	20.71	2.572		
4,800.0	4,799.9	4,799.9	4,799.9	10.3	10.8	114.94	-48.6	0.0	53.6	32.5	21.15	2.536		
4,900.0	4,899.9	4,899.9	4,899.9	10.5	11.1	115.78	-48.6	0.0	54.0	32.4	21.59	2.502		
5,000.0	4,999.9	4,999.9	4,999.9	10.8	11.3	116.60	-48.6	0.0	54.4	32.4	22.03	2.469		
5,100.0	5,099.9	5,099.9	5,099.9	11.0	11.5	117.42	-48.6	0.0	54.8	32.3	22.47	2.438		
5,200.0	5,199.9	5,199.9	5,199.9	11.2	11.7	118.22	-48.6	0.0	55.2	32.3	22.91	2.409		
5,300.0	5,299.9	5,299.9	5,299.9	11.4	11.9	119.02	-48.6	0.0	55.6	32.3	23.36	2.381		
5,400.0	5,399.9	5,399.9	5,399.9	11.6	12.2	119.80	-48.6	0.0	56.0	32.2	23.80	2.355		
5,500.0	5,499.9	5,499.9	5,499.9	11.9	12.4	120.57	-48.6	0.0	56.5	32.2	24.24	2.330		
5,600.0	5,599.9	5,599.9	5,599.9	12.1	12.6	121.32	-48.6	0.0	56.9	32.3	24.68	2.307		
5,700.0	5,699.9	5,699.9	5,699.9	12.3	12.8	122.07	-48.6	0.0	57.4	32.3	25.13	2.284		
5,800.0	5,799.9	5,799.9	5,799.9	12.5	13.0	122.80	-48.6	0.0	57.9	32.3	25.57	2.263		
5,900.0	5,899.9	5,899.9	5,899.9	12.7	13.3	123.52	-48.6	0.0	58.3	32.3	26.01	2.243		
6,000.0	5,999.9	5,999.9	5,999.9	13.0	13.5	124.23	-48.6	0.0	58.8	32.4	26.46	2.223		
6,100.0	6,099.9	6,099.9	6,099.9	13.2	13.7	124.92	-48.6	0.0	59.3	32.4	26.90	2.205		
6,200.0	6,199.8	6,199.8	6,199.8	13.4	13.9	125.61	-48.6	0.0	59.8	32.5	27.34	2.188		
6,300.0	6,299.8	6,299.8	6,299.8	13.6	14.2	126.28	-48.6	0.0	60.3	32.5	27.79	2.171		
6,400.0	6,399.8	6,399.8	6,399.8	13.9	14.4	126.94	-48.6	0.0	60.9	32.6	28.23	2.155		
6,500.0	6,499.8	6,499.8	6,499.8	14.1	14.6	127.60	-48.6	0.0	61.4	32.7	28.68	2.141		
6,600.0	6,599.8	6,599.8	6,599.8	14.3	14.8	128.24	-48.6	0.0	61.9	32.8	29.12	2.126		
6,700.0	6,699.8	6,699.8	6,699.8	14.5	15.1	128.86	-48.6	0.0	62.5	32.9	29.57	2.113		
6,800.0	6,799.8	6,799.8	6,799.8	14.7	15.3	129.48	-48.6	0.0	63.0	33.0	30.01	2.100		
6,900.0	6,899.8	6,899.8	6,899.8	15.0	15.5	130.09	-48.6	0.0	63.6	33.1	30.46	2.087		
7,000.0	6,999.8	6,999.8	6,999.8	15.2	15.7	130.69	-48.6	0.0	64.1	33.2	30.90	2.076		
7,100.0	7,099.8	7,099.8	7,099.8	15.4	15.9	131.27	-48.6	0.0	64.7	33.4	31.35	2.064		
7,200.0	7,199.8	7,199.8	7,199.8	15.6	16.2	131.85	-48.6	0.0	65.3	33.5	31.79	2.054		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Distance								Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
7,300.0	7,299.8	7,299.8	7,299.8	15.9	16.4	132.41	-48.6	0.0	65.9	33.6	32.24	2.043		
7,400.0	7,399.8	7,399.8	7,399.8	16.1	16.6	132.97	-48.6	0.0	66.5	33.8	32.69	2.034		
7,500.0	7,499.8	7,499.8	7,499.8	16.3	16.8	133.51	-48.6	0.0	67.1	33.9	33.13	2.024		
7,600.0	7,599.8	7,599.8	7,599.8	16.5	17.1	134.05	-48.6	0.0	67.7	34.1	33.58	2.015		
7,700.0	7,699.8	7,699.8	7,699.8	16.7	17.3	134.58	-48.6	0.0	68.3	34.3	34.02	2.007		
7,800.0	7,799.8	7,799.8	7,799.8	17.0	17.5	135.09	-48.6	0.0	68.9	34.4	34.47	1.999		
7,900.0	7,899.8	7,899.8	7,899.8	17.2	17.7	135.60	-48.6	0.0	69.5	34.6	34.92	1.991		
7,926.7	7,926.4	7,926.4	7,926.4	17.3	17.8	135.73	-48.6	0.0	69.7	34.6	35.04	1.989		
7,943.3	7,943.1	7,943.1	7,943.1	17.3	17.8	-134.22	-48.6	0.0	69.7	34.7	35.06	1.989		
8,000.0	7,999.8	7,999.8	7,999.8	17.4	18.0	-134.22	-48.6	0.0	69.7	34.4	35.30	1.975		
8,100.0	8,099.8	8,099.8	8,099.8	17.6	18.2	-134.22	-48.6	0.0	69.7	34.0	35.74	1.951		
8,200.0	8,199.8	8,199.8	8,199.8	17.8	18.4	-134.22	-48.6	0.0	69.7	33.6	36.17	1.928		
8,300.0	8,299.8	8,299.8	8,299.8	18.0	18.6	-134.22	-48.6	0.0	69.7	33.1	36.60	1.905		
8,400.0	8,399.8	8,399.8	8,399.8	18.2	18.8	-134.22	-48.6	0.0	69.7	32.7	37.04	1.883		
8,500.0	8,499.8	8,499.8	8,499.8	18.5	19.1	-134.22	-48.6	0.0	69.7	32.3	37.47	1.861		
8,600.0	8,599.8	8,599.8	8,599.8	18.7	19.3	-134.22	-48.6	0.0	69.7	31.8	37.90	1.840		
8,700.0	8,699.8	8,699.8	8,699.8	18.9	19.5	-134.22	-48.6	0.0	69.7	31.4	38.34	1.819		
8,800.0	8,799.8	8,799.8	8,799.8	19.1	19.7	-134.22	-48.6	0.0	69.7	31.0	38.77	1.798		
8,900.0	8,899.8	8,899.8	8,899.8	19.3	20.0	-134.22	-48.6	0.0	69.7	30.5	39.21	1.779		
9,000.0	8,999.8	8,999.8	8,999.8	19.5	20.2	-134.22	-48.6	0.0	69.7	30.1	39.65	1.759		
9,100.0	9,099.8	9,099.8	9,099.8	19.7	20.4	-134.22	-48.6	0.0	69.7	29.7	40.08	1.740		
9,200.0	9,199.8	9,199.8	9,199.8	19.9	20.6	-134.22	-48.6	0.0	69.7	29.2	40.52	1.721		
9,300.0	9,299.8	9,299.8	9,299.8	20.1	20.9	-134.22	-48.6	0.0	69.7	28.8	40.95	1.703		
9,400.0	9,399.8	9,399.8	9,399.8	20.4	21.1	-134.22	-48.6	0.0	69.7	28.3	41.39	1.685		
9,500.0	9,499.8	9,499.8	9,499.8	20.6	21.3	-134.22	-48.6	0.0	69.7	27.9	41.83	1.667		
9,600.0	9,599.8	9,599.8	9,599.8	20.8	21.5	-134.22	-48.6	0.0	69.7	27.5	42.27	1.650		
9,700.0	9,699.8	9,699.8	9,699.8	21.0	21.8	-134.22	-48.6	0.0	69.7	27.0	42.70	1.633		
9,800.0	9,799.8	9,799.8	9,799.8	21.2	22.0	-134.22	-48.6	0.0	69.7	26.6	43.14	1.616		
9,900.0	9,899.8	9,899.8	9,899.8	21.4	22.2	-134.22	-48.6	0.0	69.7	26.2	43.58	1.600		
10,000.0	9,999.8	9,999.8	9,999.8	21.6	22.4	-134.22	-48.6	0.0	69.7	25.7	44.02	1.584		
10,100.0	10,099.8	10,099.8	10,099.8	21.9	22.6	-134.22	-48.6	0.0	69.7	25.3	44.46	1.569		
10,151.8	10,151.6	10,151.6	10,151.6	22.0	22.8	-134.22	-48.6	0.0	69.7	25.1	44.68	1.561		
10,175.0	10,174.8	10,174.8	10,174.8	22.0	22.8	115.72	-48.6	0.0	70.0	25.2	44.82	1.561		
10,200.0	10,199.7	10,199.7	10,199.7	22.1	22.9	116.99	-48.6	0.0	70.8	25.9	44.91	1.577		
10,225.0	10,224.5	10,224.5	10,224.5	22.1	22.9	119.05	-48.6	0.0	72.3	27.3	44.97	1.608		
10,250.0	10,249.1	10,249.1	10,249.1	22.2	23.0	121.78	-48.6	0.0	74.6	29.6	45.00	1.658		
10,275.0	10,273.4	10,275.0	10,275.0	22.3	23.0	125.09	-48.7	0.2	77.7	32.7	44.97	1.727		
10,300.0	10,297.4	10,302.2	10,302.2	22.3	23.1	128.32	-49.2	1.8	81.0	36.1	44.87	1.805		
10,325.0	10,321.0	10,329.9	10,329.6	22.4	23.1	131.27	-50.3	4.9	84.4	39.7	44.71	1.889		
10,350.0	10,344.1	10,357.9	10,357.2	22.5	23.2	133.95	-51.8	9.6	87.9	43.5	44.47	1.977		
10,375.0	10,366.7	10,386.3	10,384.8	22.5	23.3	136.40	-53.9	15.9	91.5	47.3	44.17	2.070		
10,400.0	10,388.8	10,415.1	10,412.4	22.6	23.3	138.63	-56.6	23.8	95.0	51.1	43.80	2.168		
10,425.0	10,410.1	10,444.3	10,439.7	22.7	23.4	140.65	-59.8	33.5	98.4	55.0	43.37	2.268		
10,450.0	10,430.8	10,473.9	10,466.8	22.8	23.5	142.50	-63.6	44.9	101.7	58.8	42.88	2.372		
10,475.0	10,450.7	10,503.9	10,493.4	22.9	23.5	144.19	-68.0	58.1	104.9	62.5	42.34	2.477		
10,500.0	10,469.7	10,534.3	10,519.3	23.0	23.6	145.72	-73.0	72.9	107.9	66.1	41.75	2.584		
10,525.0	10,487.9	10,565.0	10,544.5	23.1	23.7	147.13	-78.5	89.6	110.7	69.6	41.12	2.692		
10,550.0	10,505.2	10,596.0	10,568.8	23.3	23.8	148.41	-84.6	107.9	113.3	72.8	40.47	2.800		
10,575.0	10,521.5	10,627.4	10,592.0	23.4	23.9	149.58	-91.3	127.9	115.6	75.8	39.80	2.906		
10,600.0	10,536.8	10,659.0	10,614.0	23.6	24.1	150.64	-98.5	149.5	117.7	78.6	39.12	3.009		
10,625.0	10,551.0	10,690.9	10,634.6	23.7	24.2	151.61	-106.2	172.6	119.4	81.0	38.44	3.107		
10,650.0	10,564.2	10,723.0	10,653.6	23.9	24.4	152.48	-114.4	197.2	120.9	83.1	37.78	3.200		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,675.0	10,576.2	10,755.4	10,670.9	24.1	24.6	153.28	-123.1	223.1	122.0	84.9	37.14	3.285		
10,700.0	10,587.0	10,787.8	10,686.4	24.3	24.9	153.99	-132.1	250.1	122.8	86.2	36.55	3.359		
10,725.0	10,596.7	10,820.4	10,700.0	24.6	25.1	154.62	-141.5	278.2	123.2	87.2	36.01	3.421		
10,750.0	10,605.1	10,853.1	10,711.6	24.8	25.4	155.18	-151.1	307.2	123.3	87.7	35.54	3.468		
10,775.0	10,612.3	10,885.8	10,721.0	25.1	25.8	155.67	-161.0	336.9	122.9	87.8	35.15	3.498		
10,800.0	10,618.2	10,918.5	10,728.3	25.4	26.1	156.09	-171.1	367.1	122.3	87.4	34.84	3.509		
10,825.0	10,622.9	10,951.1	10,733.4	25.7	26.5	156.44	-181.3	397.6	121.2	86.6	34.64	3.500		
10,850.0	10,626.2	10,983.6	10,736.3	26.0	26.9	156.71	-191.5	428.3	119.8	85.3	34.55	3.468		
10,875.0	10,628.3	11,013.8	10,737.0	26.4	27.3	156.93	-201.1	457.0	118.1	83.6	34.56	3.418		
10,898.5	10,629.0	11,037.6	10,737.2	26.7	27.6	157.31	-208.5	479.7	117.2	82.6	34.58	3.390		
10,900.0	10,629.0	11,039.2	10,737.2	26.7	27.6	157.34	-209.0	481.1	117.2	82.6	34.58	3.389		
11,000.0	10,629.7	11,140.6	10,737.9	28.2	29.2	159.46	-238.2	578.2	115.5	80.8	34.67	3.331		
11,100.0	10,630.4	11,241.8	10,738.6	29.9	30.9	161.67	-264.0	676.1	113.9	79.2	34.72	3.282		
11,200.0	10,631.2	11,342.9	10,739.3	31.8	32.8	163.96	-286.3	774.7	112.6	77.8	34.70	3.243		
11,300.0	10,631.9	11,443.8	10,740.0	33.9	34.9	166.31	-305.0	873.8	111.3	76.7	34.62	3.216		
11,400.0	10,632.6	11,544.5	10,740.7	36.0	37.0	168.72	-320.2	973.4	110.3	75.8	34.49	3.198		
11,500.0	10,633.3	11,645.1	10,741.5	38.3	39.3	171.19	-331.9	1,073.3	109.5	75.1	34.36	3.186		
11,600.0	10,634.0	11,745.5	10,742.2	40.6	41.6	173.70	-340.1	1,173.3	108.8	74.5	34.30	3.173		
11,700.0	10,634.7	11,845.7	10,742.9	43.0	44.0	176.24	-344.7	1,273.4	108.4	74.0	34.41	3.150		
11,800.0	10,635.4	11,945.7	10,743.6	45.5	46.4	178.77	-345.9	1,373.4	108.2	73.4	34.82	3.107		
11,878.0	10,635.9	12,023.7	10,744.1	47.4	48.3	179.96	-345.9	1,451.4	108.2	72.8	35.40	3.055		
11,900.0	10,636.1	12,045.7	10,744.3	47.9	48.8	-179.90	-345.9	1,473.4	108.2	72.6	35.59	3.040		
11,920.9	10,636.3	12,066.6	10,744.4	48.4	49.3	-179.86	-345.9	1,494.3	108.2	72.4	35.76	3.024		
12,000.0	10,636.8	12,145.7	10,745.0	50.4	51.3	-179.86	-345.9	1,573.4	108.2	71.7	36.45	2.967		
12,039.4	10,637.1	12,185.1	10,745.2	51.4	52.3	-179.86	-345.9	1,612.8	108.2	71.4	36.80	2.939		
12,100.0	10,637.5	12,245.7	10,745.7	53.0	53.9	-179.86	-345.9	1,673.4	108.2	70.8	37.35	2.896		
12,172.4	10,638.0	12,318.1	10,746.2	54.8	55.8	-179.86	-345.9	1,745.9	108.2	70.1	38.03	2.844		
12,200.0	10,638.2	12,345.7	10,746.4	55.5	56.5	-179.86	-345.9	1,773.4	108.2	69.9	38.29	2.825		
12,300.0	10,638.9	12,445.7	10,747.1	58.2	59.1	-179.86	-345.9	1,873.4	108.2	68.9	39.26	2.755		
12,339.4	10,639.2	12,485.1	10,747.3	59.2	60.1	-179.86	-345.9	1,912.8	108.2	68.5	39.65	2.728		
12,400.0	10,639.6	12,545.7	10,747.8	60.8	61.8	-179.86	-345.9	1,973.4	108.2	67.9	40.26	2.686		
12,477.4	10,640.1	12,623.1	10,748.3	62.9	63.8	-179.86	-345.9	2,050.8	108.2	67.1	41.06	2.634		
12,500.0	10,640.3	12,645.7	10,748.5	63.5	64.4	-179.86	-345.9	2,073.4	108.2	66.9	41.29	2.620		
12,600.0	10,641.0	12,745.7	10,749.2	66.2	67.1	-179.86	-345.9	2,173.4	108.2	65.8	42.35	2.554		
12,635.5	10,641.2	12,781.2	10,749.4	67.1	68.1	-179.86	-345.9	2,208.9	108.2	65.4	42.73	2.532		
12,700.0	10,641.7	12,845.7	10,749.9	68.9	69.8	-179.86	-345.9	2,273.4	108.2	64.7	43.42	2.491		
12,777.4	10,642.2	12,923.1	10,750.4	71.0	72.0	-179.86	-345.9	2,350.8	108.2	63.9	44.27	2.443		
12,800.0	10,642.4	12,945.7	10,750.5	71.6	72.6	-179.86	-345.9	2,373.4	108.2	63.6	44.52	2.429		
12,900.0	10,643.1	13,045.7	10,751.2	74.4	75.3	-179.86	-345.9	2,473.4	108.2	62.5	45.64	2.370		
12,935.5	10,643.3	13,081.2	10,751.5	75.3	76.3	-179.86	-345.9	2,508.9	108.2	62.1	46.05	2.349		
13,000.0	10,643.8	13,145.7	10,751.9	77.1	78.1	-179.86	-345.9	2,573.4	108.2	61.4	46.78	2.312		
13,072.7	10,644.3	13,218.4	10,752.5	79.1	80.1	-179.86	-345.9	2,646.0	108.2	60.5	47.62	2.271		
13,100.0	10,644.5	13,245.7	10,752.6	79.9	80.9	-179.86	-345.9	2,673.4	108.2	60.2	47.94	2.256		
13,200.0	10,645.2	13,345.7	10,753.3	82.7	83.6	-179.86	-345.9	2,773.4	108.2	59.1	49.11	2.202		
13,235.5	10,645.4	13,381.2	10,753.6	83.7	84.6	-179.86	-345.9	2,808.9	108.2	58.6	49.53	2.184		
13,300.0	10,645.9	13,445.7	10,754.0	85.5	86.4	-179.86	-345.9	2,873.4	108.2	57.9	50.30	2.150		
13,372.4	10,646.4	13,518.1	10,754.5	87.5	88.5	-179.86	-345.9	2,945.8	108.2	57.0	51.17	2.114		
13,400.0	10,646.6	13,545.7	10,754.7	88.3	89.2	-179.86	-345.9	2,973.4	108.2	56.7	51.51	2.100		
13,500.0	10,647.3	13,645.7	10,755.4	91.1	92.0	-179.86	-345.9	3,073.4	108.2	55.4	52.72	2.052		
13,535.5	10,647.5	13,681.2	10,755.7	92.1	93.0	-179.86	-345.9	3,108.9	108.2	55.0	53.16	2.035		
13,600.0	10,648.0	13,745.7	10,756.1	93.9	94.9	-179.86	-345.9	3,173.4	108.2	54.2	53.95	2.005		
13,678.2	10,648.5	13,823.9	10,756.7	96.1	97.1	-179.86	-345.9	3,251.5	108.2	53.2	54.92	1.970		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
13,700.0	10,648.7	13,845.7	10,756.8	96.7	97.7	-179.86	-345.9	3,273.4	108.2	53.0	55.19	1.960		
13,800.0	10,649.4	13,945.7	10,757.5	99.5	100.5	-179.86	-345.9	3,373.4	108.2	51.7	56.44	1.916		
13,835.5	10,649.6	13,981.2	10,757.8	100.5	101.5	-179.86	-345.9	3,408.8	108.2	51.3	56.89	1.901		
13,900.0	10,650.1	14,045.7	10,758.2	102.4	103.4	-179.86	-345.9	3,473.4	108.2	50.5	57.70	1.875		
14,000.0	10,650.8	14,145.7	10,758.9	105.2	106.2	-179.86	-345.9	3,573.4	108.2	49.2	58.97	1.834		
14,035.5	10,651.0	14,181.2	10,759.2	106.2	107.2	-179.86	-345.9	3,608.8	108.2	48.7	59.43	1.820		
14,100.0	10,651.5	14,245.7	10,759.6	108.1	109.0	-179.86	-345.9	3,673.4	108.2	47.9	60.25	1.795		
14,177.4	10,652.0	14,323.1	10,760.2	110.3	111.2	-179.86	-345.9	3,750.8	108.2	46.9	61.25	1.766		
14,200.0	10,652.2	14,345.7	10,760.3	110.9	111.9	-179.86	-345.9	3,773.4	108.2	46.6	61.54	1.758		
14,300.0	10,652.9	14,445.7	10,761.0	113.8	114.7	-179.86	-345.9	3,873.3	108.2	45.3	62.84	1.721		
14,335.5	10,653.1	14,481.2	10,761.3	114.8	115.8	-179.86	-345.9	3,908.8	108.2	44.9	63.30	1.709		
14,400.0	10,653.6	14,545.7	10,761.7	116.6	117.6	-179.86	-345.9	3,973.3	108.2	44.0	64.14	1.686		
14,472.4	10,654.1	14,618.1	10,762.2	118.7	119.7	-179.86	-345.9	4,045.8	108.2	43.1	65.09	1.662		
14,500.0	10,654.3	14,645.7	10,762.4	119.5	120.5	-179.86	-345.9	4,073.3	108.2	42.7	65.45	1.653		
14,600.0	10,655.0	14,745.7	10,763.1	122.4	123.3	-179.86	-345.9	4,173.3	108.2	41.4	66.77	1.620		
14,635.5	10,655.2	14,781.2	10,763.4	123.4	124.3	-179.86	-345.9	4,208.8	108.2	40.9	67.24	1.609		
14,700.0	10,655.7	14,845.7	10,763.8	125.2	126.2	-179.86	-345.9	4,273.3	108.2	40.1	68.09	1.589		
14,772.4	10,656.2	14,918.1	10,764.3	127.3	128.3	-179.86	-345.9	4,345.8	108.2	39.1	69.05	1.566		
14,800.0	10,656.3	14,945.7	10,764.5	128.1	129.1	-179.86	-345.9	4,373.3	108.2	38.7	69.42	1.558		
14,900.0	10,657.0	15,045.7	10,765.2	131.0	131.9	-179.86	-345.9	4,473.3	108.2	37.4	70.75	1.529		
14,935.5	10,657.3	15,081.2	10,765.5	132.0	133.0	-179.86	-345.9	4,508.8	108.2	36.9	71.23	1.519		
15,000.0	10,657.7	15,145.7	10,765.9	133.8	134.8	-179.86	-345.9	4,573.3	108.2	36.1	72.09	1.500		
15,077.4	10,658.3	15,223.1	10,766.4	136.1	137.0	-179.86	-345.9	4,650.8	108.2	35.0	73.13	1.479 Level 3	Level 3	
15,100.0	10,658.4	15,245.7	10,766.6	136.7	137.7	-179.86	-345.9	4,673.3	108.2	34.7	73.44	1.473 Level 3	Level 3	
15,200.0	10,659.1	15,345.7	10,767.3	139.6	140.6	-179.86	-345.9	4,773.3	108.2	33.4	74.78	1.446 Level 3	Level 3	
15,235.5	10,659.4	15,381.2	10,767.6	140.6	141.6	-179.86	-345.9	4,808.8	108.2	32.9	75.26	1.437 Level 3	Level 3	
15,300.0	10,659.8	15,445.7	10,768.0	142.5	143.5	-179.86	-345.9	4,873.3	108.2	32.0	76.14	1.421 Level 3	Level 3	
15,378.2	10,660.4	15,523.9	10,768.5	144.7	145.7	-179.86	-345.9	4,951.5	108.2	31.0	77.20	1.401 Level 3	Level 3	
15,400.0	10,660.5	15,545.7	10,768.7	145.4	146.3	-179.86	-345.9	4,973.3	108.2	30.7	77.49	1.396 Level 3	Level 3	
15,500.0	10,661.2	15,645.7	10,769.4	148.2	149.2	-179.86	-345.9	5,073.3	108.2	29.3	78.85	1.372 Level 3	Level 3	
15,535.5	10,661.5	15,681.2	10,769.6	149.3	150.2	-179.86	-345.9	5,108.8	108.2	28.8	79.34	1.363 Level 3	Level 3	
15,600.0	10,661.9	15,745.7	10,770.1	151.1	152.1	-179.86	-345.9	5,173.3	108.2	27.9	80.22	1.348 Level 3	Level 3	
15,672.4	10,662.4	15,818.1	10,770.6	153.2	154.2	-179.86	-345.9	5,245.8	108.2	27.0	81.21	1.332 Level 3	Level 3	
15,700.0	10,662.6	15,845.7	10,770.8	154.0	155.0	-179.86	-345.9	5,273.3	108.2	26.6	81.59	1.326 Level 3	Level 3	
15,800.0	10,663.3	15,945.7	10,771.5	156.9	157.9	-179.86	-345.9	5,373.3	108.2	25.2	82.96	1.304 Level 3	Level 3	
15,836.5	10,663.6	15,982.2	10,771.7	158.0	158.9	-179.86	-345.9	5,409.8	108.2	24.7	83.46	1.296 Level 3	Level 3	
15,900.0	10,664.0	16,045.7	10,772.2	159.8	160.8	-179.86	-345.9	5,473.3	108.2	23.8	84.33	1.283 Level 3	Level 3	
15,972.7	10,664.5	16,118.4	10,772.7	161.9	162.9	-179.86	-345.9	5,546.0	108.2	22.8	85.33	1.268 Level 3	Level 3	
16,000.0	10,664.7	16,145.7	10,772.9	162.7	163.7	-179.86	-345.9	5,573.3	108.2	22.5	85.71	1.262 Level 3	Level 3	
16,100.0	10,665.4	16,245.7	10,773.6	165.6	166.6	-179.86	-345.9	5,673.3	108.2	21.1	87.09	1.242 Level 2		
16,135.5	10,665.7	16,281.2	10,773.8	166.6	167.6	-179.86	-345.9	5,708.8	108.2	20.6	87.58	1.235 Level 2		
16,200.0	10,666.1	16,345.7	10,774.3	168.5	169.5	-179.86	-345.9	5,773.3	108.2	19.7	88.48	1.223 Level 2		
16,300.0	10,666.8	16,445.7	10,775.0	171.4	172.4	-179.86	-345.9	5,873.3	108.2	18.3	89.86	1.204 Level 2		
16,335.5	10,667.1	16,481.2	10,775.2	172.4	173.4	-179.86	-345.9	5,908.8	108.2	17.8	90.35	1.197 Level 2		
16,400.0	10,667.5	16,545.7	10,775.7	174.3	175.3	-179.86	-345.9	5,973.3	108.2	16.9	91.25	1.185 Level 2		
16,472.4	10,668.0	16,618.1	10,776.2	176.4	177.4	-179.86	-345.9	6,045.7	108.2	15.9	92.26	1.172 Level 2		
16,500.0	10,668.2	16,645.7	10,776.4	177.2	178.2	-179.86	-345.9	6,073.3	108.2	15.5	92.64	1.168 Level 2		
16,600.0	10,668.9	16,745.7	10,777.1	180.1	181.1	-179.86	-345.9	6,173.3	108.2	14.1	94.03	1.150 Level 2		
16,635.5	10,669.2	16,781.2	10,777.3	181.1	182.1	-179.86	-345.9	6,208.8	108.2	13.6	94.53	1.144 Level 2		
16,700.0	10,669.6	16,845.7	10,777.8	183.0	184.0	-179.86	-345.9	6,273.3	108.2	12.7	95.43	1.133 Level 2		
16,777.4	10,670.2	16,923.1	10,778.3	185.2	186.2	-179.86	-345.9	6,350.7	108.2	11.7	96.51	1.121 Level 2		
16,800.0	10,670.3	16,945.7	10,778.5	185.9	186.9	-179.86	-345.9	6,373.3	108.2	11.3	96.83	1.117 Level 2		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
16,900.0	10,671.0	17,045.7	10,779.2	188.8	189.8	-179.86	-345.9	6,473.3	108.2	9.9	98.23	1.101	Level 2	
16,935.5	10,671.3	17,081.2	10,779.4	189.8	190.8	-179.86	-345.9	6,508.8	108.2	9.4	98.72	1.096	Level 2	
17,000.0	10,671.7	17,145.7	10,779.9	191.7	192.7	-179.86	-345.9	6,573.3	108.2	8.5	99.63	1.086	Level 2	
17,078.2	10,672.3	17,223.9	10,780.4	194.0	194.9	-179.86	-345.9	6,651.5	108.2	7.4	100.73	1.074	Level 2	
17,100.0	10,672.4	17,245.7	10,780.6	194.6	195.6	-179.86	-345.9	6,673.3	108.2	7.1	101.03	1.071	Level 2	
17,200.0	10,673.1	17,345.7	10,781.3	197.5	198.5	-179.86	-345.9	6,773.3	108.2	5.7	102.44	1.056	Level 2	
17,235.5	10,673.4	17,381.2	10,781.5	198.5	199.5	-179.86	-345.9	6,808.8	108.2	5.2	102.94	1.051	Level 2	
17,300.0	10,673.8	17,445.7	10,782.0	200.4	201.4	-179.86	-345.9	6,873.3	108.2	4.3	103.84	1.042	Level 2	
17,377.4	10,674.3	17,523.1	10,782.5	202.7	203.6	-179.86	-345.9	6,950.7	108.2	3.2	104.93	1.031	Level 2	
17,400.0	10,674.5	17,545.7	10,782.7	203.3	204.3	-179.86	-345.9	6,973.3	108.2	2.9	105.25	1.028	Level 2	
17,500.0	10,675.2	17,645.7	10,783.4	206.2	207.2	-179.86	-345.9	7,073.3	108.2	1.5	106.66	1.014	Level 2	
17,535.5	10,675.4	17,681.2	10,783.6	207.3	208.2	-179.86	-345.9	7,108.8	108.2	1.0	107.16	1.009	Level 2	
17,600.0	10,675.9	17,745.7	10,784.1	209.1	210.1	-179.86	-345.9	7,173.3	108.2	0.1	108.07	1.001	Level 2	
17,672.7	10,676.4	17,818.4	10,784.6	211.3	212.2	-179.86	-345.9	7,245.9	108.2	-0.9	109.10	0.991	Level 1	
17,700.0	10,676.6	17,845.7	10,784.8	212.1	213.0	-179.86	-345.9	7,273.3	108.2	-1.3	109.49	0.988	Level 1	
17,800.0	10,677.3	17,945.7	10,785.5	215.0	215.9	-179.86	-345.9	7,373.3	108.2	-2.7	110.90	0.975	Level 1	
17,835.5	10,677.5	17,981.2	10,785.7	216.0	217.0	-179.86	-345.9	7,408.7	108.2	-3.2	111.40	0.971	Level 1	
17,900.0	10,678.0	18,045.7	10,786.2	217.9	218.8	-179.86	-345.9	7,473.3	108.2	-4.2	112.32	0.963	Level 1	
18,000.0	10,678.7	18,145.7	10,786.9	220.8	221.8	-179.86	-345.9	7,573.3	108.2	-5.6	113.73	0.951	Level 1	
18,035.5	10,678.9	18,181.2	10,787.1	221.8	222.8	-179.86	-345.9	7,608.7	108.2	-6.1	114.24	0.947	Level 1	
18,100.0	10,679.4	18,245.7	10,787.6	223.7	224.7	-179.86	-345.9	7,673.3	108.2	-7.0	115.15	0.939	Level 1	
18,172.4	10,679.9	18,318.1	10,788.1	225.8	226.8	-179.86	-345.9	7,745.7	108.2	-8.0	116.18	0.931	Level 1	
18,200.0	10,680.1	18,345.7	10,788.2	226.6	227.6	-179.86	-345.9	7,773.3	108.2	-8.4	116.57	0.928	Level 1	
18,300.0	10,680.8	18,445.7	10,788.9	229.5	230.5	-179.86	-345.9	7,873.3	108.2	-9.8	117.99	0.917	Level 1	
18,335.5	10,681.0	18,481.2	10,789.2	230.6	231.5	-179.86	-345.9	7,908.7	108.2	-10.3	118.50	0.913	Level 1	
18,400.0	10,681.5	18,545.7	10,789.6	232.4	233.4	-179.86	-345.9	7,973.2	108.2	-11.2	119.41	0.906	Level 1	
18,472.4	10,682.0	18,618.1	10,790.2	234.5	235.5	-179.86	-345.9	8,045.7	108.2	-12.3	120.44	0.898	Level 1	
18,500.0	10,682.2	18,645.7	10,790.3	235.3	236.3	-179.86	-345.9	8,073.2	108.2	-12.7	120.84	0.895	Level 1	
18,600.0	10,682.9	18,745.7	10,791.0	238.3	239.2	-179.86	-345.9	8,173.2	108.2	-14.1	122.26	0.885	Level 1	
18,646.4	10,683.2	18,792.1	10,791.4	239.6	240.6	-179.86	-345.9	8,219.7	108.2	-14.8	122.92	0.880	Level 1	
18,700.0	10,683.6	18,845.7	10,791.7	241.2	242.1	-179.86	-345.9	8,273.2	108.2	-15.5	123.69	0.875	Level 1	
18,773.0	10,684.1	18,918.7	10,792.2	243.3	244.3	-179.86	-345.9	8,346.2	108.2	-16.6	124.73	0.867	Level 1	
18,800.0	10,684.3	18,945.7	10,792.4	244.1	245.1	-179.86	-345.9	8,373.2	108.2	-16.9	125.11	0.865	Level 1	
18,900.0	10,685.0	19,045.7	10,793.1	247.0	248.0	-179.86	-345.9	8,473.2	108.2	-18.4	126.54	0.855	Level 1	
18,941.1	10,685.3	19,086.8	10,793.4	248.2	249.2	-179.86	-345.9	8,514.3	108.2	-19.0	127.12	0.851	Level 1	
19,000.0	10,685.7	19,145.7	10,793.8	249.9	250.9	-179.86	-345.9	8,573.2	108.2	-19.8	127.97	0.845	Level 1	
19,073.0	10,686.2	19,218.7	10,794.3	252.1	253.0	-179.86	-345.9	8,646.2	108.2	-20.8	129.01	0.838	Level 1	
19,100.0	10,686.4	19,245.7	10,794.5	252.8	253.8	-179.86	-345.9	8,673.2	108.2	-21.2	129.39	0.836	Level 1	
19,200.0	10,687.1	19,345.7	10,795.2	255.8	256.7	-179.86	-345.9	8,773.2	108.2	-22.7	130.82	0.827	Level 1	
19,244.5	10,687.4	19,390.2	10,795.5	257.1	258.0	-179.86	-345.9	8,817.7	108.2	-23.3	131.46	0.823	Level 1	
19,300.0	10,687.8	19,445.7	10,795.9	258.7	259.6	-179.86	-345.9	8,873.2	108.2	-24.1	132.25	0.818	Level 1	
19,400.0	10,688.5	19,545.7	10,796.6	261.6	262.6	-179.86	-345.9	8,973.2	108.2	-25.5	133.68	0.809	Level 1	
19,444.5	10,688.8	19,590.2	10,796.9	262.9	263.8	-179.86	-345.9	9,017.7	108.2	-26.2	134.32	0.805	Level 1	
19,500.0	10,689.2	19,645.7	10,797.3	264.5	265.5	-179.86	-345.9	9,073.2	108.2	-27.0	135.12	0.801	Level 1	
19,573.3	10,689.7	19,719.0	10,797.8	266.6	267.6	-179.86	-345.9	9,146.5	108.2	-28.0	136.17	0.794	Level 1	
19,600.0	10,689.9	19,745.7	10,798.0	267.4	268.4	-179.86	-345.9	9,173.2	108.2	-28.4	136.55	0.792	Level 1	
19,700.0	10,690.6	19,845.7	10,798.7	270.3	271.3	-179.86	-345.9	9,273.2	108.2	-29.8	137.98	0.784	Level 1	
19,744.5	10,690.9	19,890.2	10,799.0	271.6	272.6	-179.86	-345.9	9,317.7	108.2	-30.5	138.62	0.780	Level 1	
19,800.0	10,691.3	19,945.7	10,799.4	273.3	274.2	-179.86	-345.9	9,373.2	108.2	-31.3	139.42	0.776	Level 1	
19,878.3	10,691.8	20,024.0	10,800.0	275.5	276.5	-179.86	-345.9	9,451.5	108.2	-32.4	140.54	0.770	Level 1	
19,900.0	10,692.0	20,045.7	10,800.1	276.2	277.1	-179.86	-345.9	9,473.2	108.2	-32.7	140.85	0.768	Level 1	
20,000.0	10,692.7	20,145.7	10,800.8	279.1	280.1	-179.86	-345.9	9,573.2	108.2	-34.1	142.29	0.760	Level 1	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5300 44-24 4T2R - Chalmers 5300 44-24 4T2R - Design #1												Offset Site Error:	0.0 usft	
Survey Program: 0-MWD												Offset Well Error:	2.0 usft	
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Centre +E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
20,039.5	10,692.9	20,185.2	10,801.1	280.2	281.2	-179.86	-345.9	9,612.7	108.2	-34.7	142.85	0.757 Level 1		
20,100.0	10,693.4	20,245.7	10,801.5	282.0	283.0	-179.86	-345.9	9,673.2	108.2	-35.6	143.72	0.753 Level 1		
20,178.3	10,693.9	20,324.0	10,802.1	284.3	285.3	-179.86	-345.9	9,751.5	108.2	-36.7	144.85	0.747 Level 1		
20,200.0	10,694.0	20,345.7	10,802.2	284.9	285.9	-179.86	-345.9	9,773.2	108.2	-37.0	145.16	0.745 Level 1		
20,300.0	10,694.7	20,445.7	10,802.9	287.9	288.8	-179.86	-345.9	9,873.2	108.2	-38.4	146.60	0.738 Level 1		
20,339.5	10,695.0	20,485.2	10,803.2	289.0	290.0	-179.86	-345.9	9,912.7	108.2	-39.0	147.16	0.735 Level 1		
20,400.0	10,695.4	20,545.7	10,803.6	290.8	291.7	-179.86	-345.9	9,973.2	108.2	-39.9	148.03	0.731 Level 1		
20,478.3	10,696.0	20,624.0	10,804.2	293.1	294.0	-179.86	-345.9	10,051.5	108.2	-41.0	149.16	0.725 Level 1		
20,500.0	10,696.1	20,645.7	10,804.3	293.7	294.7	-179.86	-345.9	10,073.2	108.2	-41.3	149.47	0.724 Level 1		
20,600.0	10,696.8	20,745.7	10,805.0	296.6	297.6	-179.86	-345.9	10,173.2	108.2	-42.7	150.91	0.717 Level 1		
20,641.1	10,697.1	20,786.8	10,805.3	297.8	298.8	-179.86	-345.9	10,214.3	108.2	-43.3	151.50	0.714 Level 1		
20,700.0	10,697.5	20,845.7	10,805.7	299.5	300.5	-179.86	-345.9	10,273.2	108.2	-44.2	152.35	0.710 Level 1		
20,800.0	10,698.2	20,945.7	10,806.4	302.5	303.4	-179.86	-345.9	10,373.2	108.2	-45.6	153.79	0.703 Level 1		
20,887.1	10,698.8	21,032.8	10,807.0	305.0	306.0	-179.86	-345.9	10,460.3	108.2	-46.9	155.04	0.698 Level 1		
20,900.0	10,698.9	21,045.7	10,807.1	305.4	306.3	-179.86	-345.9	10,473.2	108.2	-47.1	155.23	0.697 Level 1		
20,960.1	10,699.4	21,105.8	10,807.5	307.1	308.1	-179.86	-345.9	10,533.2	108.2	-47.9	156.09	0.693 Level 1, ES, SF		

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
0.0	0.0	0.0	0.0	0.0	2.0	-180.00	-82.1	0.0	82.1				
100.0	100.0	100.0	100.0	0.1	2.0	-180.00	-82.1	0.0	82.1	80.0	2.09	39.346	
200.0	200.0	200.0	200.0	0.3	2.0	-180.00	-82.1	0.0	82.1	79.7	2.33	35.185	
300.0	300.0	300.0	300.0	0.5	2.1	-180.00	-82.1	0.0	82.1	79.5	2.60	31.522	
400.0	400.0	400.0	400.0	0.8	2.1	-180.00	-82.1	0.0	82.1	79.2	2.90	28.326	
500.0	500.0	500.0	500.0	1.0	2.2	-180.00	-82.1	0.0	82.1	78.9	3.21	25.554	
600.0	600.0	600.0	600.0	1.2	2.3	-180.00	-82.1	0.0	82.1	78.5	3.54	23.155	
700.0	700.0	700.0	700.0	1.4	2.5	-180.00	-82.1	0.0	82.1	78.2	3.89	21.083	
800.0	800.0	800.0	800.0	1.7	2.6	-180.00	-82.1	0.0	82.1	77.8	4.26	19.289	
900.0	900.0	900.0	900.0	1.9	2.7	-180.00	-82.1	0.0	82.1	77.4	4.63	17.732	
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.9	-180.00	-82.1	0.0	82.1	77.1	5.01	16.375	
1,100.0	1,100.0	1,100.0	1,100.0	2.3	3.1	-180.00	-82.1	0.0	82.1	76.7	5.40	15.188	
1,200.0	1,200.0	1,200.0	1,200.0	2.6	3.2	-180.00	-82.1	0.0	82.1	76.3	5.80	14.145	
1,300.0	1,300.0	1,300.0	1,300.0	2.8	3.4	-180.00	-82.1	0.0	82.1	75.9	6.21	13.223	
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.6	-180.00	-82.1	0.0	82.1	75.5	6.62	12.404	
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.8	-180.00	-82.1	0.0	82.1	75.0	7.03	11.674	
1,600.0	1,600.0	1,600.0	1,600.0	3.5	4.0	-180.00	-82.1	0.0	82.1	74.6	7.45	11.019	
1,700.0	1,700.0	1,700.0	1,700.0	3.7	4.2	-180.00	-82.1	0.0	82.1	74.2	7.87	10.430	
1,800.0	1,800.0	1,800.0	1,800.0	3.9	4.4	-180.00	-82.1	0.0	82.1	73.8	8.29	9.897	
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.6	-180.00	-82.1	0.0	82.1	73.4	8.72	9.414	
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.8	-180.00	-82.1	0.0	82.1	72.9	9.15	8.973	
2,100.0	2,100.0	2,100.0	2,100.0	4.6	5.0	-180.00	-82.1	0.0	82.1	72.5	9.58	8.570	
2,200.0	2,200.0	2,200.0	2,200.0	4.8	5.2	-180.00	-82.1	0.0	82.1	72.1	10.01	8.201 CC	
2,216.7	2,216.7	2,216.0	2,216.0	4.8	5.2	90.05	-82.1	0.0	82.1	72.1	10.07	8.156	
2,300.0	2,300.0	2,299.3	2,299.3	5.0	5.4	90.55	-82.9	0.0	82.9	72.5	10.37	7.995	
2,400.0	2,400.0	2,399.3	2,399.3	5.2	5.5	91.14	-83.7	0.0	83.8	73.0	10.73	7.810	
2,500.0	2,500.0	2,499.3	2,499.3	5.4	5.7	91.72	-84.6	0.0	84.7	73.6	11.09	7.633	
2,600.0	2,600.0	2,599.3	2,599.2	5.6	5.9	92.29	-85.5	0.0	85.6	74.1	11.46	7.465	
2,700.0	2,700.0	2,699.2	2,699.2	5.8	6.0	92.84	-86.4	0.0	86.5	74.6	11.84	7.304	
2,800.0	2,800.0	2,799.2	2,799.2	6.0	6.2	93.38	-87.2	0.0	87.4	75.2	12.22	7.151	
2,900.0	2,900.0	2,899.2	2,899.2	6.2	6.4	93.91	-88.1	0.0	88.3	75.7	12.61	7.006	
3,000.0	3,000.0	2,999.2	2,999.2	6.4	6.6	94.44	-89.0	0.0	89.3	76.3	13.00	6.867	
3,100.0	3,100.0	3,099.2	3,099.2	6.6	6.7	94.94	-89.9	0.0	90.2	76.8	13.39	6.735	
3,200.0	3,200.0	3,199.2	3,199.2	6.9	6.9	95.44	-90.7	0.0	91.1	77.4	13.79	6.610	
3,300.0	3,300.0	3,299.2	3,299.2	7.1	7.1	95.93	-91.6	0.0	92.1	77.9	14.19	6.491	
3,400.0	3,400.0	3,399.2	3,399.1	7.3	7.3	96.41	-92.5	0.0	93.1	78.5	14.59	6.377	
3,500.0	3,500.0	3,499.2	3,499.1	7.5	7.5	96.88	-93.3	0.0	94.0	79.0	15.00	6.269	
3,600.0	3,599.9	3,599.2	3,599.1	7.7	7.7	97.34	-94.2	0.0	95.0	79.6	15.41	6.166	
3,700.0	3,699.9	3,699.2	3,699.1	7.9	7.9	97.79	-95.1	0.0	96.0	80.2	15.82	6.068	
3,800.0	3,799.9	3,799.2	3,799.1	8.1	8.1	98.23	-96.0	0.0	97.0	80.7	16.23	5.975	
3,900.0	3,899.9	3,899.2	3,899.1	8.4	8.3	98.66	-96.8	0.0	98.0	81.3	16.64	5.885	
4,000.0	3,999.9	3,999.1	3,999.1	8.6	8.5	99.09	-97.7	0.0	99.0	81.9	17.06	5.800	
4,100.0	4,099.9	4,099.1	4,099.1	8.8	8.7	99.50	-98.6	0.0	100.0	82.5	17.48	5.719	
4,200.0	4,199.9	4,199.1	4,199.1	9.0	8.9	99.91	-99.5	0.0	101.0	83.1	17.90	5.641	
4,300.0	4,299.9	4,299.1	4,299.0	9.2	9.1	100.31	-100.3	0.0	102.0	83.7	18.32	5.567	
4,400.0	4,399.9	4,399.1	4,399.0	9.4	9.3	100.70	-101.2	0.0	103.0	84.3	18.74	5.495	
4,500.0	4,499.9	4,499.1	4,499.0	9.7	9.5	101.08	-102.1	0.0	104.0	84.8	19.17	5.427	
4,600.0	4,599.9	4,599.1	4,599.0	9.9	9.7	101.46	-102.9	0.0	105.0	85.4	19.59	5.362	
4,700.0	4,699.9	4,699.1	4,699.0	10.1	9.9	101.82	-103.8	0.0	106.1	86.1	20.02	5.299	
4,800.0	4,799.9	4,799.1	4,799.0	10.3	10.1	102.19	-104.7	0.0	107.1	86.7	20.44	5.239	
4,900.0	4,899.9	4,899.1	4,899.0	10.5	10.3	102.54	-105.6	0.0	108.1	87.3	20.87	5.181	
5,000.0	4,999.9	4,999.1	4,999.0	10.8	10.6	102.89	-106.4	0.0	109.2	87.9	21.30	5.126	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Distance								Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
5,100.0	5,099.9	5,099.1	5,099.0	11.0	10.8	103.23	-107.3	0.0	110.2	88.5	21.73	5.073		
5,200.0	5,199.9	5,199.1	5,198.9	11.2	11.0	103.56	-108.2	0.0	111.3	89.1	22.16	5.022		
5,300.0	5,299.9	5,299.0	5,298.9	11.4	11.2	103.89	-109.0	0.0	112.3	89.7	22.59	4.972		
5,400.0	5,399.9	5,399.0	5,398.9	11.6	11.4	104.21	-109.9	0.0	113.4	90.4	23.03	4.925		
5,500.0	5,499.9	5,499.0	5,498.9	11.9	11.6	104.53	-110.8	0.0	114.5	91.0	23.46	4.879		
5,600.0	5,599.9	5,599.0	5,598.9	12.1	11.8	104.84	-111.7	0.0	115.5	91.6	23.89	4.835		
5,700.0	5,699.9	5,699.0	5,698.9	12.3	12.0	105.14	-112.5	0.0	116.6	92.3	24.33	4.793		
5,800.0	5,799.9	5,799.0	5,798.9	12.5	12.3	105.44	-113.4	0.0	117.7	92.9	24.76	4.752		
5,900.0	5,899.9	5,899.0	5,898.9	12.7	12.5	105.74	-114.3	0.0	118.7	93.5	25.20	4.713		
6,000.0	5,999.9	5,999.0	5,998.9	13.0	12.7	106.03	-115.2	0.0	119.8	94.2	25.63	4.675		
6,100.0	6,099.9	6,099.0	6,098.8	13.2	12.9	106.31	-116.0	0.0	120.9	94.8	26.07	4.638		
6,200.0	6,199.8	6,199.0	6,198.8	13.4	13.1	106.59	-116.9	0.0	122.0	95.5	26.51	4.602		
6,300.0	6,299.8	6,299.0	6,298.8	13.6	13.3	106.86	-117.8	0.0	123.1	96.1	26.94	4.568		
6,400.0	6,399.8	6,399.0	6,398.8	13.9	13.6	107.13	-118.6	0.0	124.2	96.8	27.38	4.535		
6,500.0	6,499.8	6,499.0	6,498.8	14.1	13.8	107.39	-119.5	0.0	125.3	97.4	27.82	4.503		
6,600.0	6,599.8	6,598.9	6,598.8	14.3	14.0	107.65	-120.4	0.0	126.4	98.1	28.26	4.471		
6,700.0	6,699.8	6,698.9	6,698.8	14.5	14.2	107.91	-121.3	0.0	127.4	98.8	28.70	4.441		
6,800.0	6,799.8	6,798.9	6,798.8	14.7	14.4	108.16	-122.1	0.0	128.5	99.4	29.14	4.412		
6,900.0	6,899.8	6,898.9	6,898.7	15.0	14.6	108.40	-123.0	0.0	129.7	100.1	29.58	4.384		
7,000.0	6,999.8	6,998.9	6,998.7	15.2	14.9	108.65	-123.9	0.0	130.8	100.7	30.02	4.356		
7,100.0	7,099.8	7,098.9	7,098.7	15.4	15.1	108.88	-124.8	0.0	131.9	101.4	30.46	4.330		
7,200.0	7,199.8	7,198.9	7,198.7	15.6	15.3	109.12	-125.6	0.0	133.0	102.1	30.90	4.304		
7,300.0	7,299.8	7,298.9	7,298.7	15.9	15.5	109.35	-126.5	0.0	134.1	102.7	31.34	4.279		
7,400.0	7,399.8	7,398.9	7,398.7	16.1	15.7	109.58	-127.4	0.0	135.2	103.4	31.78	4.254		
7,500.0	7,499.8	7,498.9	7,498.7	16.3	16.0	109.80	-128.2	0.0	136.3	104.1	32.22	4.231		
7,600.0	7,599.8	7,598.9	7,598.7	16.5	16.2	110.02	-129.1	0.0	137.4	104.8	32.66	4.208		
7,700.0	7,699.8	7,698.9	7,698.7	16.7	16.4	110.23	-130.0	0.0	138.5	105.4	33.10	4.185		
7,800.0	7,799.8	7,798.9	7,798.6	17.0	16.6	110.44	-130.9	0.0	139.7	106.1	33.55	4.164		
7,900.0	7,899.8	7,898.9	7,898.6	17.2	16.8	110.65	-131.7	0.0	140.8	106.8	33.99	4.142		
7,926.7	7,926.4	7,925.5	7,925.3	17.3	16.9	110.71	-132.0	0.0	141.1	107.0	34.11	4.137		
7,943.3	7,943.1	7,943.3	7,943.1	17.3	16.9	-159.27	-132.1	0.0	141.2	107.0	34.18	4.131		
8,000.0	7,999.8	8,000.0	7,999.8	17.4	17.1	-159.27	-132.1	0.0	141.2	106.8	34.41	4.103		
8,100.0	8,099.8	8,100.0	8,099.8	17.6	17.2	-159.27	-132.1	0.0	141.2	106.4	34.81	4.056		
8,200.0	8,199.8	8,200.0	8,199.8	17.8	17.4	-159.27	-132.1	0.0	141.2	106.0	35.21	4.010		
8,300.0	8,299.8	8,300.0	8,299.8	18.0	17.6	-159.27	-132.1	0.0	141.2	105.6	35.61	3.965		
8,400.0	8,399.8	8,400.0	8,399.8	18.2	17.8	-159.27	-132.1	0.0	141.2	105.2	36.01	3.921		
8,500.0	8,499.8	8,500.0	8,499.8	18.5	18.0	-159.27	-132.1	0.0	141.2	104.8	36.42	3.877		
8,600.0	8,599.8	8,600.0	8,599.8	18.7	18.2	-159.27	-132.1	0.0	141.2	104.4	36.82	3.834		
8,700.0	8,699.8	8,700.0	8,699.8	18.9	18.4	-159.27	-132.1	0.0	141.2	104.0	37.23	3.793		
8,800.0	8,799.8	8,800.0	8,799.8	19.1	18.6	-159.27	-132.1	0.0	141.2	103.6	37.63	3.752		
8,900.0	8,899.8	8,900.0	8,899.8	19.3	18.8	-159.27	-132.1	0.0	141.2	103.2	38.04	3.712		
9,000.0	8,999.8	9,000.0	8,999.8	19.5	19.0	-159.27	-132.1	0.0	141.2	102.7	38.45	3.672		
9,100.0	9,099.8	9,100.0	9,099.8	19.7	19.2	-159.27	-132.1	0.0	141.2	102.3	38.86	3.633		
9,200.0	9,199.8	9,200.0	9,199.8	19.9	19.4	-159.27	-132.1	0.0	141.2	101.9	39.27	3.595		
9,300.0	9,299.8	9,300.0	9,299.8	20.1	19.6	-159.27	-132.1	0.0	141.2	101.5	39.68	3.558		
9,400.0	9,399.8	9,400.0	9,399.8	20.4	19.8	-159.27	-132.1	0.0	141.2	101.1	40.09	3.522		
9,500.0	9,499.8	9,500.0	9,499.8	20.6	20.0	-159.27	-132.1	0.0	141.2	100.7	40.51	3.486		
9,600.0	9,599.8	9,600.0	9,599.8	20.8	20.2	-159.27	-132.1	0.0	141.2	100.3	40.92	3.450		
9,700.0	9,699.8	9,700.0	9,699.8	21.0	20.4	-159.27	-132.1	0.0	141.2	99.9	41.33	3.416		
9,800.0	9,799.8	9,800.0	9,799.8	21.2	20.6	-159.27	-132.1	0.0	141.2	99.4	41.75	3.382		
9,900.0	9,899.8	9,900.0	9,899.8	21.4	20.8	-159.27	-132.1	0.0	141.2	99.0	42.16	3.349		
10,000.0	9,999.8	10,000.0	9,999.8	21.6	21.0	-159.27	-132.1	0.0	141.2	98.6	42.58	3.316		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,100.0	10,099.8	10,100.0	10,099.8	21.9	21.2	-159.27	-132.1	0.0	141.2	98.2	43.00	3.284		
10,151.8	10,151.6	10,151.8	10,151.6	22.0	21.3	-159.27	-132.1	0.0	141.2	98.0	43.21	3.267		
10,175.0	10,174.8	10,175.0	10,174.8	22.0	21.3	90.51	-132.1	0.0	141.2	97.9	43.33	3.259		
10,200.0	10,199.7	10,199.9	10,199.7	22.1	21.4	91.26	-132.1	0.0	141.2	97.8	43.44	3.251		
10,225.0	10,224.5	10,224.7	10,224.5	22.1	21.4	92.52	-132.1	0.0	141.3	97.8	43.55	3.246		
10,250.0	10,249.1	10,248.1	10,247.9	22.2	21.5	93.96	-132.4	0.4	141.7	98.0	43.65	3.246		
10,275.0	10,273.4	10,271.6	10,271.3	22.3	21.5	95.38	-133.3	1.8	142.5	98.7	43.76	3.256		
10,300.0	10,297.4	10,295.2	10,294.8	22.3	21.6	96.78	-135.0	4.1	143.7	99.8	43.86	3.276		
10,325.0	10,321.0	10,319.0	10,318.2	22.4	21.6	98.13	-137.3	7.4	145.3	101.4	43.97	3.306		
10,350.0	10,344.1	10,343.0	10,341.6	22.5	21.7	99.43	-140.3	11.7	147.4	103.3	44.06	3.345		
10,375.0	10,366.7	10,367.1	10,364.8	22.5	21.8	100.66	-144.0	17.0	149.9	105.7	44.16	3.394		
10,400.0	10,388.8	10,391.4	10,387.9	22.6	21.8	101.81	-148.4	23.3	152.7	108.5	44.25	3.452		
10,425.0	10,410.1	10,415.8	10,410.6	22.7	21.9	102.88	-153.5	30.5	156.0	111.7	44.33	3.519		
10,450.0	10,430.8	10,440.4	10,433.1	22.8	21.9	103.85	-159.3	38.8	159.7	115.2	44.41	3.595		
10,475.0	10,450.7	10,465.2	10,455.1	22.9	22.0	104.73	-165.8	48.1	163.7	119.2	44.50	3.678		
10,500.0	10,469.7	10,490.1	10,476.6	23.0	22.1	105.51	-173.1	58.3	168.0	123.4	44.58	3.768		
10,525.0	10,487.9	10,515.2	10,497.6	23.1	22.1	106.19	-181.0	69.6	172.7	128.0	44.68	3.864		
10,550.0	10,505.2	10,540.5	10,518.0	23.3	22.2	106.77	-189.6	81.8	177.6	132.8	44.79	3.966		
10,575.0	10,521.5	10,565.9	10,537.7	23.4	22.3	107.25	-198.8	95.0	182.9	137.9	44.92	4.071		
10,600.0	10,536.8	10,591.6	10,556.6	23.6	22.4	107.62	-208.7	109.1	188.4	143.3	45.07	4.179		
10,625.0	10,551.0	10,617.4	10,574.7	23.7	22.5	107.91	-219.3	124.2	194.1	148.8	45.25	4.289		
10,650.0	10,564.2	10,643.4	10,591.9	23.9	22.7	108.10	-230.5	140.1	200.1	154.6	45.47	4.400		
10,675.0	10,576.2	10,669.5	10,608.1	24.1	22.8	108.20	-242.3	156.9	206.2	160.5	45.72	4.510		
10,700.0	10,587.0	10,695.9	10,623.3	24.3	23.0	108.22	-254.7	174.5	212.5	166.5	46.02	4.618		
10,725.0	10,596.7	10,722.5	10,637.4	24.6	23.1	108.16	-267.7	193.0	219.0	172.6	46.36	4.723		
10,750.0	10,605.1	10,749.2	10,650.3	24.8	23.3	108.03	-281.1	212.1	225.6	178.8	46.75	4.825		
10,775.0	10,612.3	10,776.2	10,661.9	25.1	23.6	107.83	-295.1	232.0	232.3	185.1	47.19	4.922		
10,800.0	10,618.2	10,803.4	10,672.2	25.4	23.8	107.56	-309.6	252.6	239.1	191.4	47.69	5.013		
10,825.0	10,622.9	10,830.8	10,681.2	25.7	24.1	107.23	-324.5	273.8	245.9	197.7	48.23	5.099		
10,850.0	10,626.2	10,858.4	10,688.7	26.0	24.4	106.84	-339.7	295.5	252.8	204.0	48.81	5.179		
10,875.0	10,628.3	10,886.2	10,694.7	26.4	24.7	106.41	-355.4	317.8	259.7	210.3	49.45	5.252		
10,898.5	10,629.0	10,912.5	10,698.9	26.7	25.0	105.96	-370.3	339.0	266.2	216.1	50.08	5.315		
10,900.0	10,629.0	10,914.2	10,699.1	26.7	25.0	105.98	-371.3	340.4	266.6	216.5	50.11	5.320		
11,000.0	10,629.7	11,024.4	10,703.4	28.2	26.4	105.21	-434.1	430.6	293.1	240.2	52.92	5.538		
11,100.0	10,630.4	11,131.8	10,704.3	29.9	28.1	103.98	-492.6	520.8	319.1	263.0	56.16	5.683		
11,200.0	10,631.2	11,240.5	10,705.2	31.8	30.0	102.94	-548.3	614.1	345.0	285.3	59.72	5.777		
11,300.0	10,631.9	11,350.5	10,706.1	33.9	32.1	102.06	-601.0	710.6	370.7	307.2	63.57	5.832		
11,400.0	10,632.6	11,461.6	10,707.0	36.0	34.4	101.30	-650.4	810.2	396.1	328.5	67.63	5.858		
11,500.0	10,633.3	11,574.1	10,707.9	38.3	36.8	100.64	-696.5	912.8	421.2	349.3	71.86	5.861		
11,600.0	10,634.0	11,687.9	10,708.8	40.6	39.4	100.07	-739.0	1,018.3	445.9	369.7	76.22	5.850		
11,700.0	10,634.7	11,803.0	10,709.8	43.0	42.2	99.57	-777.6	1,126.7	470.1	389.4	80.68	5.827		
11,800.0	10,635.4	11,919.4	10,710.7	45.5	45.0	99.13	-812.3	1,237.8	493.9	408.7	85.22	5.796		
11,900.0	10,636.1	12,037.2	10,711.7	47.9	47.9	98.73	-842.7	1,351.6	517.1	427.3	89.78	5.759		
11,920.9	10,636.3	12,062.0	10,711.9	48.4	48.5	98.66	-848.5	1,375.7	521.8	431.1	90.74	5.751		
12,000.0	10,636.8	12,156.6	10,712.7	50.4	50.9	98.34	-868.8	1,468.1	538.6	443.5	95.12	5.662		
12,100.0	10,637.5	12,278.0	10,713.7	53.0	54.0	98.03	-890.3	1,587.5	556.1	455.3	100.81	5.517		
12,200.0	10,638.2	12,401.0	10,714.7	55.5	57.1	97.81	-906.8	1,709.5	569.5	462.9	106.63	5.341		
12,300.0	10,638.9	12,525.3	10,715.7	58.2	60.3	97.67	-918.3	1,833.2	578.7	466.1	112.57	5.140		
12,400.0	10,639.6	12,650.3	10,716.7	60.8	63.5	97.60	-924.3	1,958.1	583.5	464.9	118.58	4.921		
12,500.0	10,640.3	12,765.8	10,717.6	63.5	66.4	97.60	-925.4	2,073.5	584.3	459.9	124.36	4.698		
12,600.0	10,641.0	12,865.8	10,718.4	66.2	69.0	97.61	-925.4	2,173.5	584.3	454.6	129.77	4.503		
12,700.0	10,641.7	12,965.8	10,719.1	68.9	71.6	97.62	-925.4	2,273.5	584.3	449.1	135.21	4.322		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Distance								Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
12,800.0	10,642.4	13,065.8	10,719.9	71.6	74.2	97.63	-925.4	2,373.5	584.3	443.7	140.68	4.154		
12,900.0	10,643.1	13,165.8	10,720.7	74.4	76.9	97.63	-925.4	2,473.5	584.4	438.2	146.18	3.998		
13,000.0	10,643.8	13,265.8	10,721.5	77.1	79.6	97.64	-925.4	2,573.5	584.4	432.7	151.69	3.852		
13,100.0	10,644.5	13,365.8	10,722.3	79.9	82.3	97.65	-925.4	2,673.5	584.4	427.1	157.24	3.717		
13,200.0	10,645.2	13,465.8	10,723.1	82.7	85.0	97.66	-925.4	2,773.5	584.4	421.6	162.79	3.590		
13,300.0	10,645.9	13,565.8	10,723.9	85.5	87.7	97.67	-925.4	2,873.5	584.4	416.0	168.37	3.471		
13,400.0	10,646.6	13,665.8	10,724.6	88.3	90.4	97.68	-925.4	2,973.5	584.4	410.5	173.96	3.359		
13,500.0	10,647.3	13,765.8	10,725.4	91.1	93.1	97.68	-925.4	3,073.5	584.4	404.9	179.57	3.255		
13,600.0	10,648.0	13,865.8	10,726.2	93.9	95.9	97.69	-925.4	3,173.5	584.4	399.3	185.18	3.156		
13,700.0	10,648.7	13,965.8	10,727.0	96.7	98.7	97.70	-925.4	3,273.5	584.4	393.6	190.81	3.063		
13,800.0	10,649.4	14,065.8	10,727.8	99.5	101.4	97.71	-925.4	3,373.5	584.5	388.0	196.45	2.975		
13,900.0	10,650.1	14,165.8	10,728.6	102.4	104.2	97.72	-925.4	3,473.5	584.5	382.4	202.10	2.892		
14,000.0	10,650.8	14,265.8	10,729.3	105.2	107.0	97.73	-925.4	3,573.5	584.5	376.7	207.76	2.813		
14,100.0	10,651.5	14,365.8	10,730.1	108.1	109.8	97.74	-925.4	3,673.5	584.5	371.1	213.43	2.739		
14,200.0	10,652.2	14,465.8	10,730.9	110.9	112.6	97.74	-925.4	3,773.5	584.5	365.4	219.10	2.668		
14,300.0	10,652.9	14,565.8	10,731.7	113.8	115.4	97.75	-925.4	3,873.5	584.5	359.7	224.78	2.600		
14,400.0	10,653.6	14,665.8	10,732.5	116.6	118.2	97.76	-925.4	3,973.5	584.5	354.1	230.46	2.536		
14,500.0	10,654.3	14,765.8	10,733.3	119.5	121.1	97.77	-925.4	4,073.5	584.5	348.4	236.16	2.475		
14,600.0	10,655.0	14,865.8	10,734.1	122.4	123.9	97.78	-925.4	4,173.5	584.6	342.7	241.85	2.417		
14,700.0	10,655.7	14,965.8	10,734.8	125.2	126.7	97.79	-925.4	4,273.5	584.6	337.0	247.55	2.361		
14,800.0	10,656.3	15,065.8	10,735.6	128.1	129.6	97.79	-925.4	4,373.5	584.6	331.3	253.26	2.308		
14,900.0	10,657.0	15,165.8	10,736.4	131.0	132.4	97.80	-925.4	4,473.5	584.6	325.6	258.97	2.257		
15,000.0	10,657.7	15,265.8	10,737.2	133.8	135.3	97.81	-925.4	4,573.5	584.6	319.9	264.69	2.209		
15,100.0	10,658.4	15,365.8	10,738.0	136.7	138.1	97.82	-925.4	4,673.5	584.6	314.2	270.41	2.162		
15,200.0	10,659.1	15,465.8	10,738.8	139.6	141.0	97.83	-925.4	4,773.5	584.6	308.5	276.13	2.117		
15,300.0	10,659.8	15,565.8	10,739.6	142.5	143.8	97.84	-925.4	4,873.5	584.6	302.8	281.85	2.074		
15,400.0	10,660.5	15,665.8	10,740.3	145.4	146.7	97.85	-925.4	4,973.4	584.6	297.1	287.58	2.033		
15,500.0	10,661.2	15,765.8	10,741.1	148.2	149.5	97.85	-925.4	5,073.4	584.7	291.3	293.31	1.993		
15,600.0	10,661.9	15,865.8	10,741.9	151.1	152.4	97.86	-925.4	5,173.4	584.7	285.6	299.05	1.955		
15,700.0	10,662.6	15,965.8	10,742.7	154.0	155.3	97.87	-925.4	5,273.4	584.7	279.9	304.79	1.918		
15,800.0	10,663.3	16,065.8	10,743.5	156.9	158.1	97.88	-925.4	5,373.4	584.7	274.2	310.52	1.883		
15,900.0	10,664.0	16,165.8	10,744.3	159.8	161.0	97.89	-925.4	5,473.4	584.7	268.4	316.27	1.849		
16,000.0	10,664.7	16,265.8	10,745.1	162.7	163.9	97.90	-925.4	5,573.4	584.7	262.7	322.01	1.816		
16,100.0	10,665.4	16,365.8	10,745.8	165.6	166.7	97.91	-925.4	5,673.4	584.7	257.0	327.75	1.784		
16,200.0	10,666.1	16,465.8	10,746.6	168.5	169.6	97.91	-925.4	5,773.4	584.7	251.2	333.50	1.753		
16,300.0	10,666.8	16,565.8	10,747.4	171.4	172.5	97.92	-925.4	5,873.4	584.8	245.5	339.25	1.724		
16,400.0	10,667.5	16,665.8	10,748.2	174.3	175.4	97.93	-925.4	5,973.4	584.8	239.8	345.00	1.695		
16,500.0	10,668.2	16,765.8	10,749.0	177.2	178.3	97.94	-925.4	6,073.4	584.8	234.0	350.75	1.667		
16,600.0	10,668.9	16,865.8	10,749.8	180.1	181.1	97.95	-925.4	6,173.4	584.8	228.3	356.51	1.640		
16,700.0	10,669.6	16,965.8	10,750.6	183.0	184.0	97.96	-925.4	6,273.4	584.8	222.5	362.26	1.614		
16,800.0	10,670.3	17,065.8	10,751.3	185.9	186.9	97.96	-925.4	6,373.4	584.8	216.8	368.02	1.589		
16,900.0	10,671.0	17,165.8	10,752.1	188.8	189.8	97.97	-925.4	6,473.4	584.8	211.1	373.78	1.565		
17,000.0	10,671.7	17,265.8	10,752.9	191.7	192.7	97.98	-925.4	6,573.4	584.8	205.3	379.54	1.541		
17,100.0	10,672.4	17,365.8	10,753.7	194.6	195.6	97.99	-925.4	6,673.4	584.9	199.6	385.30	1.518		
17,200.0	10,673.1	17,465.8	10,754.5	197.5	198.5	98.00	-925.4	6,773.4	584.9	193.8	391.06	1.496 Level 3	Level 3	
17,300.0	10,673.8	17,565.8	10,755.3	200.4	201.4	98.01	-925.4	6,873.4	584.9	188.1	396.82	1.474 Level 3	Level 3	
17,400.0	10,674.5	17,665.8	10,756.1	203.3	204.3	98.02	-925.4	6,973.4	584.9	182.3	402.58	1.453 Level 3	Level 3	
17,500.0	10,675.2	17,765.8	10,756.8	206.2	207.2	98.02	-925.4	7,073.4	584.9	176.6	408.34	1.432 Level 3	Level 3	
17,600.0	10,675.9	17,865.8	10,757.6	209.1	210.1	98.03	-925.4	7,173.4	584.9	170.8	414.11	1.412 Level 3	Level 3	
17,700.0	10,676.6	17,965.8	10,758.4	212.1	212.9	98.04	-925.4	7,273.4	584.9	165.1	419.88	1.393 Level 3	Level 3	
17,800.0	10,677.3	18,065.8	10,759.2	215.0	215.8	98.05	-925.4	7,373.4	584.9	159.3	425.64	1.374 Level 3	Level 3	
17,900.0	10,678.0	18,165.8	10,760.0	217.9	218.7	98.06	-925.4	7,473.4	585.0	153.5	431.41	1.356 Level 3	Level 3	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance						
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
18,000.0	10,678.7	18,265.8	10,760.8	220.8	221.6	98.07	-925.4	7,573.4	585.0	147.8	437.18	1.338 Level 3	Level 3
18,100.0	10,679.4	18,365.8	10,761.5	223.7	224.5	98.07	-925.4	7,673.4	585.0	142.0	442.94	1.321 Level 3	Level 3
18,200.0	10,680.1	18,465.8	10,762.3	226.6	227.4	98.08	-925.4	7,773.4	585.0	136.3	448.71	1.304 Level 3	Level 3
18,300.0	10,680.8	18,565.8	10,763.1	229.5	230.3	98.09	-925.4	7,873.4	585.0	130.5	454.48	1.287 Level 3	Level 3
18,400.0	10,681.5	18,665.8	10,763.9	232.4	233.3	98.10	-925.4	7,973.4	585.0	124.8	460.25	1.271 Level 3	Level 3
18,500.0	10,682.2	18,765.8	10,764.7	235.3	236.2	98.11	-925.4	8,073.4	585.0	119.0	466.02	1.255 Level 3	Level 3
18,600.0	10,682.9	18,865.8	10,765.5	238.3	239.1	98.12	-925.4	8,173.4	585.0	113.2	471.79	1.240 Level 2	
18,700.0	10,683.6	18,965.8	10,766.3	241.2	242.0	98.13	-925.4	8,273.3	585.0	107.5	477.57	1.225 Level 2	
18,800.0	10,684.3	19,065.8	10,767.0	244.1	244.9	98.13	-925.4	8,373.3	585.1	101.7	483.34	1.210 Level 2	
18,900.0	10,685.0	19,165.8	10,767.8	247.0	247.8	98.14	-925.4	8,473.3	585.1	96.0	489.11	1.196 Level 2	
19,000.0	10,685.7	19,265.8	10,768.6	249.9	250.7	98.15	-925.4	8,573.3	585.1	90.2	494.88	1.182 Level 2	
19,100.0	10,686.4	19,365.8	10,769.4	252.8	253.6	98.16	-925.4	8,673.3	585.1	84.4	500.66	1.169 Level 2	
19,200.0	10,687.1	19,465.8	10,770.2	255.8	256.5	98.17	-925.4	8,773.3	585.1	78.7	506.43	1.155 Level 2	
19,300.0	10,687.8	19,565.8	10,771.0	258.7	259.4	98.18	-925.4	8,873.3	585.1	72.9	512.20	1.142 Level 2	
19,400.0	10,688.5	19,665.8	10,771.8	261.6	262.3	98.18	-925.4	8,973.3	585.1	67.2	517.98	1.130 Level 2	
19,500.0	10,689.2	19,765.8	10,772.5	264.5	265.2	98.19	-925.4	9,073.3	585.1	61.4	523.75	1.117 Level 2	
19,600.0	10,689.9	19,865.8	10,773.3	267.4	268.1	98.20	-925.4	9,173.3	585.2	55.6	529.53	1.105 Level 2	
19,700.0	10,690.6	19,965.8	10,774.1	270.3	271.0	98.21	-925.4	9,273.3	585.2	49.9	535.30	1.093 Level 2	
19,800.0	10,691.3	20,065.8	10,774.9	273.3	274.0	98.22	-925.4	9,373.3	585.2	44.1	541.08	1.082 Level 2	
19,900.0	10,692.0	20,165.8	10,775.7	276.2	276.9	98.23	-925.4	9,473.3	585.2	38.3	546.85	1.070 Level 2	
20,000.0	10,692.7	20,265.8	10,776.5	279.1	279.8	98.24	-925.4	9,573.3	585.2	32.6	552.63	1.059 Level 2	
20,100.0	10,693.4	20,365.8	10,777.3	282.0	282.7	98.24	-925.4	9,673.3	585.2	26.8	558.40	1.048 Level 2	
20,200.0	10,694.0	20,465.8	10,778.0	284.9	285.6	98.25	-925.4	9,773.3	585.2	21.1	564.18	1.037 Level 2	
20,300.0	10,694.7	20,565.8	10,778.8	287.9	288.5	98.26	-925.4	9,873.3	585.2	15.3	569.96	1.027 Level 2	
20,400.0	10,695.4	20,665.8	10,779.6	290.8	291.4	98.27	-925.4	9,973.3	585.3	9.5	575.73	1.017 Level 2	
20,500.0	10,696.1	20,765.8	10,780.4	293.7	294.3	98.28	-925.4	10,073.3	585.3	3.8	581.51	1.006 Level 2	
20,600.0	10,696.8	20,865.8	10,781.2	296.6	297.3	98.29	-925.4	10,173.3	585.3	-2.0	587.29	0.997 Level 1	
20,700.0	10,697.5	20,965.8	10,782.0	299.5	300.2	98.29	-925.4	10,273.3	585.3	-7.8	593.06	0.987 Level 1	
20,800.0	10,698.2	21,065.8	10,782.8	302.5	303.1	98.30	-925.4	10,373.3	585.3	-13.5	598.84	0.977 Level 1	
20,900.0	10,698.9	21,165.8	10,783.5	305.4	306.0	98.31	-925.4	10,473.3	585.3	-19.3	604.62	0.968 Level 1	
20,960.1	10,699.4	21,225.9	10,784.0	307.1	307.5	98.32	-925.4	10,533.3	585.3	-22.5	607.83	0.963 Level 1, ES, SF	

Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Reference Depths are relative to Well @ 1967.0usft

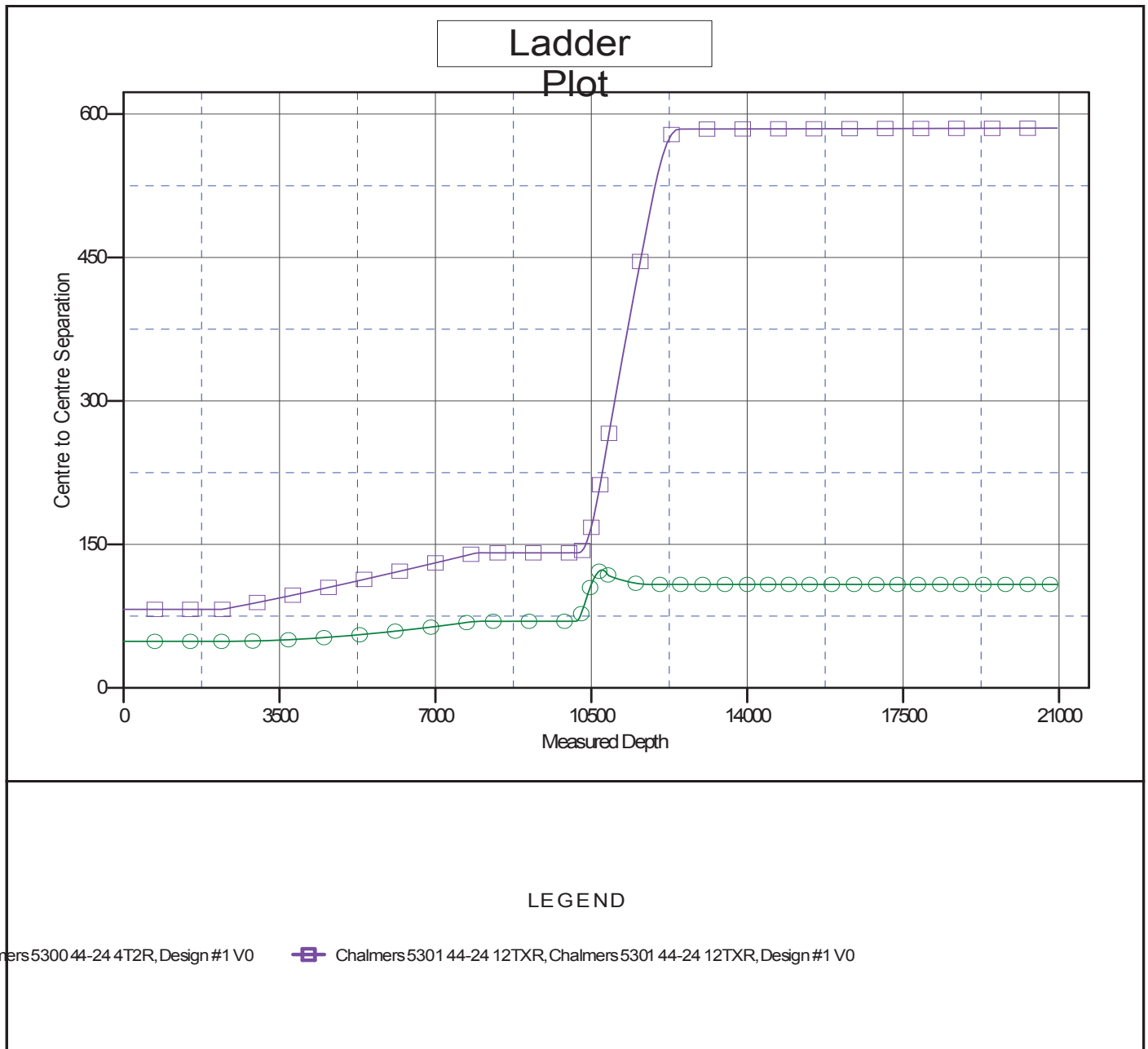
Offset Depths are relative to Offset Datum

Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Chalmers 5301 44-24 3BR

Coordinate System is US State Plane 1983, North Dakota Northern Zone

Grid Convergence at Surface is: -2.31°



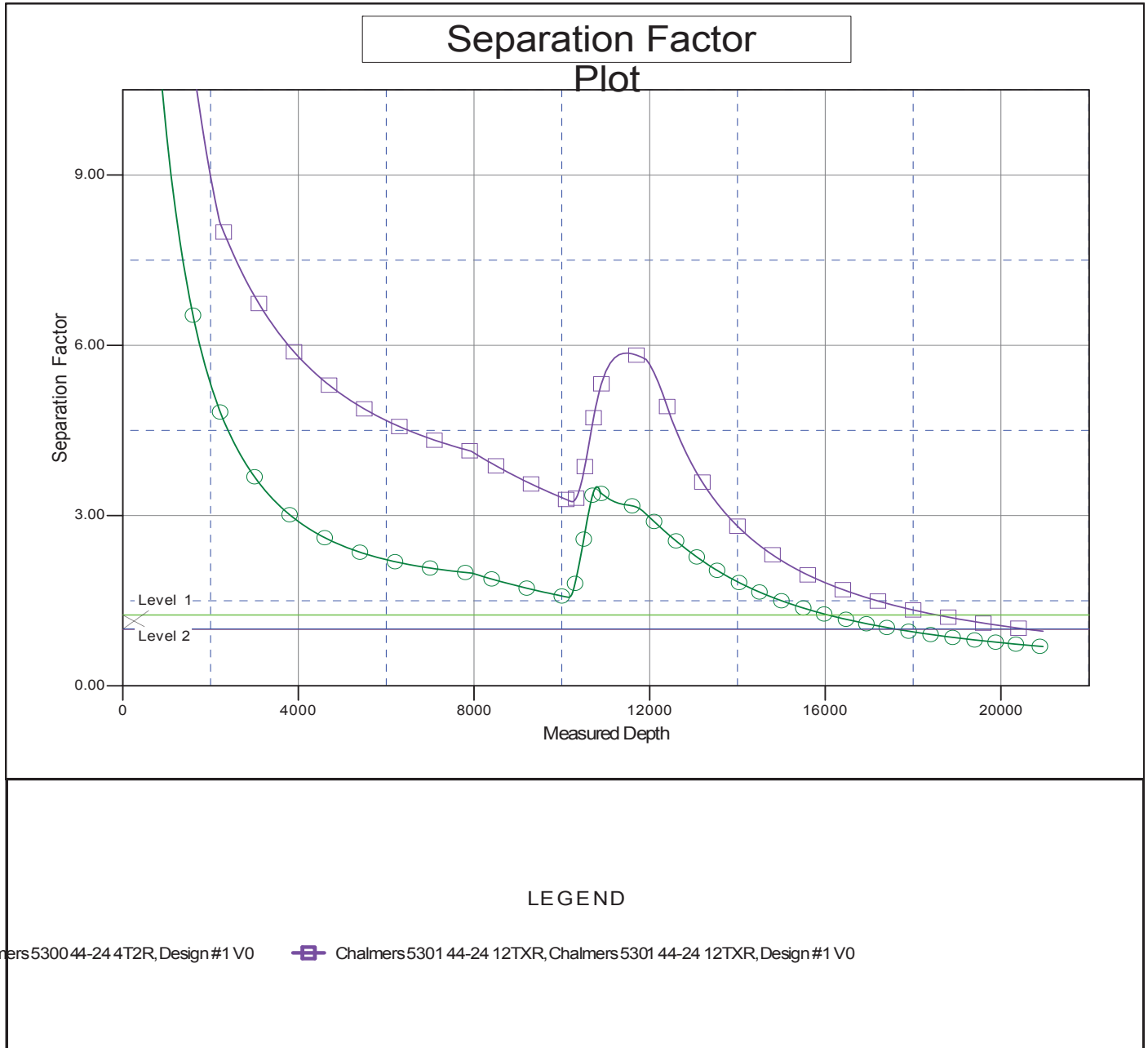
Ryan Directional Services

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Chalmers 5301 44-24 3BR
Project:	Indian Hills	TVD Reference:	Well @ 1967.0usft
Reference Site:	153N-100W-19/20_Altered	MD Reference:	Well @ 1967.0usft
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Chalmers 5301 44-24 3BR	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Chalmers 5301 44-24 3BR	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Reference Depths are relative to Well @ 1967.0usft
 Offset Depths are relative to Offset Datum
 Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Chalmers 5301 44-24 3BR
 Coordinate System is US State Plane 1983, North Dakota Northern Zone
 Grid Convergence at Surface is: -2.31°

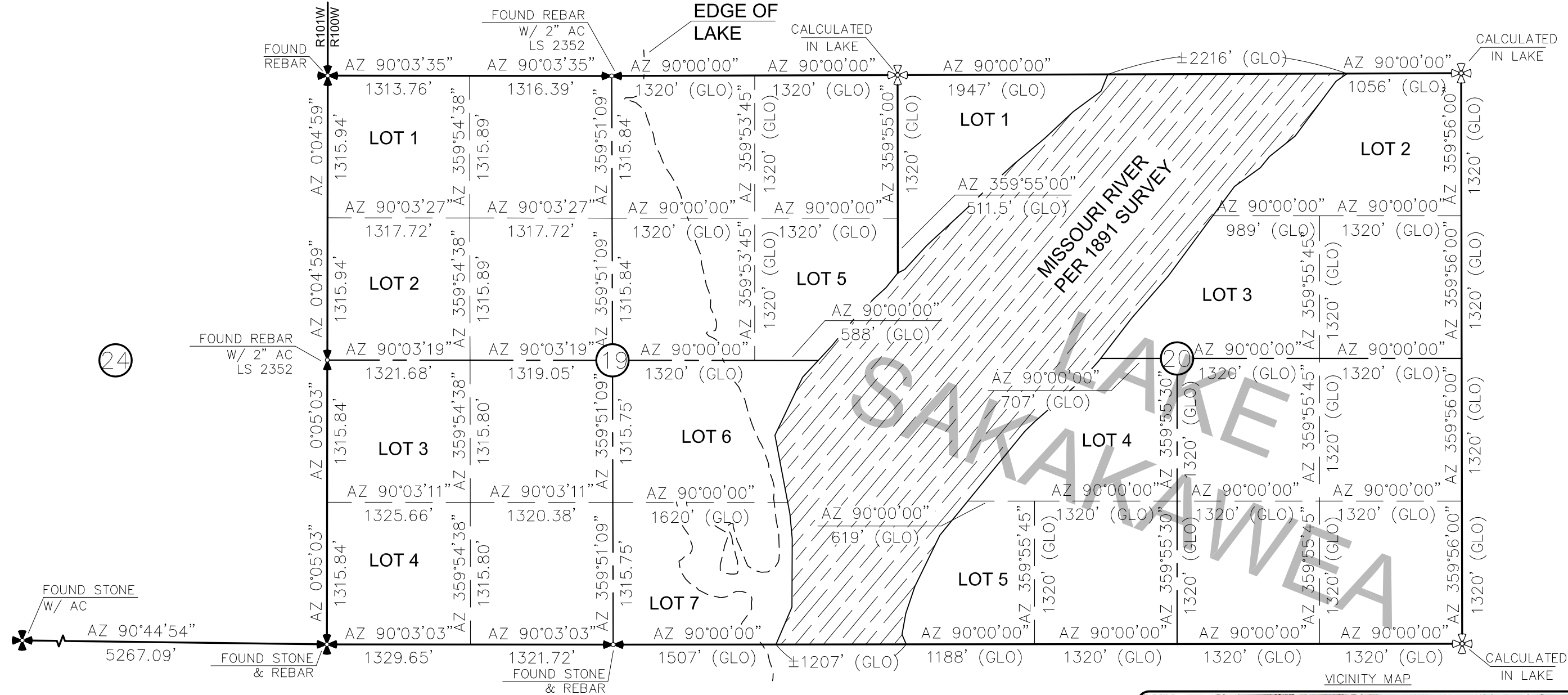


SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

REV	DATE	DESCRIPTION	ADDED DIMENSION
6	6/11/14	JDM	
		Oasis Petroleum - 2 of 3	

2: 2013\513-09-235.01 Oasis Petroleum - 2 of 3 Infill Wells for Chalmers 5300
31-19H\CAD\CHALMERS 3BR UPDATED 6-11-14.dwg - 6/11/2014 9:22 AM jockie miller

SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5301 44-24 3BR"
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
SECTION 24, T153N, R101W, & SECTION 19 & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



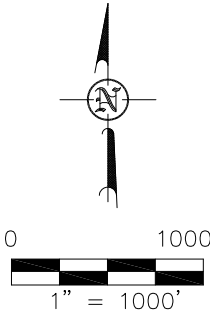
24

FOUND REBAR
W/ 2" AC
LS 2352

FOUND STONE
W/ AC

FOUND STONE
& REBAR

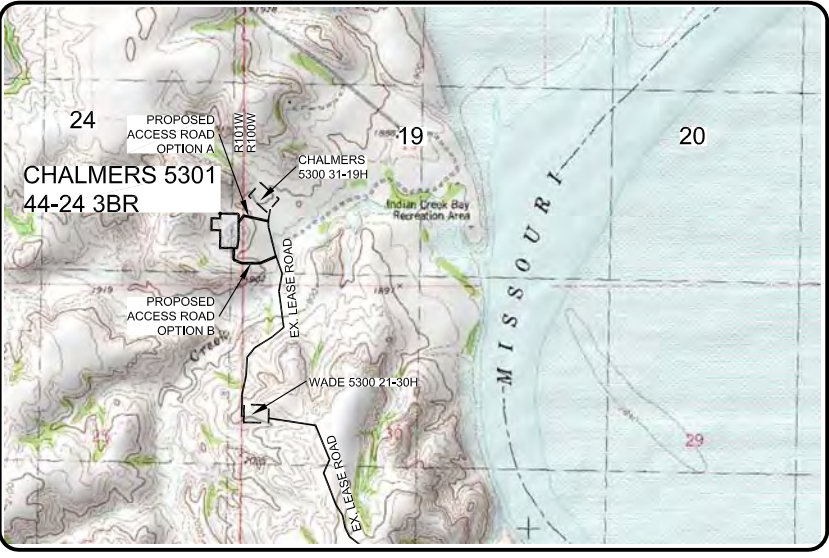
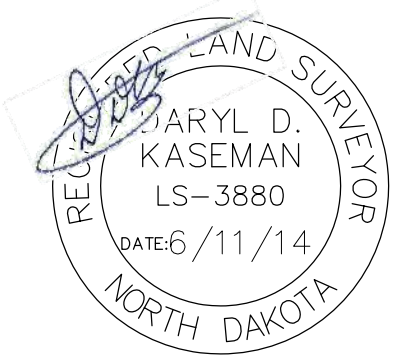
FOUND STONE
& REBAR



- ⊕ - MONUMENT - RECOVERED
- ⊗ - MONUMENT - NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY ISSUED
AND SEALED BY DARYL D. KASEMAN,
PLS, REGISTRATION NUMBER 3880 ON
6/11/14 AND THE ORIGINAL
DOCUMENTS ARE STORED AT THE
OFFICES OF INTERSTATE ENGINEERING,
INC.

ALL AZIMUTHS ARE BASED ON G.P.S.
OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
WAS 1891. THE CORNERS FOUND ARE AS
INDICATED AND ALL OTHERS ARE COMPUTED FROM
THOSE CORNERS FOUND AND BASED ON G.L.O.
DATA. THE MAPPING ANGLE FOR THIS AREA IS
APPROXIMATELY -0°03'.



© 2014, INTERSTATE ENGINEERING, INC.

Revision No.	Date	By	Description
REV 3	1/20/14	BHH	CHANGED PAD LAYOUT
REV 4	5/09/14	JJS	ADDED WELL TO PAD
REV 5	5/20/14	JJS	CHANGED NAME, MOVED WELL
REV 6	6/11/14	JDM	ADDED DIMENSIONS

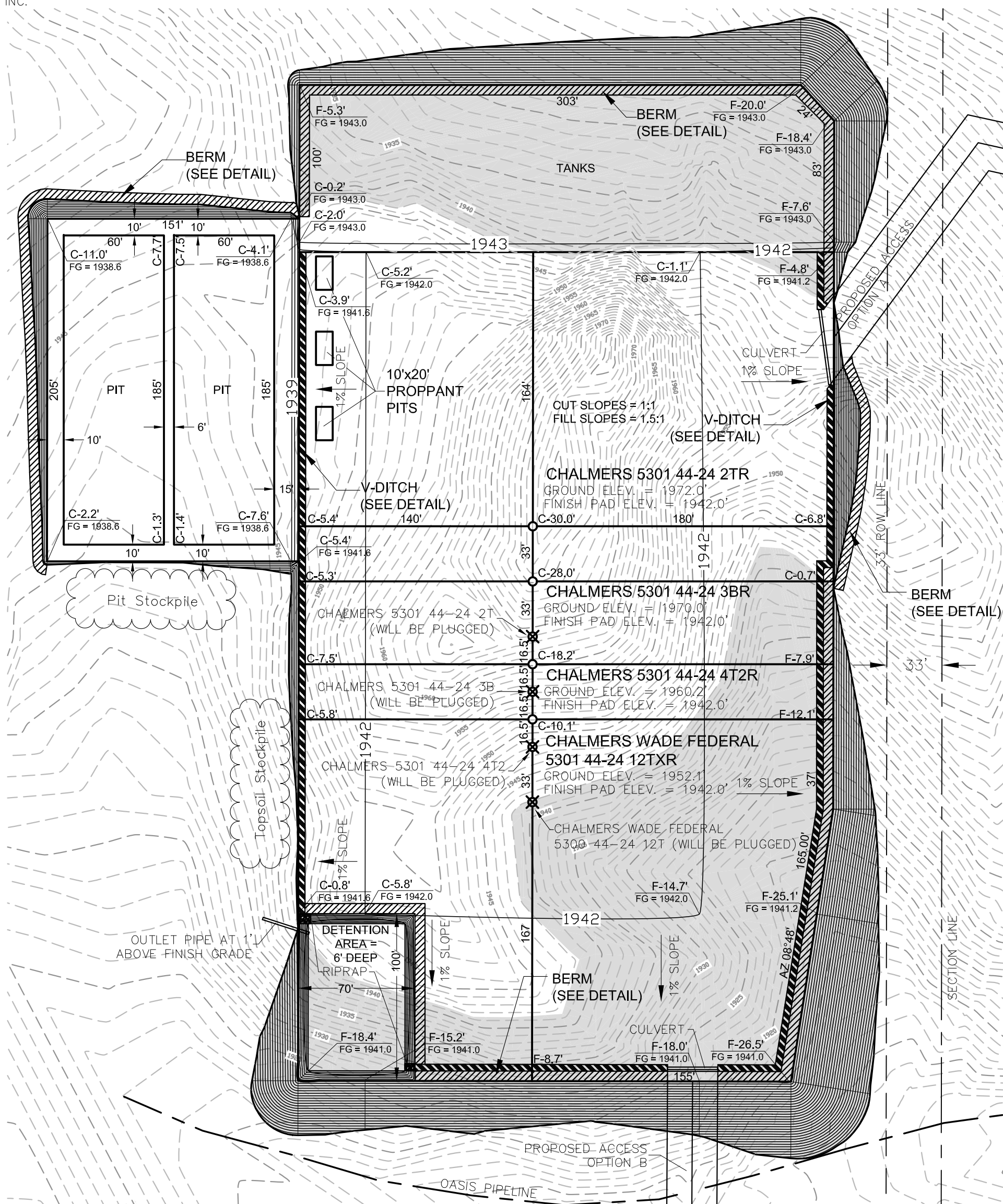
OASIS PETROLEUM NORTH AMERICA, LLC SECTION BREAKDOWN SECTION 24, T153N, R101W, & SECTIONS 19 & 20, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: S13-09-23501 Date: MAY 2014
--	---

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

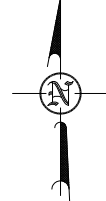


2/8
SHEET NO.

PAD LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5301 44-24 3BR"
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



 — BERM
 — DITCH
 — Proposed Contours
 — Original Contours



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>J.J.S.</u>	Project No.: <u>S13-09-235.0</u>
Checked By: <u>D.D.K.</u>	Date: <u>MAY 2014</u>

Revision No.	Date	By	Description
REV 3	1/20/14	BHH	CHANGED PAD LAYOUT
REV 4	5/09/14	JJS	ADDED WELL TO PAD
REV 5	5/20/14	JJS	CHANGED NAME, MOVED WELL
REV 6	6/11/14	JDM	ADDED DIMENSIONS

Q:\2013\S13-09-235.01 Oasis Petroleum - 2 of 3 Infill Wells for Chalmers 5300
31-10M\CAD\CHALMERS 3BR UPDATED 6-11-14.dwg - 6/11/2014 9:23 AM jackie miller

3/8

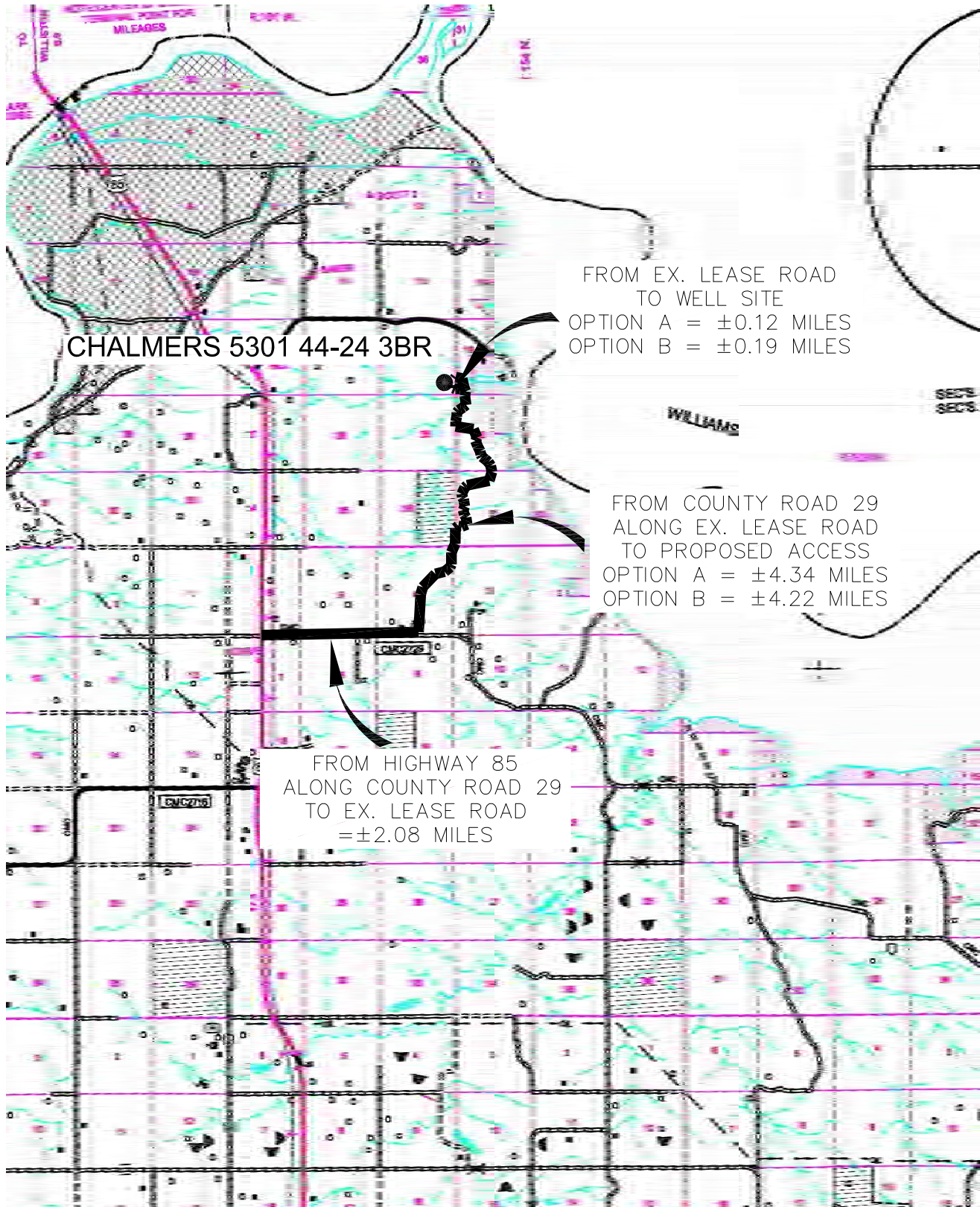
SHEET NO.



Professionals you need, people you trust

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CHALMERS 5301 44-24 3BR"
 926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
 SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

6/8

SHEET NO.



Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.Interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 COUNTY ROAD MAP
 SECTION 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235.01
 Checked By: D.D.K. Date: MAY 2014

Revision No.	Date	By	Description
REV 2	12/5/13	JJS	CHANGED PAD LAYOUT
REV 3	1/20/14	BHH	CHANGED PAD LAYOUT
REV 4	5/09/14	JJS	ADDED WELL TO PAD
REV 5	5/20/14	JJS	CHANGED NAME, MOVED WELL
REV 6	6/11/14	JDM	ADDED DIMENSIONS

WELL LOCATION SITE QUANTITIES
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CHALMERS 5301 44-24 3BR"
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

WELL SITE ELEVATION	1970.0
WELL PAD ELEVATION	1942.0
EXCAVATION PLUS PIT	55,136 9,450 64,586
EMBANKMENT PLUS SHRINKAGE (30%)	37,558 11,267 48,825
STOCKPILE PIT	9,450
STOCKPILE TOP SOIL (6")	4,770
BERMS	1,711 LF = 554 CY
DITCHES	1,029 LF = 157 CY
DETENTION AREA	1,428 CY
STOCKPILE MATERIAL	2,572
DISTURBED AREA FROM PAD	5.91 ACRES

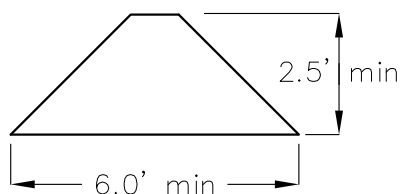
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

926' FSL

245' FEL

BERM DETAIL



DITCH DETAIL



© 2014, INTERSTATE ENGINEERING, INC.

8/8



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.Interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
QUANTITIES
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235.01
Checked By: D.D.K. Date: MAY 2014

Revision No.	Date	By	Description
REV 2	12/5/13	JJS	CHANGED PAD LAYOUT
REV 3	1/20/14	BH	CHANGED PAD LAYOUT
REV 4	5/09/14	JJS	ADDED WELL TO PAD
REV 5	5/20/14	JJS	CHANGED NAME, MOVED WELL
REV 6	6/11/14	JDM	ADDED DIMENSIONS

CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5301 44-24 3BR"

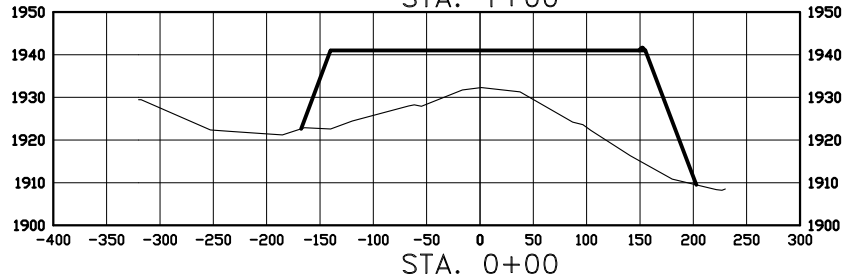
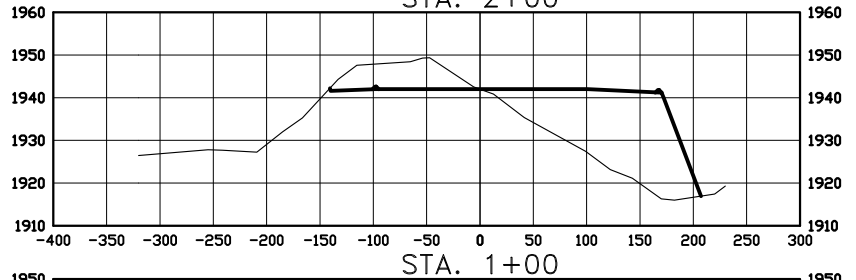
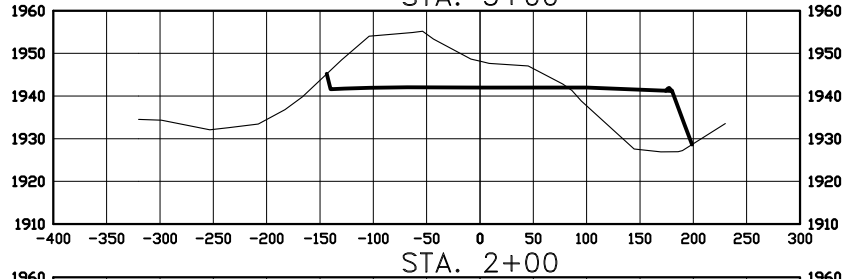
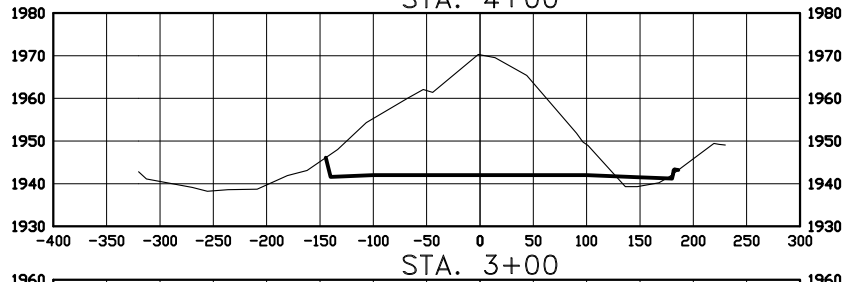
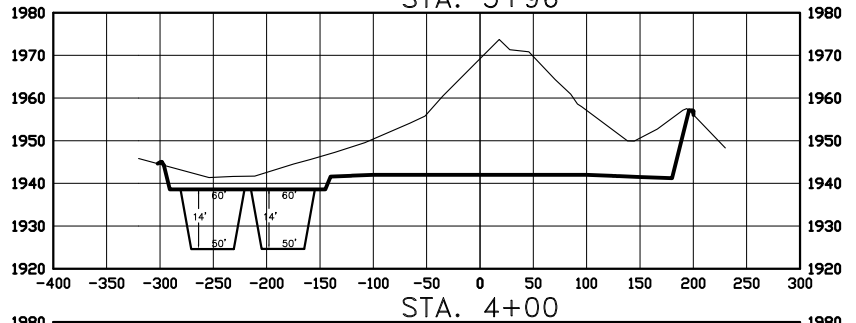
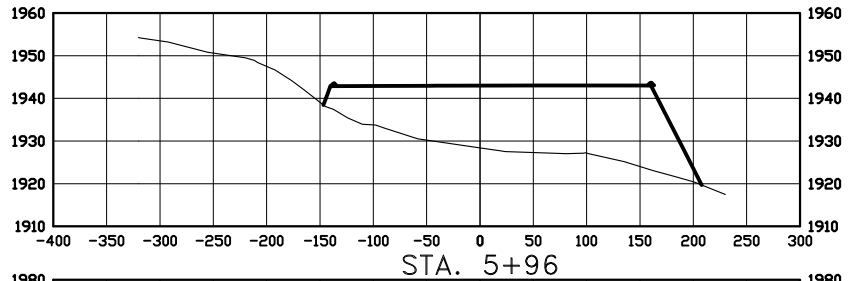
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE

SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED
AND SEALED BY DARYL D. KASEMAN,
PLS, REGISTRATION NUMBER 3880 ON
6/11/14 AND THE ORIGINAL
DOCUMENTS ARE STORED AT THE
OFFICES OF INTERSTATE ENGINEERING,
INC.



SCALE
HORIZ 1"=180'
VERT 1"=45'



© 2014, INTERSTATE ENGINEERING, INC.

7/8
SHEET NO.



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
PAD CROSS SECTIONS
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.S. Project No.: S13-09-235,01
Checked By: D.D.K. Date: MAY 2014

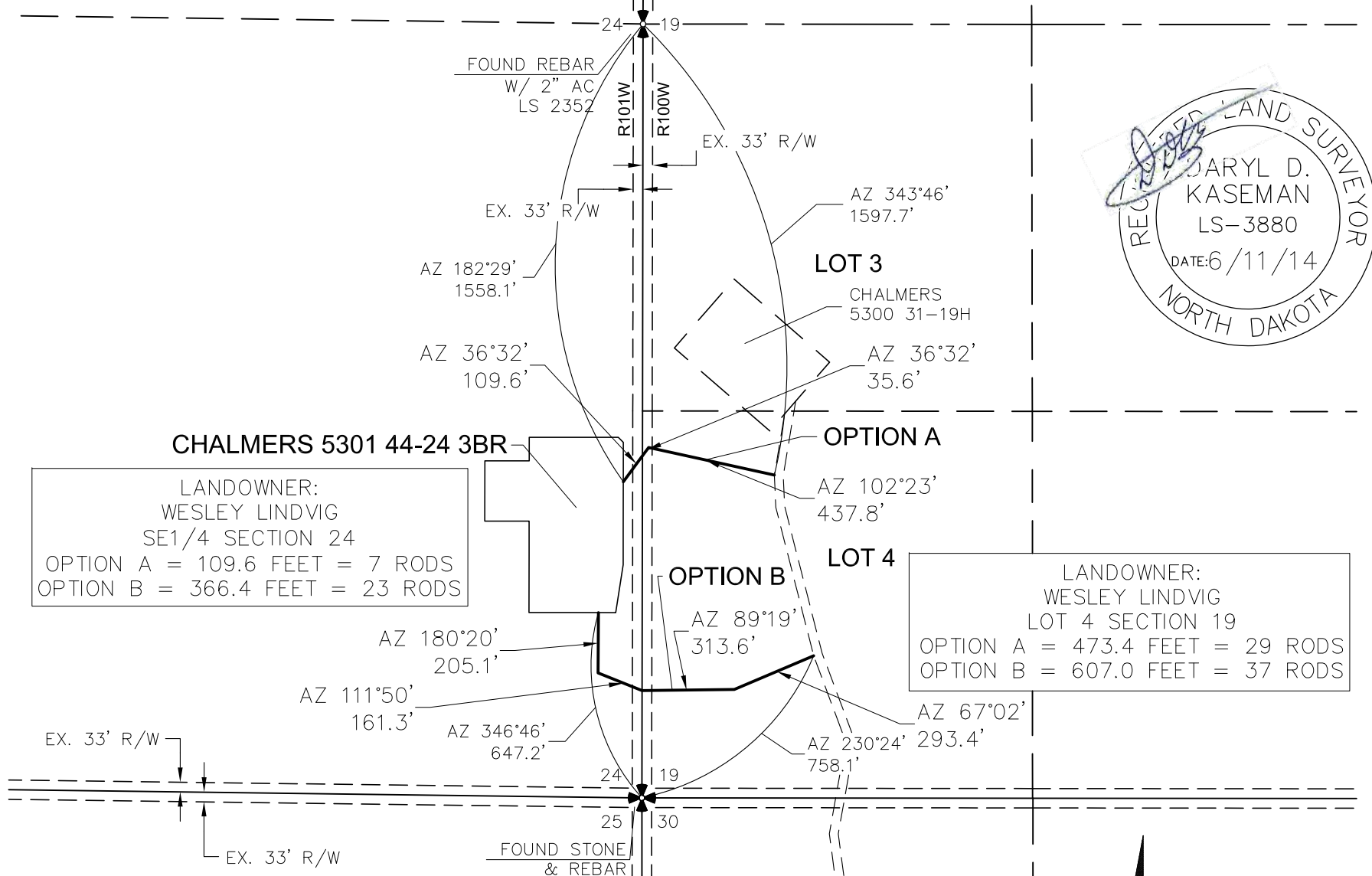
Revision No.	Date	By	Description
REV 2	12/5/13	JJS	CHANGED PAD LAYOUT
REV 3	1/20/14	BHR	CHANGED PAD LAYOUT
REV 4	5/09/14	JJS	ADDED WELL TO PAD
REV 5	5/20/14	JJS	CHANGED NAME, MOVED WELL
REV 6	6/11/14	JDM	ADDED DIMENSIONS

ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5301 44-24 3BR"

926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

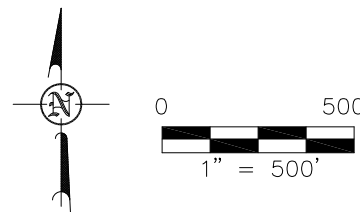


LANDOWNER:
WESLEY LINDVIG
SE1/4 SECTION 24
OPTION A = 109.6 FEET = 7 RODS
OPTION B = 366.4 FEET = 23 RODS

LANDOWNER:
WESLEY LINDVIG
LOT 4 SECTION 19
OPTION A = 473.4 FEET = 29 RODS
OPTION B = 607.0 FEET = 37 RODS

THIS DOCUMENT WAS ORIGINALLY ISSUED AND
SEALED BY DARYL D. KASEMAN, PLS,
REGISTRATION NUMBER 3880 ON 6/11/14
AND THE ORIGINAL DOCUMENTS ARE STORED AT
THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete
utilities location is recommended before construction.



Revision No.	By	Date	Description
REV 1	JLS	12/5/13	CHANGED PAD LAYOUT
REV 2	BH	1/20/14	CHANGED PAD LAYOUT
REV 3	JLS	5/29/14	CHANGED WELL TO PAD
REV 4	JLS	5/29/14	CHANGED NAME, MOVED WELL
REV 5	JLS	6/11/14	ADDED DIMENSIONS

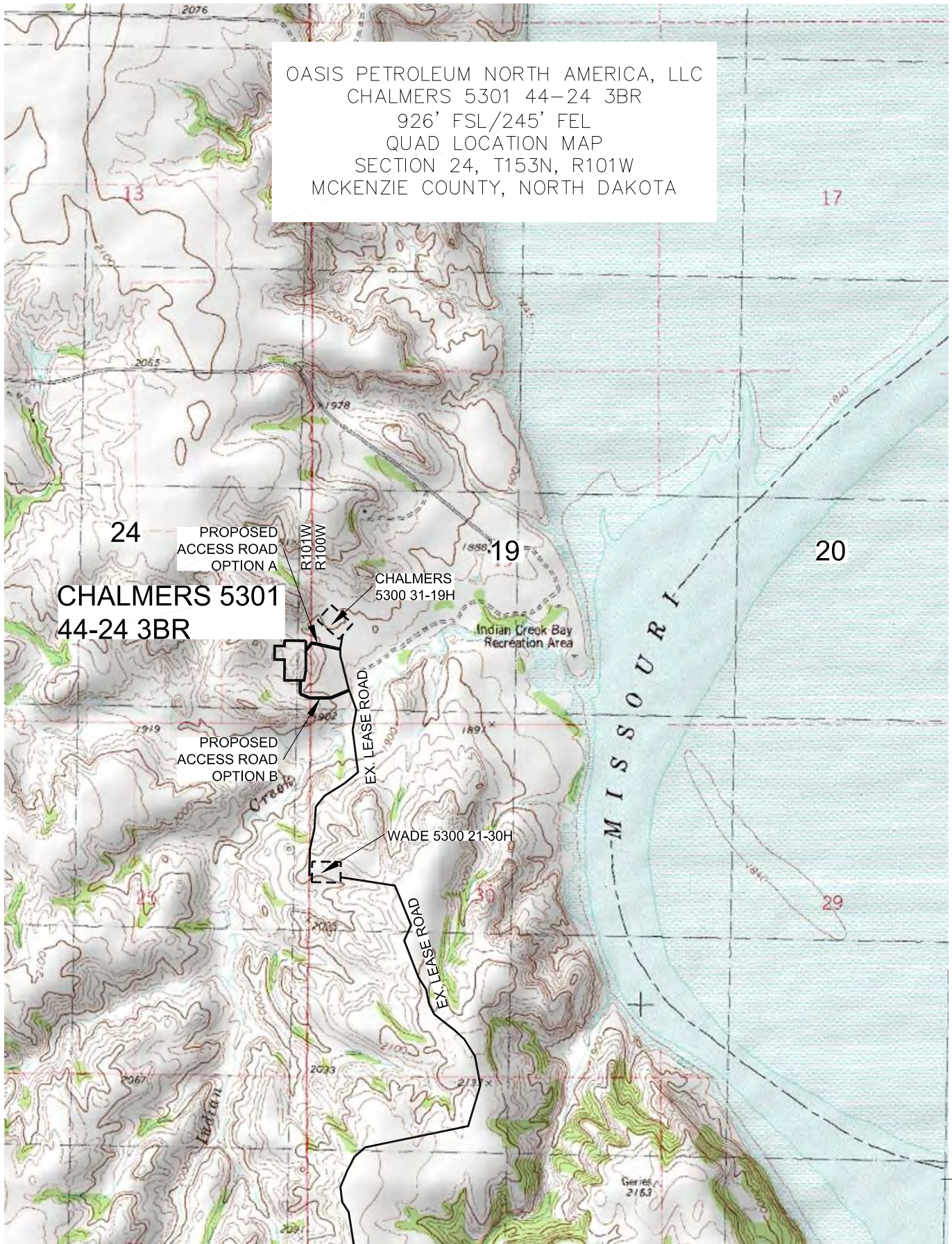
OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: JLS
Checked By: DDK
Project No: S13-09-235.01
Date: MAY 2014

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota



© 2014, INTERSTATE ENGINEERING, INC.
4/8
SHEET NO.

OASIS PETROLEUM NORTH AMERICA, LLC
 CHALMERS 5301 44-24 3BR
 926' FSL/245' FEL
 QUAD LOCATION MAP
 SECTION 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

5/8

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235.01
 Checked By: D.D.K. Date: MAY 2014

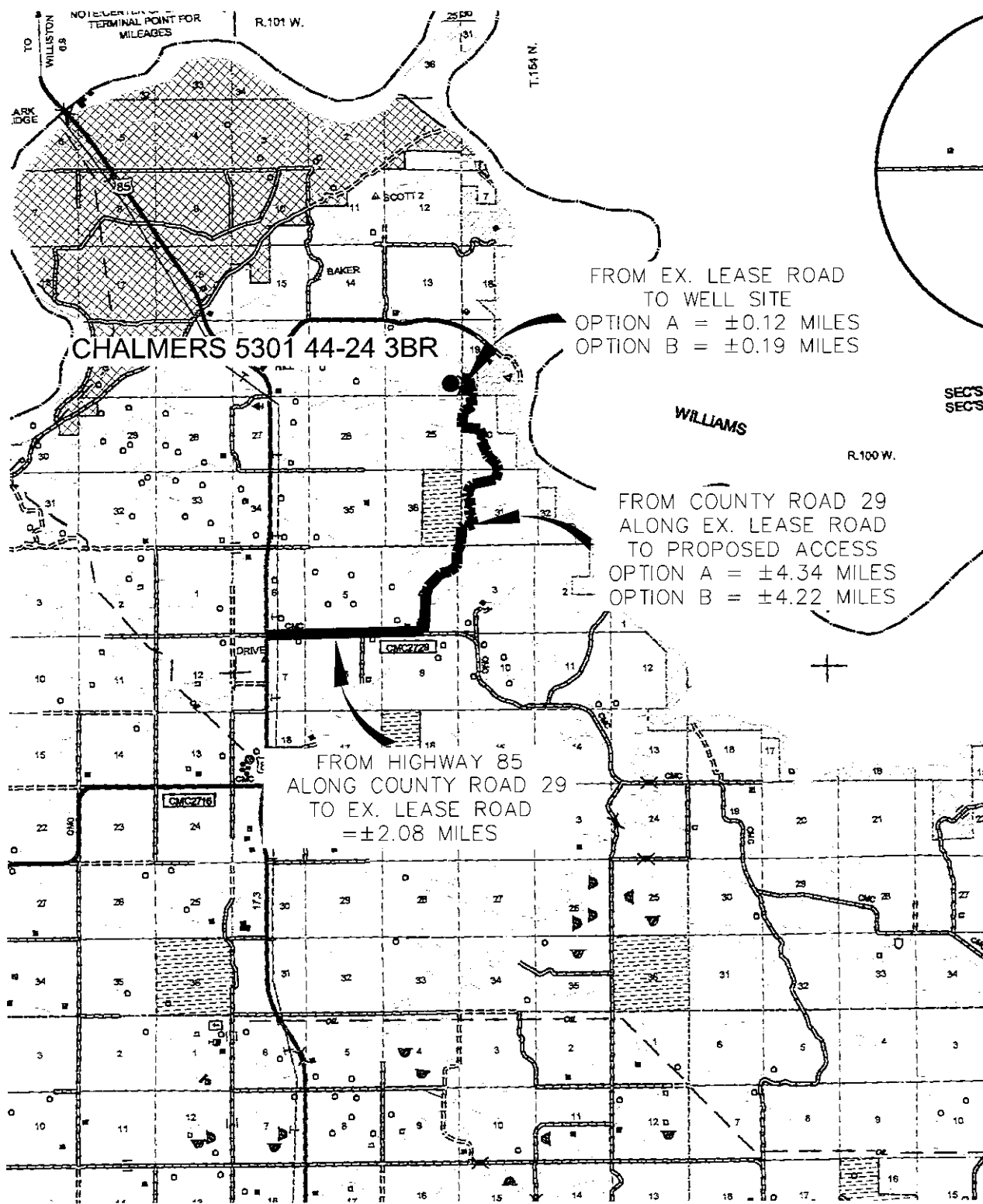
Revision No.	Date	By	Description
REV 2	12/5/13	JJS	CHANGED PAD LAYOUT
REV 3	1/20/14	BHJ	CHANGED PAD LAYOUT
REV 4	5/09/14	JJS	ADDED WELL TO PAD
REV 5	5/20/14	JJS	CHANGED NAME, MOVED WELL
REV 6	6/11/14	JDM	ADDED DIMENSIONS

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5301 44-24 3BR"

926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

6/8



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No: S13-09-235.01
Checked By: D.D.K. Date: MAY 2014

Revision No.	Date	By	Description
REV 1	10/09/13	JJS	CHANGED WELL NAME
REV 2	12/5/13	JJS	CHANGED PAD LAYOUT
REV 3	1/20/14	BNH	CHANGED PAD LAYOUT
REV 4	5/06/14	JJS	ADDED WELL TO PAD
REV 5	5/20/14	JJS	CHANGED NAME, MOVED WELL



6/1/2014

Mineral Resources Permit Manager
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: Chalmers 5301 44-24 3BR
Chalmers 5301 44-24 4T2R
Request for a legal street address

Dear NDIC:

Oasis Petroleum has requested a physical street address for the Chalmers 5301 44-24 3BR and Chalmers 5301 44-24 4T2R. The request was made to Aaron Chisolm, in McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

A handwritten signature in blue ink, appearing to read "Heather McCowan".

Heather McCowan
Regulatory Assistant
Oasis Petroleum North America, LLC

SURFACE DAMAGE SETTLEMENT AND RELEASE

This Surface Damage Settlement and Release amends that certain Surface Use and Access Agreement, dated on that 15th day of October, 2013, by and between Wesley and Barbara Lindvig, wife and husband, "Owner" and together with Oasis Petroleum North America LLC ("Oasis") for the 4.60 acre drill site of the Chalmers 5301 44-24 2T, Chalmers 5301 44-24 3B & Chalmers 5301 44-24 4T wells.

Now Thereof and In consideration for the sum of !

paid by Oasis Petroleum North America LLC ("Oasis") to the undersigned surface owner(s), Wesley and Barbara Lindvig, ("Owner") and together with Oasis, the "Parties") administrators and assigns, who hereby acknowledge the receipt and sufficiency of said payment as a full and complete settlement for and as a release of all claims for loss, damage or injury to the Subject Lands (as defined herein) arising out of the Operations (as defined herein) of the initial well and any number of future infill wells (the "Wells") located on the approximately (5.91) Five and nine one tenths acre tract of land identified on the plat attached hereto as Exhibit "A" (the "Subject Lands") and which is situated on the following described real property located in McKenzie County, State of North Dakota, to wit:

Township 153 North, Range 101 West, 5th P.M.
Section 24: SE¼SE¼

This pad shall accommodate the drilling of the **Chalmers 5301 44-24 2T, Chalmers 5301 44-24 3B, Chalmers 5301 44-24 4T & Chalmers Wade 5301 44-24 12T** well(s).

The undersigned is fully aware that the cuttings generated from the drill of the above described wells will be buried on site on the above described locations.

Grantee hereby agrees to controlling dust during fracing and drilling operations and will police up and control trash along the access road and on the well(s) site.

The Parties agree that the settlement and release described herein does not include any claims by any third party against the Owner for personal injury or property damage arising directly out of Oasis's Operations, and Oasis agrees to indemnify, defend and hold harmless Owner against all liabilities arising from such claim (except as such claim arises from the gross negligence or willful misconduct of Owner).

In further consideration of the payments specified herein, Oasis is hereby specifically granted the right to construct, install and operate, replace or remove pads, pits, pumps, compressors, tanks, roads, pipelines, equipment or other facilities on the above described tract of land necessary for its drilling, completion, operation and/or plugging and abandonment of the Wells (the "Operations"), and to the extent such facilities are maintained by Oasis for use on the Subject Lands, this Agreement shall permit Oasis's use of such facilities for the Operations on the Subject Lands.

Should commercial production be established from the Wells, Oasis agrees to pay Owner an annual amount of \$ _____ per year beginning one year after the completion of the first Well bore and for each additional well bore thereafter until the Wells are plugged and abandoned.

The Parties expressly agree and acknowledge that the payments described herein to be made by Oasis to the Owners constitute full satisfaction of the requirements of Chapter 38.11.1 of the North Dakota Century Code and, once in effect, the amended Chapter 38.11.1 of the North Dakota Century Code enacted by House Bill 1241. The Parties further expressly agree and acknowledge that the payment set forth above constitutes full and adequate consideration for damage and disruption required under Section 38.11.1-04 of the North Dakota Century Code, and that the payment set forth above constitutes full and adequate consideration for loss of production payments under Section 38.11.1-08.1 of the North Dakota Century Code.

Oasis shall keep the Site free of noxious weeds, and shall take reasonable steps to control erosion and washouts on the Site. Oasis shall restore the Site to a condition as near to the original condition of the Site as is reasonably possible, including the re-contouring, replacing of topsoil and re-seeding of the Site (such actions, the "Restoration").

Owner grants Oasis access to the Wells in the location(s) shown on the plat attached hereto as Exhibit "A".

It is agreed that within (1) year following abandonment of the Road, Operator will restore the surface of the Road, as nearly as practicable, to its condition prior to the Operator's operations. This shall include filling same, grading and leveling the Road including re-seeding as directed by Landowner. However, at the written request of Landowner, the Road may remain in place for use of the Landowner. Any gates, cattle guards, and culvert will become the property of the Landowner.

The Owner grants Oasis a perpetual twenty five (25) foot easement for the installation of all flow lines and fresh water lines needed for Operator's operations. Upon written request all flow lines will be removed after the wells are considered abandoned.

This Agreement shall apply to the Parties and their respective its successors, assigns, parent and subsidiary companies, affiliates and related companies, trusts and partnerships, as well as their contractors, subcontractors, officers, directors, agents and employees.

This Agreement may be executed in multiple counterparts, each of which shall be an original, but all of which shall constitute one instrument.

Replacement well names:

- ① Chalmers 5301 44-24 3B is now Chalmers 5301 44-24 3BR
- ② Chalmers 5301 44-24 4T is now Chalmers 5301 44-24 4TR
- ③ Chalmers Wade 5301 44-24 12T is now Chalmers Wade 44-24 12TR

HOW
u/11/14

DATED this 30 day of April, 2014

SURFACE OWNER

Wesley Lindvig
Wesley Lindvig

Barbara J. Lindvig
Barbara Lindvig

Address: 14075 41st Street NW

Alexander, ND 58831

Phone: 701-770-1152

ACKNOWLEDGMENT INDIVIDUAL

STATE OF North Dakota SS.
COUNTY OF McKenzie

BE IT REMEMBERED, That on this 30 day of April, 2014, before me, a Notary Public, in and for said County and State, personally appeared Wesley and Barbara Lindvig, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me that they executed the same as their free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year first above written.

My commission expires

JAMES DEMORRETT
Notary Public
State of North Dakota
NOTARY STAMP My Commission Expires Aug. 16, 2018

James Demorrett

Notary Public